

LWE with Quantum Amplitudes: Algorithm, Hardness, and Oblivious Sampling

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Abstract

The learning with errors problem (LWE) is one of the most important building blocks for post-quantum cryptography. To better understand the quantum hardness of LWE, it is crucial to explore quantum variants of LWE. To this end, Chen, Liu, and Zhandry [Eurocrypt 2022] defined $S|LWE\rangle$ and $C|LWE\rangle$ problems by encoding the error of LWE samples into quantum amplitudes, and showed efficient quantum algorithms for a few interesting amplitudes. However, algorithms or hardness results of the most interesting amplitude, Gaussian, were not addressed before.

In this paper, we show new algorithms, hardness results and applications for $S|LWE\rangle$ and $C|LWE\rangle$ with real Gaussian, Gaussian with linear or quadratic phase terms, and other related amplitudes. Let n be the dimension, q be the modulus of LWE samples. Our main results are

1. There is a $2^{\Theta(\sqrt{n \log q})}$ -time algorithm for $S|LWE\rangle$ with Gaussian amplitude with *known* phase, given $2^{\Theta(\sqrt{n \log q})}$ many quantum samples. The algorithm is modified from Kuperberg's sieve, and in fact works for more general amplitudes as long as the amplitudes and phases are completely *known*.
2. There is a polynomial time quantum algorithm for solving $S|LWE\rangle$ and $C|LWE\rangle$ for Gaussian with quadratic phase amplitudes, where the sample complexity is as small as $\tilde{O}(n)$. As an application, we give a quantum oblivious LWE sampler where the core quantum sampler requires only quasi-linear sample complexity. This improves upon the previous oblivious LWE sampler given by Debris-Alazard, Fallahpour, Stehlé [STOC 2024], whose core quantum sampler requires $\tilde{O}(nr)$ sample complexity, where r is the standard deviation of the error.
3. There exist polynomial time quantum reductions from standard LWE or worst-case GapSVP to $S|LWE\rangle$ with Gaussian amplitude with small *unknown* phase, and arbitrarily many samples. Compared to the first two items, the appearance of the unknown phase term places a barrier in designing efficient quantum algorithm for solving standard LWE via $S|LWE\rangle$.

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1 Introduction

The learning with errors problem asks to learn a secret vector given many noisy linear samples.

Definition 1 (Learning with errors (LWE) [Reg09]). *Let n, m, q be positive integers. Let $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector. The search LWE problem $\text{LWE}_{n,m,q,\alpha}$ asks to find the secret $\mathbf{s}$ given access to an oracle that outputs $\mathbf{a}_i, \langle \mathbf{s}, \mathbf{a}_i \rangle + e_i \pmod{q}$ on its i^{th} query, for $i = 1, \dots, m$. Here each $\mathbf{a}_i$ is a uniformly random vector in $\mathbb{Z}_q^n$, and each error term e_i is sampled from the Gaussian distribution over $\mathbb{Z}$ with standard deviation $\alpha q / \sqrt{2\pi}$.*

The LWE problem is extremely versatile, leading to advanced encryption schemes such as fully homomorphic encryptions [Gen09, BV11, Mah18]. It is also shown by Regev to be quantumly as hard as the approximate short vector problems for all lattices [Reg09]. LWE and lattice problems in general (e.g. [HPS98, Reg09]) are also popular candidates for the NIST post-quantum cryptography standardization, due to their conjectured hardness against quantum computers. However, the conjectured quantum hardness of lattice problem is still lacking solid evidences. Finding quantum algorithms for lattice problems has therefore been a major open problem in the area of quantum computation and cryptography in the past decade.

One way of exploring the quantum power for solving LWE is to consider the quantum variants of LWE, by encoding quantum states into the LWE problem (henceforth, we refer to the original LWE problem as “classical LWE” or “standard LWE” to distinct them from the quantum variant mentioned below). To this end, Chen, Liu, and Zhandry [CLZ22] define two quantum variants of LWE.

Definition 2 (Solve $|\text{LWE}\rangle$, $\text{S}|\text{LWE}\rangle$). *Let n, m, q be positive integers. Let f be a normalized function from $\mathbb{Z}_q$ to $\mathbb{C}$. Let $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector. The problem $\text{S}|\text{LWE}\rangle_{n,m,q,f}$ asks to find $\mathbf{s}$ given access to an oracle that outputs independent samples of form $\mathbf{a}_i, \sum_{e_i \in \mathbb{Z}_q} f(e_i) |\langle \mathbf{s}, \mathbf{a}_i \rangle + e_i \pmod{q}\rangle$ on its i^{th} query, for $i = 1, \dots, m$. Here each $\mathbf{a}_i$ is a uniformly random vector in $\mathbb{Z}_q^n$.*

Definition 3 (Construct $|\text{LWE}\rangle$ states, $\text{C}|\text{LWE}\rangle$). *Let n, m, q be positive integers. Let f be a normalized function from $\mathbb{Z}_q$ to $\mathbb{C}$. The problem of constructing LWE states $\text{C}|\text{LWE}\rangle_{n,m,q,f}$ asks to construct a quantum state of the form*

$$\frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \bigotimes_{i=1}^m \left(\sum_{e_i \in \mathbb{Z}_q} f(e_i) |\langle \mathbf{s}, \mathbf{a}_i \rangle + e_i \pmod{q}\rangle \right),$$

given the input $\{\mathbf{a}_i\}_{i=1,\dots,m}$ where each $\mathbf{a}_i$ is a uniformly random vector in $\mathbb{Z}_q^n$, where N is the normalization factor.

Here, the consideration of using complex amplitudes and adding precise normalization factors in the definition are due to Debris-Alazard, Fallahpour, and Stehlé [DFS24].

Note that if there is a quantum algorithm that solves $\text{S}|\text{LWE}\rangle_{n,m,q,f}$ without collapsing the input state, then there is a quantum algorithm that solves $\text{C}|\text{LWE}\rangle_{n,m,q,f}$ for the same n, m, q, f . So in the introduction we will say more about $\text{S}|\text{LWE}\rangle$ since it is more fundamental than $\text{C}|\text{LWE}\rangle$. We will mention $\text{C}|\text{LWE}\rangle$ when $\text{C}|\text{LWE}\rangle$ is necessary for the application.

Chen, Liu, and Zhandry [CLZ22] then show a quantum filtering technique to solve $S|LWE\rangle$ when the DFT of the error amplitude is non-negligible everywhere over $\mathbb{Z}_q$. Two interesting error amplitudes covered by their result are the bounded uniform amplitude and Laplacian amplitude. In particular, the classical LWE problem with bounded uniform error distribution is proven to be as hard as worst-case lattice problems [DM13, MP13]. Thus, their work gives a strong indication that $S|LWE\rangle$ is easier than classical LWE for certain error distributions.

However, the most interesting amplitude, Gaussian, was not addressed in [CLZ22]. For classical LWE, Gaussian distribution is the default error distribution since Regev [Reg09] shows a quantum reduction from worst-case lattice problems to LWE with Gaussian error, given *arbitrarily* many LWE samples. Interestingly, it was implicitly shown in [SSTX09] and [BKS18] that $S|LWE\rangle$ with Gaussian amplitude is as hard as standard LWE when the number of samples is *very small*. But if we are given *arbitrarily* many samples, is the $S|LWE\rangle$ problem with Gaussian amplitude still hard?

Not only did understanding $S|LWE\rangle$ with Gaussian amplitude shed light on our knowledge of the standard LWE, but it was also considered a bedrock on which interesting quantum protocols can be based, especially unclonable cryptography. The idea was first shown by Zhandry [Zha19] for a potential approach to construct a very powerful cryptographic primitive called quantum lightning, while its concrete and secure instantiation still remains unknown. After that, Khesin, Lu, and Shor proposed another lightning construction [KLS22] based on $S|LWE\rangle$ but later was broken by Liu, Montgomery, and Zhandry [LMZ23]. Poremba [Por23], and Ananth, Poremba, and Vaikuntanathan [APV23] build certifiable deletion based on $S|LWE\rangle$, basing on certain conjectured security. If $S|LWE\rangle$ with Gaussian amplitude was not secure, then all schemes mentioned before may not even have semantic security, let alone its unclonability.

In addition, recently, Debris-Alazard, Fallahpour, and Stehlé [DFS24] showed that solving the $C|LWE\rangle$ problem for proper amplitudes implies a quantum *oblivious LWE sampler*. Roughly speaking, an oblivious LWE sampler is able to take as input the public matrix $\mathbf{A}$, output an LWE sample $\mathbf{A}^T \mathbf{s} + \mathbf{e} \bmod q$ without knowing the secret $\mathbf{s}$. If an oblivious LWE sampler exists, it refutes the typical knowledge assumption for LWE used in several lattice-based SNARK constructions (e.g. [GMNO18], see [DFS24] for more discussions) which assume that anyone who generates an LWE sample must know the LWE secret. While we don't know of any classical oblivious LWE sampler, Debris-Alazard et al. showed that solving $C|LWE\rangle$ is perfect for the task of oblivious LWE sampling, since if we are able to generate a state like $\frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \bigotimes_{i=1}^m \left(\sum_{e_i \in \mathbb{Z}_q} f(e_i) |\langle \mathbf{s}, \mathbf{a}_i \rangle + e_i \bmod q \rangle\right)$, and then measure it, we will get an LWE sample $\mathbf{A}^T \mathbf{s} + \mathbf{e} \bmod q$ with a random and unknown secret $\mathbf{s}$. Debris-Alazard et al. then design an amplitude f whose norm is Gaussian, and whose phase is cleverly chosen, so that the algorithm of quantum unambiguous measurement [CB98] is able to solve $C|LWE\rangle$ for such an amplitude with relatively few samples.

In sum, the motivations of this work are addressing the following questions:

What amplitudes are useful for $S|LWE\rangle$ or $C|LWE\rangle$ towards solving standard LWE or improving quantum applications like oblivious LWE sampling?

Can we show algorithms or hardness for solving $S|LWE\rangle$ or $C|LWE\rangle$ for Gaussian and other related amplitudes?

Error Amplitude	# Samples	Algorithm or Hardness	Reference
Gaussian	$\tilde{O}(n)$	As hard as LWE or approx-GapSVP	[SSTX09, BKS18]
Gaussian	$2^{O(\sqrt{n \log q})}$	$2^{O(\sqrt{n \log q})}$ -time quantum algorithm	This work §3
Complex Gaussian	$\tilde{O}(n)$	$\text{poly}(n)$ -time quantum algorithm	This work §4
Gaussian with unknown phase	Arbitrary	As hard as LWE or approx-GapSVP	This work §5,6
Known with non-negl DFT	$\text{poly}(n, q)$	$\text{poly}(n, q)$ -time quantum algorithm	[CLZ22]
Known with non-negl DFT	$\text{poly}(n, \log q)$	$\text{poly}(n, \log q)$ -time quantum algorithm	[DFS24]

Table 1: Algorithm and hardness of $\mathbf{S}|\text{LWE}\rangle$ for various error amplitudes

1.1 Main results

In this paper, we show new quantum algorithms, hardness results, and applications for $\mathbf{S}|\text{LWE}\rangle$ and $\mathbf{C}|\text{LWE}\rangle$ with Gaussian and other related amplitudes. A summary of the results is given in Table 1.

Before introducing our new algorithms for solving $\mathbf{S}|\text{LWE}\rangle$, let us take a step back and review the existing methods of solving $\mathbf{S}|\text{LWE}\rangle$ for a general amplitude f . Fix some normalized amplitude $f : \mathbb{Z}_q \rightarrow \mathbb{C}$ and suppose for now that f is centered at 0. Then each sample of $\mathbf{S}|\text{LWE}\rangle_{n,m,q,f}$ is given by

$$\mathbf{a} \in \mathbb{Z}_q^n, \quad |\phi_{f,\langle \mathbf{a}, \mathbf{s} \rangle}\rangle := \sum_{x \in \mathbb{Z}_q} f(x) |x + \langle \mathbf{a}, \mathbf{s} \rangle \bmod q\rangle. \quad (1)$$

The state $|\phi_{f,\langle \mathbf{a}, \mathbf{s} \rangle}\rangle$ can be seen as a shift of $|\phi_{f,0}\rangle$ where the center of the state is shifted from 0 to $\langle \mathbf{a}, \mathbf{s} \rangle$. A general approach of solving $\mathbf{S}|\text{LWE}\rangle$ is then: try to design a quantum algorithm that predicts the center $\langle \mathbf{a}, \mathbf{s} \rangle$ of each state $|\phi_{f,\langle \mathbf{a}, \mathbf{s} \rangle}\rangle$ correctly, and if we predict the centers of n states correctly, use Gaussian elimination to find out $\mathbf{s}$.

However, a difficulty faced by the general approach is: for a “typical” error amplitude, like Gaussian of some width $r \in (\sqrt{n}, q/\sqrt{n})$, i.e., for $\rho_r(x) := \exp\left(-\pi \frac{x^2}{r^2}\right)$, the overlap between $|\phi_{\rho_r,0}\rangle$ and $|\phi_{\rho_r,1}\rangle$ is quite large, which means it is inherently hard to distinguish states with adjacent shifts with high confidence. The quantum algorithms used in [CLZ22, DFS24] have to use sophisticated measurements depending on f to guarantee that once the center c of $|\phi_{f,c}\rangle$ is predicted correctly for some samples, we know it is correct. But the probability of predicting correctly in the algorithms used in [CLZ22, DFS24] is proportional to the square of the minimum of $|\text{DFT}_q(f)|$ and is in general very low (the success probability of [DFS24] is better than [CLZ22] in its dependency on q). Although the success probability is non-negligible for certain f , it is exponentially small for the typical Gaussian amplitude. For example, in [DFS24], the required sample complexity is $\frac{n \cdot \omega(\log n)}{q \cdot \min_{x \in \mathbb{Z}_q} |\text{DFT}_q(f)(x)|^2}$. By taking $\min_{x \in \mathbb{Z}_q} |\text{DFT}_q(\rho_r)(x)|^2 \approx \frac{1}{\sqrt{2r}} \cdot \rho_{\frac{q}{\sqrt{2r}}}(\frac{q}{2}) \approx \frac{\sqrt{2r}}{q} \cdot e^{-\pi \frac{r^2}{2}}$, the sample complexity for solving $\mathbf{S}|\text{LWE}\rangle$ for Gaussian amplitude ρ_r is greater than $O(n/r) \cdot e^{\pi \frac{r^2}{2}}$, which is exponential in n when $r > \sqrt{n}$ (when r is in $O(n^{0.5-c})$ for any $c > 0$, the classical Arora-Ge algorithm [AG11] has solved classical LWE with error distribution ρ_r in subexponential time and sample complexity already). Therefore [CLZ22, DFS24] are not able to provide subexponential time algorithms for solving $\mathbf{S}|\text{LWE}\rangle$ for the Gaussian amplitude ρ_r when $r > \sqrt{n}$.

Subexponential time algorithm for general amplitudes. Our first result is a sub-exponential time quantum algorithm for solving $\mathbf{S}|\text{LWE}\rangle$ with Gaussian amplitude, given sub-exponentially

many $S|LWE\rangle$ samples. In fact, it works for a more general amplitude f as long as two points in the discrete Fourier transform of f are more than subexponentially small (the DFT of Gaussian certainly has two non-negligible points). The algorithm combines Kuperberg's sieve [Kup05] and quantum rejection sampling [ORR13].

Theorem 4 (Theorem 31, informal). *Let $f : \mathbb{Z} \rightarrow \mathbb{C}$ be a known, normalized error amplitude function for $S|LWE\rangle$ such that for the DFT $_q$ of f , denoted by $\hat{f}_q$, there exists two distinct values $j_1, j_2 \in \mathbb{Z}_q$ such that $\gcd(j_1 - j_2, q) = 1$ and $|\hat{f}_q(j_1)|, |\hat{f}_q(j_2)| \geq 2^{-\sqrt{n \log q}}$, and $\hat{f}_q(j_1), \hat{f}_q(j_2)$ are computable in time $2^{\Theta(\sqrt{n \log q})}$. Then there exists a quantum algorithm that, given $m = 2^{\Theta(\sqrt{n \log q})}$ samples of $S|LWE\rangle$ with amplitude f , finds the secret within a time complexity of $2^{\Theta(\sqrt{n \log q})}$.*

Polynomial time algorithm for complex Gaussian amplitudes. Our second result shows when the amplitude f is Gaussian with quadratic imaginary phase (henceforth complex Gaussian), i.e., $f_{r,t}(x) = \exp(-\pi(\frac{1}{r^2} + \frac{i}{t})x^2)$ for some $r \in \mathbb{R}$ and $t \in \mathbb{N}^+$, there are $\text{poly}(n, \log q)$ time quantum algorithms for solving $S|LWE\rangle$ and $C|LWE\rangle$.

Our algorithm does not rely on the non-negligible DFT of $f_{r,t}$ like [CLZ22, DFS24] (in fact, the DFT of $f_{r,t}$ is negligible somewhere, see Figure 1). Instead, it makes use of an interesting property of the complex Gaussian function $f_{r,t}(x) = \exp(-\pi(\frac{1}{r^2} + \frac{i}{t})x^2)$: when $r > t \cdot \Omega(\sqrt{n})$, the overlap between $|\phi_{f_{r,t},c_1}\rangle$ and $|\phi_{f_{r,t},c_2}\rangle$ (following the notation in Eqn. (1)) is negligible whenever $c_1 \not\equiv c_2 \pmod{t}$. In other words, given a state $|\phi_{f_{r,t},c}\rangle$, it should in principle be easy to extract the value of $c \bmod t$. Indeed, we show when $r > \tilde{O}(n) \cdot t$, if we simply measure the least significant bits of $|\phi_{f_{r,t},c}\rangle$ by the purely imaginary Gaussian basis $\{|\psi_d\rangle\}_{d \in \mathbb{Z}_t}$, where

$$|\psi_d\rangle = \frac{1}{\sqrt{t}} \sum_{x \in \mathbb{Z}_t} e^{-\frac{\pi i(x-d)^2}{t}} |x\rangle,$$

and output the measurement result $d \in \mathbb{Z}_t$, then $d = c \bmod t$ with probability $1 - O(1/n)$, which finds the center $c \bmod t$ of $|\phi_{f_{r,t},c}\rangle$ with high probability. Utilizing this center finding procedure, we can solve a variant of $S|LWE\rangle$ where $q = q_1 q_2$ with coprime factors q_1, q_2 , such that the first half of samples can be viewed as from $S|LWE\rangle_{n, \tilde{O}(n), q_1, f_{r, q_1}}$, while the second half can be viewed as samples from $S|LWE\rangle_{n, \tilde{O}(n), q_2, f_{r, q_2}}$ with the same secret $\mathbf{s} \bmod q$. Then we employ Guess-then-Gaussian-Elimination on the two halves to recover $\mathbf{s} \bmod q_1$ and $\mathbf{s} \bmod q_2$, and finally use the Chinese Remainder Theorem to recover the whole secret in $\mathbb{Z}_q$.

The idea can be generalized to the case where q has more than two factors, and it directly works for solving $C|LWE\rangle$. Our main theorem is

Theorem 5. *Let n, m, q, ℓ be positive integers and r be a real number. Suppose that $m = \ell n \cdot \omega(\log n)$, q is a composite number satisfying $q = q_1 q_2 \cdots q_\ell$ where $q_1, q_2, \dots, q_\ell$ are coprime, r satisfies $\frac{q}{2\sqrt{n}} > r > 30n \log n \cdot \max\{q_1, q_2, \dots, q_\ell\}$.*

There exists a quantum algorithm running in time $\text{poly}(n, \ell, \log q)$ that, takes input $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, outputs a state ϕ' such that the trace distance between ϕ' and $\phi := |\phi\rangle\langle\phi|$ is negligible, where

$$|\phi\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |(\mathbf{A}^T \mathbf{s} + \mathbf{x}) \bmod q\rangle$$

is a quantum LWE state with Gaussian amplitude and quadratic phase terms where N is the normalization factor, and here we write $\mathbf{x}$ into ℓ blocks $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_\ell)$ with $\mathbf{x}_j \in \mathbb{Z}^{m/\ell}$, $j = 1, 2, \dots, \ell$.

Error Amplitude	# Samples	Algorithm	Reference
Bounded uniform	$O(nrq^4)$	Quantum filtering	[CLZ22]
Gaussian with half phase	$\tilde{O}(nr)$	Unambiguous measurement	[DFS24]
Complex Gaussian	$\tilde{O}(n)$	Imaginary measurement and CRT	This work §4

Table 2: Algorithm and sample complexity for $\mathbb{C}|\text{LWE}\rangle$ and Oblivious LWE sampling. Here n is the dimension of the LWE secret, r is the standard deviation of the error distribution, q is the modulus. All quantum algorithms in this table run in $\text{poly}(n, m)$ time where m is the sample complexity.

Application to oblivious LWE sampling. As mentioned in [DFS24], to build a quantum oblivious LWE sampler via solving $\mathbb{C}|\text{LWE}\rangle$, the key is to choose the amplitude f where the real part is Gaussian, and the imaginary part is suitable so that an efficient quantum algorithm can find the center c of $|\phi_{f,c}\rangle$ with non-negligible probability. In [DFS24], f is chosen to be *Gaussian with half phase*: $f(x) = \exp\left(-\pi\frac{x^2}{r^2}\right) \cdot \text{sgn}^+(x)$, where $\text{sgn}^+(x)$ outputs 1 when $x \geq 0$, -1 when $x < 0$. They are able to solve $\mathbb{C}|\text{LWE}\rangle_{n,m,q,f}$ for such f with sample complexity $m \in \tilde{O}(nr)$ and with prime modulus q . Once they get an oblivious LWE sample with parameter $(n, m, q, r/q)$ (here $r/\sqrt{2\pi}$ is the standard deviation of the LWE error term), they can throw away additional samples, and apply the classical modulus switching transformation [BLP⁺13] to get oblivious LWE samples with parameter $(n, m', q', r'/q')$ for certain $m' \leq m$, $q' < q$, $r' > r$ (we omit more detailed restrictions in the introduction).

Using the complex Gaussian amplitude, we can directly use Theorem 5 with $\ell \in O(1)$ to reduce the sample complexity of $\mathbb{C}|\text{LWE}\rangle$ from $\tilde{O}(nr)$ to $\tilde{O}(n)$. Here is the main theorem for the core quantum component of our oblivious LWE sampler:

Theorem 6 (Informal version of Corollary 44). *Let n, m, q, ℓ, r satisfy the same conditions as in Theorem 5. Assume the quantum hardness of $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}$, there exists a witness-oblivious quantum sampler for $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}$.*

Once we get an oblivious LWE sampler for some composite modulus q , we can also use modulus switching to get an oblivious LWE sampler for certain prime modulus $q' < q$. The detailed parameters are given in Corollary 46 in the main body.

In Table 2 we summarize the known poly-time algorithms for solving $\mathbb{C}|\text{LWE}\rangle$ and oblivious LWE sampling. In Figure 1 we provide some examples of interesting amplitudes addressed in our paper or previous papers.

Hardness results. Given our new poly-time quantum algorithms for solving $\mathbb{S}|\text{LWE}\rangle$ for complex Gaussian amplitude, readers may wonder whether they lead to algorithmic advantages for standard LWE. In fact, the complex Gaussian function was introduced by Chen [Che24] in a failed attempt of solving standard LWE. We haven't been able to fix the problem in [Che24]. Instead, we extract the intuition from [Che24] and show that complex Gaussian is at least suitable for solving $\mathbb{S}|\text{LWE}\rangle$, $\mathbb{C}|\text{LWE}\rangle$, and reduce the quantum sample complexity for oblivious LWE sampling. We are not able to show that solving $\mathbb{S}|\text{LWE}\rangle$, $\mathbb{C}|\text{LWE}\rangle$ with complex Gaussian amplitude implies solving standard LWE.

How about the real Gaussian amplitude? Whether Theorem 4 leads to sub-exponential time quantum algorithms for standard LWE? To address this question, let us recall the quan-

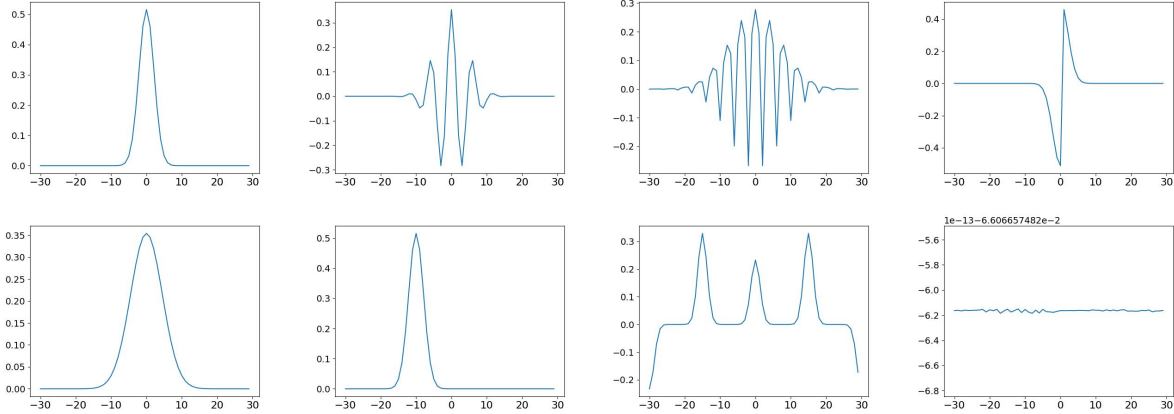


Figure 1: Interesting $S|LWE\rangle$ error amplitudes (top) and their DFTs (bottom). All pictures are depicting the real parts of the functions. The x -axis is the input (from -30 to 29 , all examples are given over $\mathbb{Z}_{60}$). The y -axis is the amplitude. Four pictures on the top from left to right are: (1) Gaussian, where our sub-exponential algorithm applies; (2) Gaussian with imaginary linear phase, where our reductions apply when the phase (or the center of the DFT) is unknown; (3) Gaussian with imaginary quadratic phase, where our oblivious LWE sampler uses; (4) Gaussian where the phase changes in the middle, where the oblivious LWE sampler in [DFS24] uses.

tum reduction from GapSVP and SIVP to classical LWE with Gaussian error distribution due to Regev [Reg09]. This reduction works even given *arbitrarily many* classical LWE samples. At first glance, one would conjecture that the reduction can be modified to a quantum reduction from worst-case lattice problems to $S|LWE\rangle$ with Gaussian amplitude.

However, we are only able to modify Regev’s reduction into a quantum reduction from worst-case lattice problem to $S|LWE\rangle$ with Gaussian amplitude with some small but *unknown* phase, with arbitrarily many quantum samples. Before stating our main theorem, let us first introduce the definition of $S|LWE\rangle$ with phase.

Definition 7 ($S|LWE\rangle^{\text{phase}}$). Let n, m be the dimensions, q be the modulus of LWE. Define the following components: (1) a normalized amplitude function $f : \text{supp}(f) \rightarrow \mathbb{R}_{\geq 0}$ where $\text{supp}(f) = \{x/Q : -q/2 < x/Q \leq q/2, x \in \mathbb{Z}\}$ for some integer Q ; (2) a mapping $\theta : \text{supp}(\theta) \rightarrow \mathbb{R}$ where $\text{supp}(\theta)$ is a subset of high-dimensional integer vectors; (3) a distribution D_θ over the set $\text{supp}(\theta)$.

We say a quantum algorithm is capable of solving $S|LWE\rangle_{n,m,q,f,\theta,D_\theta}^{\text{phase}}$, if for any hidden vector $\mathbf{s} \in \mathbb{Z}_q^n$, when provided with m samples of

$$\mathbf{a} \leftarrow U(\mathbb{Z}_q^n), \quad \mathbf{y} \leftarrow D_\theta, \quad \sum_{e \in \text{supp}(f)} f(e) \exp(2\pi i \cdot e\theta(\mathbf{y})) |(\langle \mathbf{a}, \mathbf{s} \rangle + e) \bmod q|,$$

the algorithm outputs $\mathbf{s}$ with probability at least $1 - 2^{-\Omega(n)}$.

On the first pass of the definition, readers can think of the integer vector $\mathbf{y}$ as some auxiliary information. The phase term $\theta(\mathbf{y})$ is a function of $\mathbf{y}$. As it is defined, the function θ may or may not be efficiently computable (it is not efficiently computable in our result). So we can think of $S|LWE\rangle^{\text{phase}}$ as a variant of $S|LWE\rangle$ with a phase term in the amplitude.

Theorem 8. [*Theorem 57, informal*] Let $q = q(n) > 10n$ be an integer of at most $\text{poly}(n)$ bits, $\alpha \in (0, \frac{1}{5\sqrt{n}})$ such that $\alpha q > 2\sqrt{n}$. Suppose there exists a quantum algorithm that solves $\text{S|LWE}^{\text{phase}}$ with a normalized amplitude $f(e) \propto \exp\left(-\pi \frac{e^2}{(\alpha q)^2}\right)$ and a specific set of pairs (θ, D_θ) (the choice of (θ, D_θ) is not known to the algorithm), using $m = 2^{o(n)}$ samples and in time complexity T . Then there exists a quantum algorithm that solves GapSVP_γ and SIVP_γ , where $\gamma \in \tilde{O}(n/\alpha)$, in time $\text{poly}(n, m, T)$.

The informal statement above omits the distribution of the unknown phase term $\theta(\mathbf{y})$ which are defined in [Definition 56](#), and other details can be found in the statement of [Theorem 57](#). Morally, [Theorem 57](#) says there is a quantum reduction from worst-case lattice problems to S|LWE with Gaussian amplitude with unknown phase, with arbitrarily many samples. Let us remark that the distribution of the unknown phase is *known* to be Gaussian with *small* width.

We also provide a quantum reduction directly from classical LWE to $\text{S|LWE}^{\text{phase}}$ in [Theorem 50](#). This reduction goes through the (extrapolated) dihedral coset problem, originally used in [\[Reg02, BKS18\]](#). It achieves slightly worse parameters compared to [Theorem 57](#), but is much simpler to describe. For more details we refer the readers to [Section 5](#). Although the result appears to be qualitatively similar to [Theorem 57](#), as it also says S|LWE with Gaussian amplitude with small unknown phase is as hard as classical LWE; the algorithm used in the reduction is very different, therefore it might offer a different approach for potential improvements.

Compared to the reduction of [\[BKS18\]](#), while their reduction can be seen as converting classical LWE to S|LWE samples with non-negligible probability of failure (meaning that the amplitude of S|LWE samples does not follow an expected shape; the event of failure is not efficiently detectable), our reduction converts classical LWE to S|LWE samples which always have a Gaussian amplitude, but the phase has a small unknown term. Strictly speaking, our result is incomparable with the result in [\[BKS18\]](#). But we hope that maybe some quantum algorithms in future can handle S|LWE samples with small unknown phase better than S|LWE samples with failure.

Is the unknown phase an inherent barrier? Overall, we show subexponential time quantum algorithm for S|LWE with known amplitudes. We also reduce worst-case lattice problems, or classical LWE, to S|LWE with unknown phase. Readers may wonder whether the *unknown phase* would be an inherent barrier for getting a subexponential quantum algorithm for solving standard LWE. We think it is a wonderful question for future work. On the one hand, the unknown phase appeared in our reduction is not completely random (if the unknown phase were completely random then the quantum LWE samples are as useless as classical LWE samples) – we know they are small and follow Gaussian distributions, so there are some hope of further utilizing the unknown phases (such as trying to use the filtering technique in [\[CLZ22\]](#) to guess the phase). On the other hand, all our attempts of utilizing the unknown phase failed. The reasons of failures are rather technical and entangled with the details of our algorithms, so we refer readers to the main body for discussions therein. We think it is worth to understand whether the unknown phase is an inherent obstacle towards solving LWE in quantum subexponential time, or there is a hope of finding innovative methods in handling the unknown phase appeared in our reductions.

Organization. The rest of this paper is organized as follows. In [Section 2](#) we provide the background of quantum computation and lattice problems. In [Section 3](#) we provide our sub-exponential

time quantum algorithm for $S|\text{LWE}\rangle$ with completely known amplitude. In [Section 4](#) we provide our polynomial time quantum algorithm for $S|\text{LWE}\rangle$ with complex Gaussian amplitudes, and the application to oblivious LWE sampling. In [Section 5](#) we provide the reduction from classical LWE to $S|\text{LWE}\rangle^{\text{phase}}$ via extrapolated DCP. We provide the reduction from worst-case lattice problem to $S|\text{LWE}\rangle^{\text{phase}}$ via quantizing Regev's reduction in [Section 6](#), due to page limit. We choose to present the reduction from classical LWE to $S|\text{LWE}\rangle^{\text{phase}}$ via extrapolated DCP first, since this reduction is relatively easier to follow. [Section 3](#), [Section 4](#), [Section 5](#), [Section 6](#) are all in fact self-contained and independent, containing their own overview if necessary, so readers can start from any section without reading the others.

2 Preliminaries

Notations and terminology. Let $\mathbb{C}, \mathbb{R}, \mathbb{R}_{\geq 0}, \mathbb{Z}, \mathbb{N}, \mathbb{N}^+$ be the set of complex numbers, real numbers, non-negative real numbers, integers, natural numbers (non-negative integers), and positive integers. For $n \in \mathbb{N}^+$, let $[n] := \{1, 2, \dots, n\}$. Denote $\mathbb{Z}/q\mathbb{Z}$ by $\mathbb{Z}_q$. By default we represent the elements of $\mathbb{Z}_q$ by elements in $(-q/2, q/2] \cap \mathbb{Z}$. For any integer $q \geq 2$, let $\omega_q = e^{2\pi i/q}$ denote the primitive q -th root of unity. The rounding operation $\lfloor a \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$ rounds a real number a to its nearest integer (if $a \in \mathbb{Z} + 0.5$, we round it to $a + 0.5$). For positive integer q , the rounding operation $\lfloor a \rfloor_q : \mathbb{R} \rightarrow q\mathbb{Z}$ rounds a real number a to its nearest integer which is a multiple of q , i.e. $\lfloor a \rfloor_q = q \lfloor a/q \rfloor$.

A vector in $\mathbb{R}^n$ (represented in column form by default) is written as a bold lower-case letter, e.g. $\mathbf{v}$. For a vector $\mathbf{v}$, the i^{th} component of $\mathbf{v}$ will be denoted by v_i . A matrix is written as a bold capital letter, e.g. $\mathbf{A}$. The i^{th} column vector of $\mathbf{A}$ is denoted $\mathbf{a}_i$.

For $x \in \mathbb{R}$ and $q \in \mathbb{N}^+$, let $x \bmod q$ be the unique real number $z \in (-q/2, q/2]$ such that $x - z$ is a multiple of q . For a vector $\mathbf{v} \in \mathbb{R}^n$, we do $\bmod$ coordinate-wise, i.e. the i^{th} coordinate of $\mathbf{v} \bmod q$ is given by $v_i \bmod q$. To avoid ambiguity, we give $\bmod$ lower precedence than addition/subtraction. For example, $\mathbf{a} + \mathbf{b} \bmod q$ means $(\mathbf{a} + \mathbf{b}) \bmod q$.

The length of a vector is the ℓ_2 -norm $\|\mathbf{v}\| := (\sum_i |v_i|^2)^{1/2}$, or the infinity norm given by its largest entry $\|\mathbf{v}\|_{\infty} := \max_i \{|v_i|\}$. The ℓ_2 -norm of a matrix is the norm of its longest column: $\|\mathbf{A}\| := \max_i \|\mathbf{a}_i\|$. Let $\mathbf{x} \in \mathbb{C}^n$, then $\|\mathbf{x}\|_{\infty} \leq \|\mathbf{x}\|$. Let B_2^n (respectively, B_{∞}^n) denote the open unit ball in $\mathbb{R}^n$ in the ℓ_2 -norm (respectively, ℓ_{∞} -norm). By default we use ℓ_2 -norm unless explicitly mentioned.

When a variable v is drawn uniformly random from the set S we denote as $v \leftarrow U(S)$. When a function f is applied on a set S , it means $f(S) := \sum_{x \in S} f(x)$.

Statistical distance. For two distributions over a set S with probability density functions f_1 and f_2 , we define the statistical distance between them as

$$D(f_1, f_2) = \frac{1}{2} \sum_{x \in S} |f_1(x) - f_2(x)|.$$

We say two distributions (respectively, quantum states) are ϵ -close to each other if their statistical distance (respectively, trace distance by default) is at most ϵ . We say two pure (unnormalized) states $|\phi\rangle$ and $|\psi\rangle$ are ϵ -close in ℓ_2 distance if $\| |\phi\rangle - |\psi\rangle \| \leq \epsilon \max(\| |\phi\rangle \|, \| |\psi\rangle \|)$.

Fourier transform. The continuous Fourier transform of a function $h : \mathbb{R}^n \rightarrow \mathbb{C}$ is defined to be

$$\hat{h}(\mathbf{w}) = \int_{\mathbb{R}^n} h(\mathbf{x}) \exp(-2\pi i \langle \mathbf{x}, \mathbf{w} \rangle) d\mathbf{x}.$$

We recall some formulas about Fourier transform (cf. [Gra08, P.100, Proposition 2.2.11]). If h is defined by $h(\mathbf{x}) = g(\mathbf{x} + \mathbf{v})$ for some function $g : \mathbb{R}^n \rightarrow \mathbb{C}$ and vector $\mathbf{v} \in \mathbb{R}^n$, then

$$\hat{h}(\mathbf{w}) = \hat{g}(\mathbf{w}) \cdot \exp(2\pi i \langle \mathbf{v}, \mathbf{w} \rangle). \quad (2)$$

If h is defined by $h(\mathbf{x}) = g(\mathbf{x}) \exp(2\pi i \langle \mathbf{x}, \mathbf{v} \rangle)$ for some function $g : \mathbb{R}^n \rightarrow \mathbb{C}$ and vector $\mathbf{v} \in \mathbb{R}^n$, then

$$\hat{h}(\mathbf{w}) = \hat{g}(\mathbf{w} - \mathbf{v}). \quad (3)$$

As a corollary of Eqns. (2) and (3), if h is defined by $h(\mathbf{x}) = f(\mathbf{x} + \mathbf{v}) \exp(2\pi i \langle \mathbf{x}, \mathbf{z} \rangle)$ for some function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ and vectors $\mathbf{v}, \mathbf{z} \in \mathbb{R}^n$, then

$$\hat{h}(\mathbf{w}) = \hat{f}(\mathbf{w} - \mathbf{z}) \cdot \exp(2\pi i \langle \mathbf{v}, \mathbf{w} - \mathbf{z} \rangle).$$

For any integer $q \geq 2$, the discrete Fourier transform (DFT) of a function $f : \mathbb{Z}_q^n \rightarrow \mathbb{C}$ over $\mathbb{Z}_q^n$ is defined to be

$$\hat{f}_q(\mathbf{w}) = \frac{1}{\sqrt{q^n}} \sum_{\mathbf{x} \in \mathbb{Z}_q^n} f(\mathbf{x}) \omega_q^{-\langle \mathbf{x}, \mathbf{w} \rangle}.$$

2.1 Lattices

An n -dimensional lattice $\mathcal{L}$ of rank $k \leq n$ is a discrete additive subgroup of $\mathbb{R}^n$. Given k linearly independent basis vectors $\mathbf{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_k \in \mathbb{R}^n\}$, the lattice generated by $\mathbf{B}$ is

$$\mathcal{L}(\mathbf{B}) = \mathcal{L}(\mathbf{b}_1, \dots, \mathbf{b}_k) = \left\{ \sum_{i=1}^k x_i \cdot \mathbf{b}_i, x_i \in \mathbb{Z} \right\}.$$

By default we work with full-rank lattices unless explicitly mentioned.

The minimum distance $\lambda_1(\mathcal{L})$ of a lattice $\mathcal{L}$ is the length (in the ℓ_2 -norm by default) of its shortest nonzero vector: $\lambda_1(\mathcal{L}) = \min_{\mathbf{x} \in \mathcal{L} \setminus \{0\}} \|\mathbf{x}\|$. More generally, the i^{th} successive minimum $\lambda_i(\mathcal{L})$ is the smallest radius r such that $\mathcal{L}$ contains i linearly independent vectors of norm at most r . We write λ_1^∞ as the minimum distance in the ℓ_∞ -norm.

For a point $\mathbf{y} \in \mathbb{R}^n$, its distance to $\mathcal{L}$ is given by $\text{dist}(\mathbf{y}, \mathcal{L}) = \min_{\mathbf{x} \in \mathcal{L}} \{\|\mathbf{y} - \mathbf{x}\|\}$. Define “the ball around lattice” as $B_{\mathcal{L}}(r) = \{\mathbf{x} \in \mathbb{R}^n : \text{dist}(\mathbf{x}, \mathcal{L}) < r\}$. For $\mathbf{y} \in B_{\mathcal{L}}(\lambda_1(\mathcal{L})/2)$, the (unique) closest vector to $\mathbf{y}$ in $\mathcal{L}$ is given by $\kappa_{\mathcal{L}}(\mathbf{y}) = \arg\min_{\mathbf{x} \in \mathcal{L}} \{\|\mathbf{y} - \mathbf{x}\|\}$. For convenience, we omit the $\lambda_1(\mathcal{L})/2$ term and define $B_{\mathcal{L}}$ as $B_{\mathcal{L}}(\lambda_1(\mathcal{L})/2)$, over which $\kappa_{\mathcal{L}}$ is uniquely defined.

The dual of a lattice $\mathcal{L} \in \mathbb{R}^n$ is defined as

$$\mathcal{L}^* := \{\mathbf{y} \in \mathbb{R}^n : \langle \mathbf{y}, \mathbf{x} \rangle \in \mathbb{Z} \text{ for all } \mathbf{x} \in \mathcal{L}\}.$$

Lemma 9 (Poisson Summation Formula). *For any lattice $\mathcal{L}$ and any Schwartz function $f : \mathbb{R}^n \rightarrow \mathbb{C}$, we have $f(\mathcal{L}) = \det(\mathcal{L}^*) \hat{f}(\mathcal{L}^*)$.*

Gaussians and lattices. For any $s > 0$, define the Gaussian function on $\mathbb{R}^n$ with parameter s following the convention in [MR07]

$$\forall \mathbf{x} \in \mathbb{R}^n, \rho_s(\mathbf{x}) = \exp(-\pi \|\mathbf{x}\|^2 / s^2).$$

For any $\mathbf{c} \in \mathbb{R}^n$, define $\rho_{s,\mathbf{c}}(\mathbf{x}) := \rho_s(\mathbf{x} - \mathbf{c})$. The subscripts s and $\mathbf{c}$ are taken to be 1 and $\mathbf{0}$ (respectively) when omitted. Note that the standard deviation of ρ_s is $s/\sqrt{2\pi}$. The Fourier transform for Gaussian satisfies $\hat{\rho}_s = s^n \rho_{1/s}$. From Poisson summation formula we have $\rho_s(\mathcal{L}) = s^n \cdot \det(\mathcal{L}^*) \cdot \rho_{1/s}(\mathcal{L}^*)$.

For a full-rank, symmetric, positive definite $n \times n$ matrix Σ , define the Gaussian function on $\mathbb{R}^n$ with parameter $\sqrt{\Sigma}$ following the convention in [MP12]

$$\forall \mathbf{x} \in \mathbb{R}^n, \rho_{\sqrt{\Sigma}}(\mathbf{x}) = \exp(-\pi \cdot \mathbf{x}^T \Sigma^{-1} \mathbf{x}).$$

For any dimension n , set $T \in \mathbb{R}^n$ and real $s > 0$, define the restricted Gaussian distribution $D_{T,s}$ as

$$\forall \mathbf{x} \in T, D_{T,s}(\mathbf{x}) = \frac{\rho_s(\mathbf{x})}{\rho_s(T)}.$$

Specifically, for any $\mathbf{c} \in \mathbb{R}^n$, real $s > 0$, and n -dimensional lattice $\mathcal{L}$, define the discrete Gaussian distribution $D_{\mathcal{L}+\mathbf{c},s}$ as

$$\forall \mathbf{x} \in \mathcal{L} + \mathbf{c}, D_{\mathcal{L}+\mathbf{c},s}(\mathbf{x}) = \frac{\rho_s(\mathbf{x})}{\rho_s(\mathcal{L} + \mathbf{c})}.$$

Similarly, for a full-rank, symmetric, positive definite $n \times n$ matrix Σ , define the discrete Gaussian distribution $D_{\mathcal{L}+\mathbf{c},\sqrt{\Sigma}}$ as

$$\forall \mathbf{x} \in \mathcal{L} + \mathbf{c}, D_{\mathcal{L}+\mathbf{c},\sqrt{\Sigma}}(\mathbf{x}) = \frac{\rho_{\sqrt{\Sigma}}(\mathbf{x})}{\rho_{\sqrt{\Sigma}}(\mathcal{L} + \mathbf{c})}.$$

We will use the following Gaussian tail bound over lattices for ℓ_2 -norm and ℓ_∞ -norm, due to Banaszczyk.

Lemma 10 (Banaszczyk's tail bounds [Ban93, Ban95]). *For any n -dimensional lattice $\mathcal{L}$, and $r \geq \frac{1}{\sqrt{2\pi}}$, $\mathbf{c} \in \mathbb{R}^n$,*

$$\begin{aligned} \rho((\mathcal{L} - \mathbf{c}) \setminus r\sqrt{n}B_2^n) &< 2 \left(r\sqrt{2\pi}e \cdot e^{-\pi r^2} \right)^n \rho(\mathcal{L}). \\ \rho((\mathcal{L} - \mathbf{c}) \setminus rB_\infty^n) &< (2n \cdot e^{-\pi r^2}) \rho(\mathcal{L}). \end{aligned} \tag{4}$$

Lemma 11 (Claim 8.1 [RS17]). *For any $n \geq 1$, $s > 0$,*

$$s^n(1 + 2e^{-\pi s^2})^n \leq \rho_s(\mathbb{Z}^n) \leq s^n(1 + (2 + 1/s)e^{-\pi s^2})^n.$$

Smoothing parameter. We recall the definition of smoothing parameter for Gaussian over lattices and some useful facts.

Definition 12 (Smoothing parameter [MR07]). *For any lattice $\mathcal{L}$ and positive real $\epsilon > 0$, the smoothing parameter $\eta_\epsilon(\mathcal{L})$ is the smallest real $s > 0$ such that $\rho_{1/s}(\mathcal{L}^* \setminus \{\mathbf{0}\}) \leq \epsilon$.*

Lemma 13 (Lemma 2.12 [Reg09]). *For any n -dimensional lattice $\mathcal{L}$, and any real $\epsilon > 0$,*

$$\eta_\epsilon(\mathcal{L}) \leq \lambda_n(\mathcal{L}) \cdot \sqrt{\ln(2n(1 + 1/\epsilon))/\pi}.$$

Lemma 14 (Claim 2.13 [Reg09]). *For any n -dimensional lattice $\mathcal{L}$, and any real $\epsilon > 0$,*

$$\eta_\epsilon(\mathcal{L}) \geq \sqrt{\frac{\ln 1/\epsilon}{\pi}} \frac{1}{\lambda_1(\mathcal{L}^*)}.$$

Lemma 15 (Claim 3.8 of [Reg09]). *For any n -dimensional lattice $\mathcal{L}$, $\mathbf{c} \in \mathbb{R}^n$, $\epsilon > 0$, and $r \geq \eta_\epsilon(\mathcal{L})$*

$$\rho_r(\mathcal{L} + \mathbf{c}) \in r^n \det(\mathcal{L}^*)(1 \pm \epsilon).$$

q -ary lattices. Given $n < m \in \mathbb{N}$ and a modulus $q \geq 2$, for $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, define q -ary lattices as

$$\begin{aligned} \mathcal{L}_q(\mathbf{A}) &= \{\mathbf{x} \in \mathbb{Z}^m : \exists \mathbf{s} \in \mathbb{Z}^n \text{ such that } \mathbf{x} = \mathbf{A}^T \cdot \mathbf{s} + q\mathbb{Z}^m\}; \\ \mathcal{L}_q^\perp(\mathbf{A}) &= \{\mathbf{x} \in \mathbb{Z}^m : \mathbf{A} \cdot \mathbf{x} = \mathbf{0} \pmod{q}\}. \end{aligned}$$

Those two lattices are dual of each other up to a factor of q , i.e., $\mathcal{L}_q(\mathbf{A}) = q \cdot \mathcal{L}_q^\perp(\mathbf{A})^*$. We need the following lemma for q -ary lattices, which shows that the shortest nonzero vector in a q -lattice is relatively long. The proof is deferred to [Appendix A.1](#).

Lemma 16. *Let $q \geq 2, m \geq 2n \log_2 q$, then for all but at most $q^{-0.16n}$ fraction of $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, we have*

$$\lambda_1^\infty(\mathcal{L}_q(\mathbf{A})) \geq \frac{q}{4}.$$

Lattice problems. We have formally defined the LWE , $\text{S}|\text{LWE}\rangle$, $\text{C}|\text{LWE}\rangle$, and $\text{S}|\text{LWE}\rangle^{\text{phase}}$ problems in the introduction. Now let us recall the definitions for other lattice problems.

The shortest vector problem (SVP) asks to find a lattice vector of length λ_1 . More generally, let $\gamma(n) \geq 1$ be an approximation factor, we consider the approximation version of SVP and its close variants.

Definition 17 (Approximate SVP). *Given a basis $\mathbf{B}$ of an n -dimensional lattice $\mathcal{L}$, the SVP_γ problem asks to output a non-zero lattice vector $\mathbf{B}\mathbf{x}$, $\mathbf{x} \in \mathbb{Z}^n \setminus \{\mathbf{0}\}$, such that $\|\mathbf{B}\mathbf{x}\| \leq \gamma(n) \cdot \lambda_1(\mathcal{L})$.*

Definition 18 (GapSVP). *Given a basis $\mathbf{B}$ of an n -dimensional lattice $\mathcal{L}$ and a number $d > 0$, the GapSVP_γ problem asks to decide whether $\lambda_1(\mathcal{L}) \leq d$ or $\lambda_1(\mathcal{L}) > d \cdot \gamma(n)$.*

Definition 19 (Shortest independent vector problem (SIVP)). *Given a basis $\mathbf{B}$ of an n -dimensional lattice $\mathcal{L}$, the SIVP_γ problem asks to output a set of n linearly independent vectors of length at most $\gamma(n) \cdot \lambda_n(\mathcal{L})$.*

Definition 20 (Discrete Gaussian Sampling Problem (DGS)). *Given a basis $\mathbf{B}$ of an n -dimensional lattice $\mathcal{L}$ and a parameter $s > 0$, the DGS_s problem asks to output a vector whose distribution is statistically close to $D_{\mathcal{L},s}$.*

Definition 21 (Quantum Discrete Gaussian Sampling Problem ($|\text{DGS}\rangle$)). *Given a basis $\mathbf{B}$ of an n -dimensional lattice $\mathcal{L}$ and a parameter $s > 0$, the $|\text{DGS}\rangle_s$ problem asks to output a quantum state that is $2^{-\Omega(n)}$ -close to $\sum_{\mathbf{v} \in \mathcal{L}} \rho_s(\mathbf{v}) |\mathbf{v}\rangle$ in trace distance.*

If a quantum algorithm solves $|\text{DGS}\rangle_s$, then it immediately solves $\text{DGS}_{s/\sqrt{2}}$ by simply measuring the quantum state. Let us also recall the relationships among DGS , GapSVP , and SIVP .

Lemma 22 (Lemma 3.20 of [Reg09]). *For any $\gamma = \gamma(n) \geq 1$, there exists a polynomial time reduction from $\text{GapSVP}_{100\sqrt{n}\gamma}$ for $\mathcal{L}$ to $\text{DGS}_{\sqrt{n}\gamma/\lambda_1(\mathcal{L}^*)}$ for $\mathcal{L}^*$.*

Lemma 23 (Lemma 3.17 of [Reg09]). *For any $\gamma > \omega(\sqrt{\log n})$, there exists a polynomial time reduction from $\text{SIVP}_{2\sqrt{n}\gamma}$ for $\mathcal{L}$ to $\text{DGS}_{\gamma\lambda_n(\mathcal{L})}$ for $\mathcal{L}$.*

2.2 Quantum computation

We assume readers are familiar with the basic concepts of quantum computation. All the quantum background we need in this paper are available in standard textbooks of quantum computation, e.g., [NC16].

The trace distance between two quantum states ρ and σ is defined as $\delta(\rho, \sigma) := \frac{1}{2} \text{tr} |\rho - \sigma|$. Note that when ρ and σ commute they are diagonal in the same basis,

$$\rho = \sum_i r_i |i\rangle \langle i|, \quad \sigma = \sum_i s_i |i\rangle \langle i|,$$

for some orthonormal basis $|i\rangle$, then $\delta(\rho, \sigma) = \frac{1}{2} \text{tr} |\sum_i (r_i - s_i) |i\rangle \langle i|| = \frac{1}{2} \sum_i |r_i - s_i|$. The trace distance is preserved under unitary transformations, and is contractive under trace-preserving operations.

Lemma 24. *Let $|\phi\rangle, |\psi\rangle$ be un-normalized vectors s.t. $\| |\phi\rangle \| \geq \mu$ and $\| |\phi\rangle - |\psi\rangle \| \leq \epsilon$. Then*

$$\delta\left(\frac{1}{\| |\phi\rangle \|} |\phi\rangle, \frac{1}{\| |\psi\rangle \|} |\psi\rangle\right) = \sqrt{1 - \left(\frac{|\langle \phi | \psi \rangle|}{\| |\phi\rangle \| \| |\psi\rangle \|}\right)^2} \leq O\left(\sqrt{\frac{\epsilon}{\mu}}\right).$$

Lemma 25 (Gentle measurement [Win99]). *Let ρ be a quantum state and let $(\Pi, \mathbf{I} - \Pi)$ be a two-outcome projective measurement such that $\text{tr}(\Pi\rho) \geq 1 - \epsilon$. Let $\rho' = \frac{\Pi\rho\Pi}{\text{tr}(\Pi\rho)}$ be the state after applying the measurement and post-selecting on getting the first outcome. Then $\delta(\rho, \rho') \leq 2\sqrt{\epsilon}$.*

Quantum algorithms. Let $q \geq 2$ be an integer. The Quantum Fourier Transform (QFT) [Kit95] unitary operator QFT_q can be implemented by $\text{poly}(\log q)$ elementary quantum gates. When QFT_q is applied on a quantum state $|\phi\rangle := \sum_{x \in \mathbb{Z}_q} f(x) |x\rangle$, we have

$$\text{QFT}_q |\phi\rangle = \sum_{y \in \mathbb{Z}_q} \hat{f}_q(y) |y\rangle.$$

We use the following lemmas to change the amplitude and phase of a state.

Lemma 26 (Quantum rejection sampling [ORR13]). *Let $f : D \rightarrow \mathbb{C}$ be a normalized amplitude of some quantum state $|\phi_f\rangle := \sum_{x \in D} f(x) |x\rangle$. Let $\gamma : D \rightarrow [0, 1]$ be a function that is T -time computable function within $2^{-\Omega(\lambda)}$ precision. There is a $\text{poly}(\log(|D|), T)$ time quantum algorithm that takes as input $|\phi_f\rangle$, outputs either “success” and a state $2^{-\Omega(\lambda)}$ -close to $\sum_{x \in D} \frac{1}{\sqrt{M}} \gamma(x) f(x) |x\rangle$ with probability M where $M = \sum_{x \in D} \gamma^2(x) |f(x)|^2$, or “fail” with probability $1 - M$.*

Lemma 27 (Phase kick-back [CEMM98]). *Let $q \in \mathbb{N}^+$, $f(x) \in \mathbb{Z}_q$. If the transformation $|x\rangle \mapsto |x\rangle |f(x)\rangle$ is computable in time T , then the unitary transformation $|x\rangle \mapsto e^{\frac{2\pi i f(x)}{q}} |x\rangle$ can be performed in time $\text{poly}(T, \log(q))$.*

Gaussian state preparation. In this paper we are interested in preparing the Gaussian state $|\sigma_{n,R}\rangle := \sum_{\mathbf{x} \in \mathbb{Z}^n \cap B^n(R\sqrt{n})} \rho_R(\mathbf{x}) |\mathbf{x}\rangle$ for some radius $R \leq 2^{\text{poly}(n)}$. Given Lemma 10, there is a $2^{-\Omega(n)}$ mass in the tail of $\rho_R(\mathbf{x})$ outside $B^n(R\sqrt{n})$. This means for any integer $P \in (2R\sqrt{n}, 2^{\text{poly}(n)})$, $|\sigma_{n,R}\rangle$ is $2^{-\Omega(n)}$ close to $\sum_{\mathbf{x} \in \mathbb{Z}_P^n} \rho_R(\mathbf{x}) |\mathbf{x}\rangle$. This also means we can prepare $|\sigma_{n,R}\rangle$ by generating n independent samples of one-dimensional Gaussian state $|\sigma_{1,R}\rangle$, which can be done efficiently using [GR02].

Lemma 28 (Gaussian state preparation). *Let $n, R \in \mathbb{N}$ such that $1 \leq R \leq 2^{n^c}$ for some constant $c \geq 0$. Then we can create a $\text{poly}(n)$ size unitary U that maps $|\mathbf{0}\rangle$ to a state within trace distance $2^{-\Omega(n)}$ from $|\sigma_{n,R}\rangle$ with $2n \lceil n^c \cdot \log n \rceil$ qubits.*

Lemma 29 (Complex Gaussian state preparation). *Let $n \in \mathbb{N}$, $R \in \mathbb{R}$ satisfy $1 \leq R \leq 2^{n^c}$ for some constant $c \geq 0$. Let $S > 0$ be a number such that $\frac{1}{S^2}$ can be efficiently computed within $\frac{2^{-\Omega(n)}}{R^2 n}$ precision, i.e., we can compute $\frac{1}{S^2} \in \frac{1}{S^2} \pm \frac{2^{-\Omega(n)}}{R^2 n}$ and $\frac{1}{S^2}$ is a rational number that can be represented by $\text{poly}(n)$ bits. Then the complex Gaussian states with $\mathcal{B}_2^n$ and $\mathcal{B}_\infty^n$ boundaries,*

$$|\zeta_{n,R,S}\rangle := \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \rho_R(\mathbf{y}) \cdot e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} |\mathbf{y}\rangle, \quad |\zeta_{n,R,S}^\infty\rangle := \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R \log n \mathcal{B}_\infty^n} \rho_R(\mathbf{y}) \cdot e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} |\mathbf{y}\rangle$$

can both be prepared in $\text{poly}(n)$ time within trace distances $2^{-\Omega(n)}$.

Proof. We describe how to prepare $|\zeta_{n,R,S}\rangle$. The procedure for preparing $|\zeta_{n,R,S}^\infty\rangle$ is similar.

To prepare $|\zeta_{n,R,S}\rangle$ we start from preparing the Gaussian state $|\sigma_{n,R}\rangle$ using Lemma 28, and then apply Lemma 27 to change the phase:

$$\sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \rho_R(\mathbf{y}) |\mathbf{y}\rangle \mapsto \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \rho_r(\mathbf{y}) \cdot e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} |\mathbf{y}\rangle =: |\zeta'_{n,R,S}\rangle.$$

$|\zeta'_{n,R,S}\rangle$ and $|\zeta_{n,R,S}\rangle$ are $2^{-\Omega(n)}$ -close in the ℓ_2 distance because the normalization factor of both $|\zeta'_{n,R,S}\rangle$ and $|\zeta_{n,R,S}\rangle$ is $\sqrt{\sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \rho_R^2(\mathbf{y})}$, and

$$\begin{aligned} \|\zeta'_{n,R,S} - \zeta_{n,R,S}\|_2^2 &= \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \left| \rho_R(\mathbf{y}) \cdot \left(e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} - e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} \right) \right|^2 \\ &= \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \left| \rho_R(\mathbf{y}) \cdot e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} \left(1 - e^{-\pi i \cdot (1/S^2 - 1/\tilde{S}^2) \cdot \|\mathbf{y}\|^2} \right) \right|^2 \\ &\stackrel{(a)}{\in} \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R\sqrt{n}\mathcal{B}_2^n} \rho_R^2(\mathbf{y}) \cdot 2^{-\Omega(n)}, \end{aligned}$$

where (a) holds since $\|\mathbf{y}\|^2 \leq R^2 \cdot n$, so $(1/S^2 - 1/\tilde{S}^2) \cdot \|\mathbf{y}\|^2 \in 2^{-\Omega(n)}$. $\square$

Remark 30. An implication of the lemma above with more flexibility is: for the same parameters as in Lemma 29, and any $R \leq R' \leq \infty$, the complex Gaussian states

$$|\zeta_{n,R',R,S}\rangle := \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R' \sqrt{n} \mathcal{B}_2^n} \rho_R(\mathbf{y}) \cdot e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} |\mathbf{y}\rangle, \quad |\zeta_{n,R',R,S}^\infty\rangle := \sum_{\mathbf{y} \in \mathbb{Z}^n \cap R' \log n \mathcal{B}_\infty^n} \rho_R(\mathbf{y}) \cdot e^{-\pi i \frac{\|\mathbf{y}\|^2}{S^2}} |\mathbf{y}\rangle$$

can also be prepared in $\text{poly}(n)$ times with trace distances $2^{-\Omega(n)}$, through the exact same process, since the trace distances between these states with boundary R' and the corresponding states with boundary R are $2^{-\Omega(n)}$.

3 Quantum Sub-exponential Time Algorithm for $S|\text{LWE}\rangle$

In this section, we provide a quantum sub-exponential time algorithm designed to solve $S|\text{LWE}\rangle$ instances with specific amplitudes. More precisely, we consider scenarios where the discrete Fourier transform of these amplitudes has at least $2^{-\sqrt{n \log q}}$ mass on two distinct points. Our approach is built upon two key steps: (1) generate a DCP state for each quantum $S|\text{LWE}\rangle$ sample, resulting in a sub-exponential collection of DCP states; (2) employ the Kuperberg sieve technique [Kup05] on the DCP states to successfully recover the secret vector. Formally, we state the main theorem of this section as follows:

Theorem 31 (Main theorem). *Let n, q be positive integers such that $q = 2^{o(n)}$. Let $f : \mathbb{Z}_q \rightarrow \mathbb{C}$ be a $2^{O(\sqrt{n \log q})}$ -time computable normalized amplitude function such that, there exist two distinct points $j_1, j_2 \in \mathbb{Z}_q$ with $\gcd(j_1 - j_2, q) = 1$ and $|\hat{f}_q(j_1)|$ and $|\hat{f}_q(j_2)|$ are both greater than $2^{-\sqrt{n \log q}}$.*

There exists a quantum algorithm that, given $\ell = 2^{\Theta(\sqrt{n \log q})}$ samples of vector $\mathbf{a} \leftarrow U(\mathbb{Z}_q^n)$ and $S|\text{LWE}\rangle$ state of form

$$S|\text{LWE}\rangle = \sum_{e \in \mathbb{Z}_q} f(e) |\langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q\rangle,$$

finds the secret vector $\mathbf{s} \in \mathbb{Z}_q^n$ with probability $1 - 2^{-\Omega(n)}$. The algorithm runs in time $2^{\Theta(\sqrt{n \log q})}$.

Note that when q is a prime, then $\gcd(j_1 - j_2, q) = 1$ is always satisfied for different $j_1, j_2 \in \mathbb{Z}_q$. So the condition $\gcd(j_1 - j_2, q) = 1$ is only needed to be taken care of when q is a composite number.

To prove Theorem 31, we begin by introducing the Kuperberg sieve algorithm [Kup05].

Lemma 32 (Kuperberg sieve, Theorem 7.1 of [Kup05]). *Let $\mathbf{s} \in \mathbb{Z}_q^n$ be a secret vector. There exists a quantum algorithm that given $\ell^* = 2^{\Theta(\sqrt{n \log q})}$ samples of*

$$\mathbf{a} \leftarrow U(\mathbb{Z}_q^n), \quad |\psi_{\mathbf{a}}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + \omega_q^{\langle \mathbf{a}, \mathbf{s} \rangle} |1\rangle \right),$$

finds out the secret vector $\mathbf{s}$ in time $2^{\Theta(\sqrt{n \log q})}$, with probability $1 - 2^{-\Omega(n)}$ over the randomness of $\mathbf{a}$ and the quantum mechanism.

Now we present the proof of Theorem 31 here.

Proof of Theorem 31. Suppose we have $\ell = \ell^* \cdot 2^{4\sqrt{n \log q}} = 2^{\Theta(\sqrt{n \log q})}$ instances of $S|\text{LWE}\rangle$ states. Our quantum algorithm proceeds as follows:

1. Compute $\hat{f}_q(j)$ for each $j \in \mathbb{Z}_q$ and find two distinct points j_1 and j_2 from $\mathbb{Z}_q$ with $\gcd(j_1 - j_2, q) = 1$ and $|\hat{f}_q(j_1)|$ and $|\hat{f}_q(j_2)|$ are both greater than $2^{-\sqrt{n \log q}}$.
2. Apply $\text{QFT}_q^\dagger$ to each $S|\text{LWE}\rangle$ state, resulting in the state

$$\text{QFT}_q^\dagger \cdot S|\text{LWE}\rangle = \frac{1}{\sqrt{q}} \sum_{j \in \mathbb{Z}_q} \sum_{e \in \mathbb{Z}_q} f(e) \omega_q^{j(\langle \mathbf{a}, \mathbf{s} \rangle + e)} |j\rangle = \sum_{j \in \mathbb{Z}_q} \omega_q^{\langle j, \mathbf{a}, \mathbf{s} \rangle} \hat{f}_q(j) |j\rangle, \quad (5)$$

3. Define $\gamma(j) : \mathbb{Z}_q \rightarrow [0, 1]$ as

$$\gamma(j) = \frac{\min\{|\hat{f}_q(j_1)|, |\hat{f}_q(j_2)|\}}{|\hat{f}_q(j)|} \text{ for } j = j_1, j_2$$

and $\gamma(j) = 0$ otherwise. Note that the $\gamma(j)$'s are always in $[0, 1]$ and are $2^{O(\sqrt{n \log q})}$ -time computable, so we can apply quantum rejection sampling (Lemma 26) on Equation (5) to obtain a state that is $2^{-\Omega(n)}$ -close to

$$\frac{1}{\sqrt{2}} \left(\frac{\hat{f}_q(j_1)}{|\hat{f}_q(j_1)|} \omega_q^{\langle j_1, \mathbf{a}, \mathbf{s} \rangle} |j_1\rangle + \frac{\hat{f}_q(j_2)}{|\hat{f}_q(j_2)|} \omega_q^{\langle j_2, \mathbf{a}, \mathbf{s} \rangle} |j_2\rangle \right), \quad (6)$$

in time $2^{O(\sqrt{n \log q})}$ with probability

$$M = \sum_{j \in \mathbb{Z}_q} \gamma^2(j) |\hat{f}_q(j)|^2 = 2 \min\{|\hat{f}_q(j_1)|, |\hat{f}_q(j_2)|\}^2 > 2^{-2\sqrt{n \log q}}.$$

4. For the states that have been successfully transformed to Equation (6), apply a unitary operation

$$\mathcal{U} : |j_i\rangle \rightarrow \frac{\overline{\hat{f}_q(j_i)}}{|\hat{f}_q(j_i)|} |j_i\rangle \rightarrow \frac{\overline{\hat{f}_q(j_i)}}{|\hat{f}_q(j_i)|} |i-1\rangle, i = 1, 2,$$

where the first map uses the phase kick-back trick (Lemma 27), since that $\frac{\overline{\hat{f}_q(j_i)}}{|\hat{f}_q(j_i)|}$ has norm 1 which can be approximated within any exponentially small error. This results in a state that is $2^{-\Omega(n)}$ -close to

$$|\psi_{(j_2-j_1)\mathbf{a}}\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + \omega_q^{\langle (j_2-j_1)\mathbf{a}, \mathbf{s} \rangle} |1\rangle \right),$$

where $(j_2 - j_1)\mathbf{a}$ is a known vector in $\mathbb{Z}_q^n$ that is uniformly random by the assumption that $\gcd(j_1 - j_2, q) = 1$.

5. Select ℓ^* such states obtained in step 3 and apply the Kuperberg sieve algorithm with respect to the uniformly random vectors $(j_2 - j_1)\mathbf{a}$, to recover the secret vector $\mathbf{s} \in \mathbb{Z}_q^n$, with probability $1 - 2^{-\Omega(n)}$.

In step 3, the count of states successfully transformed to Equation (6) will be at least $M^2 \ell = \ell^*$ with probability $1 - 2^{-\Omega(n)}$ by Chernoff bound. This shows that our algorithm can output the correct $\mathbf{s} \in \mathbb{Z}_q^n$ with probability at least $1 - 2^{-\Omega(n)} - \ell^* \cdot 2^{-\Omega(n)} = 1 - 2^{-\Omega(n)}$ relying on the correctness Kuperberg sieve algorithm. For the running time, it is clear that for the $2^{\Theta(\sqrt{n \log q})}$ many $S|\text{LWE}\rangle$ states, step 1 to 4 all run in $2^{\Theta(\sqrt{n \log q})}$ time; in step 5, the Kuperberg sieve runs in time $2^{\Theta(\sqrt{n \log q})}$. So the overall running time is $2^{\Theta(\sqrt{n \log q})}$. $\square$

Corollary 33. *Let n, q be positive integers such that $q = 2^{o(n)}$. There exists a quantum algorithm that, with probability $1 - 2^{-\Omega(n)}$, given $2^{\Theta(\sqrt{n} \log q)}$ samples of vector $\mathbf{a} \leftarrow \mathcal{U}(\mathbb{Z}_q^n)$ and $S | \text{LWE} \rangle$ state of form*

$$S | \text{LWE} \rangle = \frac{1}{\sqrt{N_\sigma}} \sum_{e \in \mathbb{Z}_q} \rho_\sigma(e) e^{2\pi i \cdot ce/q} |\langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q \rangle,$$

where N_σ normalizes the state, the Gaussian width σ satisfies $\sigma = \Omega(\sqrt{n})$, $q > \sigma\sqrt{n}$, and c is an arbitrary known number that can be different for different samples, solves the secret vector $\mathbf{s} \in \mathbb{Z}_q^n$ within a time complexity of $2^{\Theta(\sqrt{n} \log q)}$.

Proof. First we have

$$N_\sigma = \sum_{e \in \mathbb{Z}_q} \rho_\sigma^2(e) = \left(1 \pm 2^{-\Omega(n)}\right) \sum_{e \in \mathbb{Z}} \frac{\sigma}{\sqrt{2}} \rho_{\sqrt{2}/\sigma}(e) = \left(1 \pm 2^{-\Omega(n)}\right) \frac{\sigma}{\sqrt{2}},$$

where the former equation uses Banaszczyk's tail bound under the assumption $q > \sigma\sqrt{n}$, the latter equation holds under the assumption that $\sigma = \Omega(\sqrt{n})$. In this case, the summation of $\rho_{\sqrt{2}/\sigma}$ is concentrated at $\rho_{\sqrt{2}/\sigma}(0)$, with exponentially small weight elsewhere.

In the given problem scenario,

$$f(e) = \frac{1}{\sqrt{N_\sigma}} \rho_\sigma(e) e^{2\pi i \cdot ce/q},$$

thus

$$\begin{aligned} |\hat{f}(j)| &= \frac{1}{\sqrt{qN_\sigma}} \left| \sum_{e \in \mathbb{Z}_q} \rho_\sigma(e) e^{2\pi i \cdot (j+c)e/q} \right| \\ &=_{(a)} \frac{1 + 2^{-\Omega(n)}}{\sqrt{qN_\sigma}} \left| \sum_{e \in \mathbb{Z}} \rho_\sigma(e) e^{2\pi i \cdot (j+c)e/q} \right| \\ &=_{(b)} \frac{1 + 2^{-\Omega(n)}}{\sqrt{qN_\sigma}} \sum_{e \in \mathbb{Z}} \sigma \rho_{1/\sigma} \left(e - \frac{j+c}{q} \right) \\ &=_{(c)} \frac{1 + 2^{-\Omega(n)}}{\sqrt{qN_\sigma}} \sigma \rho_{1/\sigma} \left(\left\lfloor \frac{j+c}{q} \right\rfloor - \frac{j+c}{q} \right) \\ &= \frac{1 + 2^{-\Omega(n)}}{\sqrt{qN_\sigma}} \sigma \rho_{q/\sigma}(j+c - \lfloor j+c \rfloor_q), \end{aligned}$$

here (a) uses Banaszczyk's tail bound, (b) is from the Poisson summation formula, (c) holds due to the assumption that $\sigma = \Omega(\sqrt{n})$. In this case, the summation $\sum_{j \in \mathbb{Z}_q} \rho_{1/\sigma} \left(e - \frac{j+c}{q} \right)$ is concentrated at $\rho_{1/\sigma} \left(\left\lfloor \frac{j+c}{q} \right\rfloor - \frac{j+c}{q} \right)$, with exponentially small weight elsewhere.

Define $j_1 = \lfloor -c \rfloor \bmod q$, $j_2 = (\lfloor -c \rfloor + 1) \bmod q$, we can establish that $|j+c - \lfloor j+c \rfloor_q| \leq 1$ holds for both $j = j_1$ and $j = j_2$. This implies that

$$|\hat{f}(j)| \geq \sqrt{\frac{\sqrt{2}\sigma}{q}} \rho_{q/\sigma}(1) \left(1 - 2^{-\Omega(n)}\right) \geq \sqrt{\frac{\sqrt{2}}{q}} e^{-\pi} \left(1 - 2^{-\Omega(n)}\right) \gg 2^{-\sqrt{n} \log q},$$

for both $j = j_1, j_2$.

As a result, we can deduce the validity of the original statement by straightforwardly applying [Theorem 31](#). $\square$

Remark 34. *Readers may be curious about why our sub-exponential algorithm cannot handle $S|LWE\rangle$ instances with an unknown phase (which we refer to as $S|LWE\rangle^{\text{phase}}$). Informally speaking, when the phase term of $S|LWE\rangle^{\text{phase}}$ samples is unknown, we can only obtain a DCP-like state with varying weights in the superposition. Although the ratio of weights on $|0\rangle$ and $|1\rangle$ can be bounded by an inverse polynomial, this ratio tends to become extremely large during the sieving step of Kuperberg’s algorithm. As a result, the final state collapses into either $|0\rangle$ or $|1\rangle$, and the information about $\mathbf{s}$ is entirely lost.*

4 Complex Gaussian and Oblivious LWE Sampling

In this section, we provide a quantum polynomial time algorithm that can obviously sample LWE instances. Our approach follows the reduction from oblivious LWE sampling to the $C|LWE\rangle$ problem introduced in [\[DFS24\]](#), digs deeper into the imaginary quadratic Gaussian amplitudes introduced in [\[Che24\]](#) (we have not used any technical lemma in [\[Che24\]](#)), and uses a center finding trick. The main benefit of our algorithm is, it improves upon the sample complexity of the core quantum algorithm from $\tilde{O}(nr)$ (where r is the Gaussian width) in [\[DFS24\]](#), to only $\tilde{O}(n)$.

In [Section 4.1](#), we’ll give the overall framework of our polynomial time quantum algorithms for solving $S|LWE\rangle$, $C|LWE\rangle$ problems, while deferring the detailed descriptions of two key unitaries, center finding and Gaussian elimination, to later subsections. Full descriptions and proofs for these unitaries are provided in [Section 4.2](#) and [Section 4.3](#), respectively. In [Section 4.4](#), we explain how our algorithm for $C|LWE\rangle$ can be applied to obviously sample LWE instances in general cases.

4.1 Constructing quantum LWE states with Gaussian amplitudes

In this subsection, we’ll give the overall framework of our polynomial time quantum algorithms for solving $S|LWE\rangle$, $C|LWE\rangle$ problems, where the amplitude is Gaussian with a specific choice of phase terms. Here we will state the theorem and algorithm for $C|LWE\rangle$, since the corresponding ones for $S|LWE\rangle$ are analogous.

Theorem 35. *Let n, m, q, ℓ be positive integers which are polynomials of the security parameter, and r be a real number. Suppose that $m = \ell n \cdot \omega(\log n)$, q is a composite number satisfying $q = q_1 q_2 \cdots q_\ell$ for some coprime $q_1, q_2, \dots, q_\ell$, and r satisfies $\frac{q}{2\sqrt{n}} > r > 30n \log n \cdot \max\{q_1, q_2, \dots, q_\ell\}$.*

There exists a quantum algorithm running in time $\text{poly}(n, \ell, \log q)$ that, takes a uniformly random matrix $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$ as input, with $1 - \text{negl}(n)$ probability outputs a state $\tilde{\phi}$ such that the trace distance between $\tilde{\phi}$ and $\phi := |\phi\rangle\langle\phi|$ is negligible, where

$$|\phi\rangle := \frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \frac{1}{\sqrt{N_r^m}} \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |(\mathbf{A}^T \mathbf{s} + \mathbf{x}) \bmod q\rangle$$

is a quantum LWE state with Gaussian amplitude and quadratic phase terms. Here we write $\mathbf{x}$ into ℓ blocks $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_\ell)$ with $\mathbf{x}_j \in \mathbb{Z}^{m/\ell}$, $j = 1, 2, \dots, \ell$, N and N_r are normalization factors

such that $N_r := \sum_{x \in \mathbb{Z}_q} \rho_r^2(x)$ normalizes the Gaussian amplitude, and N normalizes the entire state $|\phi\rangle$.

To prove [Theorem 35](#), we need to prove the following lemma first.

Lemma 36. *Let $n, m, q, \ell, r, \mathbf{A}$ satisfy the same conditions as in [Theorem 35](#). Consider the following two unnormalized vectors:*

$$\begin{aligned} |\phi'\rangle &:= \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |(\mathbf{A}^T \mathbf{s} + \mathbf{x}) \bmod q\rangle, \\ |\phi''\rangle &:= \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \sum_{\mathbf{x} \in \mathbb{Z}^m, \|\mathbf{x}\|_{\infty} < r \log(m)} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |(\mathbf{A}^T \mathbf{s} + \mathbf{x}) \bmod q\rangle. \end{aligned} \quad (7)$$

Then $\| |\phi'\rangle - |\phi''\rangle \|_2^2 \leq \text{negl}(n) \cdot \| |\phi''\rangle \|_2^2$, $\| |\phi''\rangle \|_2^2 = q^n \cdot (r/\sqrt{2})^m \cdot (1 \pm \text{negl}(n))$.

Proof. Let $L := \mathcal{L}_q(\mathbf{A})$. Let $\mathcal{P}(L)$ be any fundamental parallelepiped of L . Define a lattice

$$W := \{(\mathbf{w}_1, \mathbf{w}_2) \mid \mathbf{w}_1 \in \mathbb{Z}^n, \mathbf{w}_2 \in \mathbb{Z}^m \text{ s.t. } \mathbf{w}_1 \equiv \mathbf{w}_2 \pmod{\mathcal{P}(L)}\}.$$

Consider the following inequality:

$$\begin{aligned} & \| |\phi'\rangle - |\phi''\rangle \|_2^2 \\ & \leq \sum_{\mathbf{c} \in \mathbb{Z}_q^m} \left| \sum_{\mathbf{v} \in L, \|\mathbf{c} - \mathbf{v}\|_{\infty} \geq r \log m} e^{-\pi \frac{\|\mathbf{c} - \mathbf{v}\|^2}{r^2}} \right|^2 \\ & = \sum_{\mathbf{c} \in \mathbb{Z}_q^m} \left(\sum_{\mathbf{w}_1 \in L, \|\mathbf{c} - \mathbf{w}_1\|_{\infty} \geq r \log m} e^{-\pi \frac{\|\mathbf{c} - \mathbf{w}_1\|^2}{r^2}} \right) \left(\sum_{\mathbf{w}_2 \in L, \|\mathbf{c} - \mathbf{w}_2\|_{\infty} \geq r \log m} e^{-\pi \frac{\|\mathbf{c} - \mathbf{w}_2\|^2}{r^2}} \right) \\ & = \sum_{\mathbf{c} \in \mathbb{Z}_q^m} \left(\sum_{\mathbf{w}'_1 \in L + \mathbf{c}, \|\mathbf{w}'_1\|_{\infty} \geq r \log m} e^{-\pi \frac{\|\mathbf{w}'_1\|^2}{r^2}} \right) \left(\sum_{\mathbf{w}'_2 \in L + \mathbf{c}, \|\mathbf{w}'_2\|_{\infty} \geq r \log m} e^{-\pi \frac{\|\mathbf{w}'_2\|^2}{r^2}} \right) \\ & \stackrel{(a)}{\leq} \frac{q^m}{\text{vol}(\mathcal{P}(L))} \cdot \sum_{\mathbf{w}' \in W, \|\mathbf{w}'\|_{\infty} \geq r \log m} e^{-\pi \frac{\|\mathbf{w}'\|^2}{r^2}} \\ & \stackrel{(b)}{\leq} q^n \cdot \text{negl}(m) \cdot \rho_r(W), \end{aligned} \quad (8)$$

where (a) holds by partitioning $\mathbf{c} \in \mathbb{Z}_q^m$ into $\frac{q^n}{\text{vol}(\mathcal{P}(L))}$ copies of $\mathcal{P}(L)$; (b) uses $\text{vol}(\mathcal{P}(L)) = \det(L) = q^{m-n}$, and [Lemma 10](#).

Note that by [Lemma 16](#), $\lambda_1^{\infty}(\mathcal{L}_q(\mathbf{A})) > q/4$ with $1 - 2^{-\Omega(n)}$ probability over the randomness in sampling $\mathbf{A}$, so $|\phi''\rangle$ effectively contains q^n separated complex Gaussian balls with tails truncated.

Therefore,

$$\begin{aligned}
& \| |\phi''\rangle \|_2^2 \\
&= \sum_{\mathbf{c} \in \mathbb{Z}_q^m} \left| \sum_{\mathbf{v} \in L, \|\mathbf{c}-\mathbf{v}\|_\infty < r \log m} e^{-\pi \frac{\|\mathbf{c}-\mathbf{v}\|^2}{r^2}} \right|^2 \\
&= \sum_{\mathbf{c} \in \mathbb{Z}_q^m} \left(\sum_{\mathbf{w}_1 \in L, \|\mathbf{c}-\mathbf{w}_1\|_\infty < r \log m} e^{-\pi \frac{\|\mathbf{c}-\mathbf{w}_1\|^2}{r^2}} \right) \left(\sum_{\mathbf{w}_2 \in L, \|\mathbf{c}-\mathbf{w}_2\|_\infty < r \log m} e^{-\pi \frac{\|\mathbf{c}-\mathbf{w}_2\|^2}{r^2}} \right) \\
&= \sum_{\mathbf{c} \in \mathbb{Z}_q^m} \left(\sum_{\mathbf{w}'_1 \in L+\mathbf{c}, \|\mathbf{w}'_1\|_\infty < r \log m} e^{-\pi \frac{\|\mathbf{w}'_1\|^2}{r^2}} \right) \left(\sum_{\mathbf{w}'_2 \in L+\mathbf{c}, \|\mathbf{w}'_2\|_\infty < r \log m} e^{-\pi \frac{\|\mathbf{w}'_2\|^2}{r^2}} \right) \\
&= \frac{q^m}{\text{vol}(\mathcal{P}(L))} \cdot \sum_{\mathbf{w}' \in W, \|\mathbf{w}'\|_\infty < r \log m} e^{-\pi \frac{\|\mathbf{w}'\|^2}{r^2}} \\
&\geq q^n \cdot \rho_r(W) \cdot (1 - \text{negl}(m)).
\end{aligned}$$

Therefore, $\| |\phi'\rangle - |\phi''\rangle \|_2^2 \leq \text{negl}(n) \cdot \| |\phi''\rangle \|_2^2$.

We can alternatively calculate $\| |\phi''\rangle \|_2^2$ as

$$\| |\phi''\rangle \|_2^2 = q^n \cdot \left(\sum_{x \in \mathbb{Z}, |x| < r \log m} \rho_r^2(x) \right)^m = q^n \cdot (r/\sqrt{2})^m \cdot (1 \pm \text{negl}(m)).$$

This completes the proof of [Lemma 36](#). $\square$

Note that $|\phi'\rangle$ is just $|\phi\rangle$ without normalization factors, and $|\phi''\rangle$ is the same as $|\phi'\rangle$ except that the tails of complex Gaussian balls are removed according to ℓ_∞ -norm. By [Lemma 16](#), $\lambda_1^\infty(\mathcal{L}_q(\mathbf{A})) > q/4$ with $1 - 2^{-\Omega(n)}$ probability over the randomness in sampling $\mathbf{A}$. Since $q/4 > r \log m$, $|\phi''\rangle$ contains q^n separated complex Gaussian balls with tails truncated. So [Lemma 36](#) implies that the trace distance between $|\phi\rangle$ and $|\phi''\rangle$ is $\text{negl}(n)$. Given that $N_r = \frac{r}{\sqrt{2}}(1 \pm 2^{-\Omega(n)})$ due to [Lemma 10](#). As a result of [Lemma 36](#), the normalization factor N of $|\phi\rangle$ satisfies $N = q^n(1 \pm \text{negl}(n))$.

We will explicitly present the algorithm that computes ϕ' . Our algorithm begins by applying [Lemma 29](#) and [Remark 30](#) and the fact that $\frac{q}{2} \geq r \log m$, $\mathbb{Z}^m \cap \frac{q}{2}\mathcal{B}_\infty^m = \mathbb{Z}_q^m$ to construct the state (within trace distance $2^{-\Omega(n)}$)

$$|\phi_1\rangle := \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |\mathbf{s}\rangle \otimes \frac{1}{\sqrt{N_r^m}} \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |\mathbf{x}\rangle,$$

then adding $\mathbf{A}^T \mathbf{s}$ to the second register, to obtain the state (within trace distance $2^{-\Omega(n)}$)

$$|\phi_2\rangle := \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |\mathbf{s}\rangle \otimes \frac{1}{\sqrt{N_r^m}} \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |(\mathbf{A}^T \mathbf{s} + \mathbf{x}) \bmod q\rangle.$$

To approximate ϕ , it remains to get rid of the first register in $|\phi_2\rangle$. Intuitively, the idea is to first extract the Gaussian center $\mathbf{A}^T \mathbf{s}$ of the second register, and then solve $\mathbf{s}$ by Gaussian elimination.

We'll realize this idea using the center finding unitary $\mathcal{U}_{\text{CF}}$ and the Gaussian elimination unitary $\mathcal{U}_{\text{GE}}$, constructed and proved correct later in [Section 4.2](#) and [Section 4.3](#).

We write $\mathbf{A} = (\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_\ell)$ with $\mathbf{A}_j \in \mathbb{Z}_q^{n \times (m/\ell)}$, $j = 1, 2, \dots, \ell$; and write each $\mathbf{A}_j = (\mathbf{a}_{j,1}, \mathbf{a}_{j,2}, \dots, \mathbf{a}_{j,m/\ell})$ with $\mathbf{a}_{j,k} \in \mathbb{Z}_q^n$, $k = 1, 2, \dots, m/\ell$. The algorithm applies a center finding unitary $\mathcal{U}_{\text{CF}}$ ([Theorem 37](#) in [Section 4.2](#)) in order to decode $\mathbf{a}_{j,k}^T \mathbf{s} \bmod q_j$ on each coordinate of the second register, and obtains the state (within trace distance $2^{-\Omega(n)}$)

$$|\phi_3\rangle := \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |\mathbf{s}\rangle \bigotimes_{j=1}^{\ell} \left(\bigotimes_{k=1}^{m/\ell} (\sqrt{p_{j,k,s}} |\varphi_{j,k,s}\rangle |\mathbf{a}_{j,k}^T \mathbf{s} \bmod q_j\rangle + \sqrt{1 - p_{j,k,s}} |\text{err}_{j,k,s}\rangle) \right),$$

where the weight $p_{j,k,s}$ satisfies that $p_{j,k,s} \geq 1 - 1/n$, as guaranteed by [Theorem 37](#), and the states $|\varphi_{j,k,s}\rangle$ and $|\text{err}_{j,k,s}\rangle$ are unit vectors, whose precise descriptions are irrelevant to our algorithm. For convenience, denote $|\psi_{j,s}\rangle := \bigotimes_{k=1}^{m/\ell} (\sqrt{p_{j,k,s}} |\varphi_{j,k,s}\rangle |\mathbf{a}_{j,k}^T \mathbf{s} \bmod q_j\rangle + \sqrt{1 - p_{j,k,s}} |\text{err}_{j,k,s}\rangle)$, and thus $|\phi_3\rangle$ can be rewritten as

$$|\phi_3\rangle = \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |\mathbf{s}\rangle \bigotimes_{j=1}^{\ell} |\psi_{j,s}\rangle.$$

Note that $|\psi_{j,s}\rangle$ after measured in computational basis produces a vector $\tilde{\mathbf{y}}_{j,s}$ where every coordinate agrees with $\mathbf{A}_j^T \mathbf{s} \bmod q_j$ with probability at least $1 - 1/n$. Then we can use a classical algorithm ([Theorem 39](#) in [Section 4.3](#)) whose core is Gaussian elimination to solve $\mathbf{s} \bmod q_j$ given $\mathbf{A}_j$ and $\tilde{\mathbf{y}}_{j,s}$. Applying this in superposition, for each $j = 1, 2, \dots, m/\ell$, our algorithm takes $\mathbf{A}_j \bmod q_j$ and applies the Gaussian elimination unitary $\mathcal{U}_{\text{GE}}$ ([Corollary 40](#) in [Section 4.3](#)) on $|\psi_{j,s}\rangle$ to obtain the state (within trace distance $2^{-\Omega(n)}$)

$$|\phi_4\rangle := \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |\mathbf{s}\rangle \bigotimes_{j=1}^{\ell} (\sqrt{p_{j,s}} |\psi'_{j,s}\rangle |\mathbf{s} \bmod q_j\rangle + \sqrt{1 - p_{j,s}} |\text{err}_{j,s}\rangle),$$

where the weight $p_{j,s}$ satisfies that $p_{j,s} \geq 1 - \text{negl}(n)$ as guaranteed by [Corollary 40](#), with success probability $1 - \text{negl}(n)$ over the randomness of $\mathbf{A}$, and the states $|\psi_{j,s}\rangle$ and $|\text{err}_{j,s}\rangle$ are unit vectors, whose precise descriptions are irrelevant to our algorithm.

By Chinese Remainder Theorem (CRT), we can (coherently) recover $\mathbf{s} \bmod q$ via $\mathbf{s} \bmod q_j$, $j = 1, 2, \dots, \ell$ and subtract it in the first register, to obtain the state (within trace distance $\text{negl}(n)$)

$$|\phi_5\rangle := \frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |0\rangle \bigotimes_{j=1}^{\ell} (\sqrt{p_{j,s}} |\psi'_{j,s}\rangle |\mathbf{s} \bmod q_j\rangle + \sqrt{1 - p_{j,s}} |\text{err}_{j,s}\rangle)$$

with success probability $1 - \text{negl}(n)$ over the randomness of $\mathbf{A}$. To see how it works, we have that $|\phi_4\rangle$ is negligibly close to

$$|\phi'_4\rangle := \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |\mathbf{s}\rangle \bigotimes_{j=1}^{\ell} |\psi'_{j,s}\rangle |\mathbf{s} \bmod q_j\rangle$$

in trace distance, and by subtracting $\mathbf{s}$ in the first register, we get

$$|\phi'_5\rangle := \frac{1}{\sqrt{q^n}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} |0\rangle \bigotimes_{j=1}^{\ell} |\psi'_{j,\mathbf{s}}\rangle |\mathbf{s} \bmod q_j\rangle,$$

which is negligibly close to $|\phi_5\rangle$ in trace distance. Note that the overall normalization factor of $|\phi_5\rangle$ is changed from q^n to $N = q^n(1 \pm \text{negl}(n))$, because there might be overlaps for $\sqrt{1 - p_{j,\mathbf{s}}} |\text{err}_{j,\mathbf{s}}\rangle$ on the summation of $\mathbf{s} \in \mathbb{Z}_q^n$.

Finally, the algorithm discards the first register, and applies the inverse of $\mathcal{U}_{\text{GE}}$ to obtain the state

$$|\phi_6\rangle := \frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \bigotimes_{j=1}^{\ell} |\psi_{j,\mathbf{s}}\rangle$$

and then applies the inverse of $\mathcal{U}_{\text{CF}}$ to recover the $\mathbb{C}|\text{LWE}\rangle$ state (within trace distance $\text{negl}(n)$)

$$|\phi\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{s} \in \mathbb{Z}_q^n} \frac{1}{\sqrt{N_r^m}} \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_r(\mathbf{x}) e^{-\pi i \sum_{j=1}^{\ell} \|\mathbf{x}_j\|^2 / q_j} |(\mathbf{A}^T \mathbf{s} + \mathbf{x}) \bmod q\rangle,$$

with success probability $1 - \text{negl}(n)$ over the randomness of $\mathbf{A}$.

This completes the description of the overall framework of our algorithm.

4.2 Center finding

We show an efficient quantum algorithm for finding the center c of the following complex Gaussian state, up to modulus q' :

$$|\varphi_c\rangle := \frac{1}{\sqrt{N_r}} \sum_{x \in \mathbb{Z}_q} \rho_r(x) e^{-\pi i x^2 / q'} |(x + c) \bmod q\rangle,$$

where q' is a factor of q and $N_r = \sum_{x \in \mathbb{Z}_q} \rho_r^2(x)$ is the normalization factor.

Theorem 37 (Center finding). *Let q, q', n be positive integers such that $q/q' \in \mathbb{Z}$, r be a positive real such that $q > 2r\sqrt{n}$, $r > 30q'n \log n$. There exists a center finding unitary $\mathcal{U}_{\text{CF}}$ such that for any $c \in \mathbb{Z}$,*

$$\mathcal{U}_{\text{CF}}(|\varphi_c\rangle_{\text{A}} |0\rangle_{\text{B}}) = \sqrt{p} |\varphi'\rangle_{\text{A}} |c \bmod q'\rangle_{\text{B}} + \sqrt{1 - p} |\text{err}\rangle_{\text{AB}},$$

where $p \geq 1 - 1/n$ and $|\varphi'\rangle_{\text{A}} |c \bmod q'\rangle_{\text{B}}, |\text{err}\rangle_{\text{AB}}$ are orthonormal states.

Before proving the theorem, we first give the key observation that for any $c_1, c_2 \in \mathbb{Z}$ with $c_1 \not\equiv c_2 \bmod q'$, the complex Gaussian states $|\varphi_{c_1}\rangle, |\varphi_{c_2}\rangle$ are nearly orthogonal and, therefore, almost perfectly distinguishable.

Fact 38. *For any $c_1, c_2 \in \mathbb{Z}$ with $c_1 \not\equiv c_2 \bmod q'$,*

$$|\langle \varphi_{c_1} | \varphi_{c_2} \rangle| \leq 2^{-\Omega(n)}.$$

This observation will not be used in our proof and only provides with intuition. We defer the proof to [Appendix A.2](#). Keeping the above observation in mind, we are back to the proof of [Theorem 37](#).

Proof of Theorem 37. The algorithm of center finding is straightforward: “measure” the lower order bits (up to modulus q') of $|\varphi_c\rangle$ under the basis $\{|\mu_d\rangle\}_{d \in \mathbb{Z}_{q'}}$, and output the measurement result $d \in \mathbb{Z}_{q'}$, where

$$|\mu_d\rangle = \frac{1}{\sqrt{q'}} \sum_{x \in \mathbb{Z}_{q'}} e^{-\pi i(x-d)^2/q'} |x\rangle.$$

We first show that $\{|\mu_d\rangle\}_{d \in \mathbb{Z}_{q'}}$ is a basis that can be efficiently prepared.

For $d, d' \in \mathbb{Z}_{q'}$, the overlap of $|\mu_d\rangle$ and $|\mu_{d'}\rangle$ is

$$\begin{aligned} \langle \mu_{d'} | \mu_d \rangle &= \frac{1}{q'} \sum_{x \in \mathbb{Z}_{q'}} e^{-\pi i((x-d)^2 - (x-d')^2)/q'} \\ &= \frac{1}{q'} e^{\pi i(d'^2 - d^2)/q'} \sum_{x \in \mathbb{Z}_{q'}} e^{2\pi i(d-d')x/q'} \\ &= \begin{cases} 1 & d = d' \\ 0 & d \neq d' \end{cases}. \end{aligned}$$

Therefore $\{|\mu_d\rangle\}_{d \in \mathbb{Z}_{q'}}$ is indeed a basis. For any $d \in \mathbb{Z}_{q'}$, the basis state $|\mu_d\rangle$ can be efficiently prepared by:

$$\begin{aligned} |d\rangle &\mapsto_{(a)} e^{-\pi i d^2/q'} |d\rangle \mapsto_{\text{QFT}_{q'}^\dagger} \sum_{x \in \mathbb{Z}_{q'}} e^{2\pi i x d/q'} e^{-\pi i d^2/q'} |x\rangle \\ &\mapsto_{(b)} \sum_{x \in \mathbb{Z}_{q'}} e^{-\pi i x^2/q'} e^{2\pi i x d/q'} e^{-\pi i d^2/q'} |x\rangle = \sum_{x \in \mathbb{Z}_{q'}} e^{-\pi i(x-d)^2/q'} |x\rangle, \end{aligned}$$

where (a), (b) use the phase kick-back trick ([Lemma 27](#)).

Define $\mathcal{U} = \sum_{d \in \mathbb{Z}_{q'}} |d\rangle \langle \mu_d|$. The center finding unitary $\mathcal{U}_{\text{CF}}$ acts as follows

$$\begin{aligned} |\varphi_c\rangle_{\text{A}} |0\rangle_{\text{B}} &= \frac{1}{\sqrt{N_r}} \sum_{x \in \mathbb{Z}_q} \rho_r(x) e^{-\pi i x^2/q'} |(x+c) \bmod q\rangle_{\text{A}} |0\rangle_{\text{B}} \\ &\mapsto \frac{1}{\sqrt{N_r}} \sum_{x \in \mathbb{Z}_q} \rho_r(x) e^{-\pi i x^2/q'} |[x+c] \bmod q\rangle_{q'}_{\text{A}} |(x+c) \bmod q'\rangle_{\text{B}} \\ &\mapsto_{\mathcal{I}_{\text{A}} \otimes \mathcal{U}_{\text{B}}} \frac{1}{\sqrt{N_r q'}} \sum_{x \in \mathbb{Z}_q} \sum_{d \in \mathbb{Z}_{q'}} \rho_r(x) e^{-\pi i(x^2 - (x'-d)^2)/q'} |[x+c] \bmod q\rangle_{q'}_{\text{A}} |d\rangle_{\text{B}} \end{aligned}$$

where $x' = (x+c) \bmod q'$. Recall that at the beginning of this subsection, we observed $|\varphi_0\rangle$ and $|\varphi_c\rangle$ are nearly orthogonal if $c \neq 0$. According to a similar intuition, the amplitude in the above

state should concentrate on $d = c \bmod q'$. Therefore we break the above state into two parts:

$$\underbrace{\frac{1}{\sqrt{N_r q'}} \sum_{x \in \mathbb{Z}_q} \rho_r(x) e^{-\pi i(x^2 - (x' - c')^2)/q'} \lfloor (x + c) \bmod q \rfloor_{q'} \rangle_{\mathbf{A}} |c \bmod q'\rangle_{\mathbf{B}}}_{\sqrt{p} |\varphi'\rangle_{\mathbf{A}}} + \underbrace{\frac{1}{\sqrt{N_r q'}} \sum_{x \in \mathbb{Z}_q} \sum_{d \in \mathbb{Z}_{q'}, d \neq c'} \rho_r(x) e^{-\pi i(x^2 - (x' - d)^2)/q'} \lfloor (x + c) \bmod q \rfloor_{q'} \rangle_{\mathbf{A}} |d\rangle_{\mathbf{B}}}_{\sqrt{1-p} |\text{err}\rangle_{\mathbf{AB}}},$$

where $c' = c \bmod q'$, and $|\varphi'\rangle_{\mathbf{A}}, |\text{err}\rangle_{\mathbf{AB}}$ are unit vectors. We will proceed to show $p \geq 1 - 1/n$, which will conclude the proof.

Denote $k_0 = (c - c')/q' \in \mathbb{Z}$. Fix an integer $k \in \mathbb{Z}_{q/q'}$, note that for every $x \in \mathbb{Z}_q$ such that $\lfloor (x + c) \bmod q \rfloor_{q'} = kq'$, we have $x' = x + c - kq'$, so the phase term in the expression of $\sqrt{p} |\varphi'\rangle_{\mathbf{A}}$ is

$$e^{-\pi i(x^2 - (x' - c')^2)/q'} = e^{-\pi i(x^2 - (x + (k_0 - k)q')^2)/q'} = e^{\pi i(k_0 - k)q'} = (-1)^{(k_0 - k)q'},$$

which is also fixed. Thus

$$\begin{aligned} p &= \frac{1}{N_r q'} \sum_{k \in \mathbb{Z}_{q/q'}} \left(\sum_{x \in \mathbb{Z}_q, \lfloor (x + c) \bmod q \rfloor_{q'} = kq'} \rho_r(x) \right)^2 \\ &= \frac{1}{N_r q'} \sum_{k \in \mathbb{Z}_{q/q'}} \left(\sum_{y \in \mathbb{Z}_{q'}} \rho_r((kq' + y - c) \bmod q) \right)^2. \end{aligned}$$

To lower bound p , we observe the following three facts: (1) the term c can be replaced by c' since $c' = c \bmod q'$; (2) $\bmod q$ can be dropped since for any integer x , the absolute value of $x \bmod q$ is no larger than $|x|$; (3) $q > 2r \log n$ so we can truncate the support of k to $|k| \leq r \log n/q'$. So

$$\begin{aligned} p &\geq \frac{1}{N_r q'} \sum_{k \in \mathbb{Z}_{q/q'}} \left(\sum_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c') \right)^2 \\ &\geq \frac{1}{N_r q'} \sum_{k \in \mathbb{Z}, |k| \leq r \log n/q'} \left(q' \cdot \frac{\min_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c')^2}{\max_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c')^2} \sum_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c')^2 \right) \\ &\geq_{(a)} \frac{1 - \frac{1}{3n}}{N_r} \sum_{k \in \mathbb{Z}, |k| \leq r \log n/q'} \sum_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c')^2 \\ &\geq \frac{1 - \frac{1}{3n}}{N_r} \sum_{x \in \mathbb{Z}, |x| \leq r \log n} \rho_r(x)^2 \\ &\geq_{(b)} 1 - \frac{1}{n} \end{aligned}$$

where (a) is because of the following bound on the quantity $f(k) := \frac{\min_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c')^2}{\max_{y \in \mathbb{Z}_{q'}} \rho_r(kq' + y - c')^2} \geq 1 - \frac{1}{3n}$, for all $|k| \leq r \log n/q'$. Given our assumption that $r > 30q'n \log n$, this bound is easily shown

by splitting into two cases: for $3 \leq |k| \leq r \log n / q'$, $f(k) \geq e^{-8\pi k q'^2 / r^2} \geq e^{-\frac{1}{3n}}$; for $|k| \leq 2$, $f(k) \geq e^{-32\pi q'^2 / r^2} \geq e^{-o(\frac{1}{n})}$. (b) is due to Banaszczyk's tail bound on $\dim-1$ $\mathcal{L} = \mathbb{Z}$, ball radius $\log n$ and $\frac{q}{2r} > \log n$, since $N_r = \sum_{x \in \mathbb{Z}_q} \rho_r(x)^2$. This ends the proof of [Theorem 37](#). $\square$

4.3 Gaussian elimination

In this subsection, we construct a classical polynomial time probabilistic algorithm, which uses Gaussian elimination along with a verification process to obtain $\mathbf{s} \bmod q_j$ given $\mathbf{A}_j$ and $\tilde{\mathbf{y}}$, given that any coordinate $\tilde{y}_k$ of $\tilde{\mathbf{y}}$ equals to the coordinate $\mathbf{a}_{j,k}^T \mathbf{s} \bmod q_j$ of $\mathbf{A}_j^T \mathbf{s} \bmod q_j$ with probability at least $1 - 1/n$, independently for $k = 1, 2, \dots, m/\ell$.

Theorem 39. *Let n, m, q_j, ℓ be positive integers such that $m = \ell n \cdot \omega(\log n)$. There exists a classical polynomial time probabilistic algorithm $\mathcal{B}$ that, given $\mathbf{A}_j = (\mathbf{a}_{j,1}, \mathbf{a}_{j,2}, \dots, \mathbf{a}_{j,m/\ell}) \leftarrow U(\mathbb{Z}_{q_j}^{n \times (m/\ell)})$ and $\tilde{\mathbf{y}} \in \mathbb{Z}_{q_j}^{m/\ell}$ such that independently for all $k = 1, 2, \dots, m/l$,*

$$\Pr_{\tilde{\mathbf{y}}}[\tilde{y}_k = \mathbf{a}_{j,k}^T \mathbf{s} \bmod q_j] \geq 1 - \frac{1}{n}$$

for some hidden vector $\mathbf{s} \in \mathbb{Z}_{q_j}^n$, recovers $\mathbf{s}$ with probability $1 - \text{negl}(n)$ over the randomness of $\mathbf{A}_j$, $\tilde{\mathbf{y}}$ and the randomness of $\mathcal{B}$.

Applying the algorithm in superposition, we immediately obtain the following corollary that is used in the algorithm presented in [Section 4.1](#).

Corollary 40. *Given the state*

$$|\psi_{j,s}\rangle = \bigotimes_{k=1}^{m/\ell} (\sqrt{p_{j,k,s}} |\varphi_{j,k,s}\rangle |\mathbf{a}_{j,k}^T \mathbf{s} \bmod q_j\rangle + \sqrt{1 - p_{j,k,s}} |\text{err}_{j,k,s}\rangle)$$

such that all $p_{j,k,s} \geq 1 - 1/n$, there exists a Gaussian elimination unitary $\mathcal{U}_{\text{GE}}$ satisfying that, with probability $1 - \text{negl}(n)$ over the randomness of $\mathbf{A}_j$,

$$\mathcal{U}_{\text{GE}}(|\psi_{j,s}\rangle_{\text{A}} |0\rangle_{\text{B}}) = \sqrt{p_j} |\psi'_{j,s}\rangle_{\text{A}} |\mathbf{s} \bmod q_j\rangle_{\text{B}} + \sqrt{1 - p_j} |\text{err}_{j,s}\rangle_{\text{AB}},$$

where $p_j = 1 - \text{negl}(n)$ and $|\psi'_{j,s}\rangle_{\text{A}} |\mathbf{s} \bmod q_j\rangle_{\text{B}}, |\text{err}_{j,s}\rangle$ are orthogonal states.

Proof of [Theorem 39](#). The classical algorithm $\mathcal{B}$ runs as follows:

1. Divide $\mathbf{A}_j$ into groups of n by $2n$ matrices $\mathbf{A}_{j,1}, \mathbf{A}_{j,2}, \dots, \mathbf{A}_{j,m/(2\ell n)} \in \mathbb{Z}_q^{n \times 2n}$, and write $\tilde{\mathbf{y}}$ into blocks $\tilde{\mathbf{y}} = (\tilde{\mathbf{y}}_1, \tilde{\mathbf{y}}_2, \dots, \tilde{\mathbf{y}}_{m/(2\ell n)})$ with each block belonging to $\mathbb{Z}_{q_j}^{2n}$. The algorithm uses the first half of matrices for computing $\mathbf{s}$ and the second half for verification.
2. For each $i = 1, 2, \dots, m/(4\ell n)$:
 - (a) If $\mathbf{A}_{j,i}$ is not full-rank (with rank n), go to the next iteration;
 - (b) Solve the linear equations $\mathbf{A}_{j,i}^T \mathbf{s} = \tilde{\mathbf{y}}_i \bmod q_j$ using Gaussian elimination. If the linear equations are not solvable, go to the next iteration; otherwise obtain a solution $\mathbf{s}' \in \mathbb{Z}_{q_j}^n$;

- (c) Compare $\mathbf{y}' = \mathbf{A}_{j,i+m/(4\ell n)}^T \mathbf{s}' \bmod q_j$ with $\tilde{\mathbf{y}}_{i+m/(4\ell n)}$. If there are at least 0.9 fraction of entries are matched, then output $\mathbf{s} = \mathbf{s}'$; otherwise, go to the next iteration.

By construction, $\mathcal{B}$ runs in polynomial time. We then show its correctness.

First, the verification process in step 2(c) ensures that the probability of outputting a wrong answer is negligible. Assume $\mathbf{s}' \neq \mathbf{s} \bmod q_j$ is a wrong answer obtained in step 2(b) upon the i th iteration. Since the matrix $\mathbf{A}_j$ is sampled uniformly at random, each row of $\mathbf{A}_{j,i+m/(4\ell n)}^T \mathbf{s} \bmod q_j$ and $\mathbf{y}' = \mathbf{A}_{j,i+m/(4\ell n)}^T \mathbf{s}' \bmod q_j$ are different with probability at least $\frac{1}{2}$ independently. Meanwhile, conditioning on $\mathbf{A}$, each row of $\mathbf{A}_{j,i+m/(4\ell n)}^T \mathbf{s} \bmod q_j$ and $\tilde{\mathbf{y}}_{i+m/(4\ell n)}$ agrees with probability at least $1 - \frac{1}{n}$ independently. Therefore, for each row $k = 1, 2, \dots, 2n$,

$$\begin{aligned} & \Pr_{\mathbf{A}_j, \mathbf{y}'} [(\tilde{\mathbf{y}}_{i+m/(4\ell n)})_k \neq (\mathbf{y}')_k] \\ & \geq \Pr_{\mathbf{A}_j, \mathbf{y}'} [(\tilde{\mathbf{y}}_{i+m/(4\ell n)})_k = (\mathbf{A}_{j,i+m/(4\ell n)}^T \mathbf{s} \bmod q_j)_k] \cdot \Pr_{\mathbf{A}_j, \mathbf{y}'} [(\mathbf{y}')_k \neq (\mathbf{A}_{j,i+m/(4\ell n)}^T \mathbf{s} \bmod q_j)_k] \\ & \geq \frac{1}{2}(1 - 1/n) \end{aligned}$$

By Chernoff bound, this $\mathbf{s}' \neq \mathbf{s} \bmod q_j$ only passes 2(c) with negligible probability.

Second, the probability that the algorithm outputs the correct secret $\mathbf{s} \bmod q_j$ is overwhelming. The failure probability in step 2(a) is negligible because of the following proposition:

Proposition 41. *Suppose $q = q_1^{c_1} \dots q_\ell^{c_\ell}$, where $q_1, \dots, q_\ell$ are ℓ distinct primes, $c_1, \dots, c_\ell \in \mathbb{N}^+$. With probability more than $1 - k \log q \cdot 2^{-n}$ over the randomness of sampling uniformly random $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \leftarrow U(\mathbb{Z}_q^{n+k})$, the vectors are linearly independent in $\mathbb{Z}_{q_j}$ for all $j = 1, 2, \dots, \ell$.*

Proof. The vectors are linearly independent in $\mathbb{Z}_{q_j}$ with probability

$$\left(1 - q_j^{-(n+k)}\right) \left(1 - q_j^{-(n+k-1)}\right) \dots \left(1 - q_j^{-(n+1)}\right) \geq (1 - 2^{-n})^k.$$

Note that $\ell \leq \log q$, so the probability that $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are linearly independent in $\mathbb{Z}_{q_j}$ for all $j = 1, 2, \dots, \ell$ is greater than $(1 - 2^{-n})^{k \log q} \geq 1 - k \log q \cdot 2^{-n}$. $\square$

By Proposition 41, a random n by $2n$ matrix is full-rank in $\mathbb{Z}_{q_j}$ with probability at least $1 - n \log q \cdot 2^{-n} = 1 - \text{negl}(n)$. Observe that none of the entries in $\tilde{\mathbf{y}}_i$ is wrong with probability at least $(1 - 1/n)^{2n} > 0.1$, thus, among the $m/(4\ell n) = \omega(\log n)$ linear equations $\mathbf{A}_{j,i}^T \mathbf{s} = \tilde{\mathbf{y}}_i \bmod q_j$ for $i = 1, 2, \dots, m/(4\ell n)$, except with negligible probability, at least one of them consists of rows without any error, which leads to the correct $\mathbf{s}$ in step 2(b). This correct $\mathbf{s}$ will pass the verification process in step 2(c) with $1 - \text{negl}(n)$ probability since each entry of $\mathbf{y}'$ and $\tilde{\mathbf{y}}_{i+m/(4\ell n)}$ are equal with probability at least $1 - 1/n$ in this case. This completes the proof for the correctness of algorithm $\mathcal{B}$. $\square$

4.4 Oblivious LWE sampling

While constructing a quantum LWE state with a specific choice of phase terms may not seem interesting at first glance, the problem is actually closely related to a problem called oblivious LWE sampling for a vast range of LWE parametrizations.

Roughly speaking, in oblivious LWE sampling, the sampler is asked to sample from the LWE distribution $(\mathbf{A}, \mathbf{A}^T \mathbf{s} + \mathbf{e})$, while any extractor, observing the sampler (but not disturbing the sampler's state), cannot extract the secret $\mathbf{s}$.

Here we abuse the notation $\text{LWE}_{n,m,q,\alpha}$ to represent the distribution of valid LWE samples $\mathbf{A}, \mathbf{b} = \mathbf{A}^T \mathbf{s} + \mathbf{e} \bmod q$, where $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$, $\mathbf{s} \leftarrow U(\mathbb{Z}_q^n)$, and $\mathbf{e} \leftarrow D_{\mathbb{Z}, \alpha q}^m$. Later we will use two variants of $\text{LWE}_{n,m,q,\alpha}$. Let $\text{LWE}_{n,m,q,\leq \alpha}$ denote the case where the error is sampled from $D_{\mathbb{Z}, \beta q}$ for some $\beta \leq \alpha$. Let $\text{LWE}_{n,m,q,\alpha}(D_{\mathbb{Z}^n, s})$ denote the distribution that produces LWE samples where the secret is sampled from $D_{\mathbb{Z}^n, s}$ for some $s > 0$.

Definition 42 (Witness-Oblivious Quantum Samplers, [DFS24]). *Let n, m, q, α be LWE parameters following the same definition as in Definition 1. A quantum polynomial time algorithm $\mathcal{S}$ is called a witness-oblivious quantum sampler for $\text{LWE}_{n,m,q,\alpha}$ if it has the following properties:*

1. *Given as input the security parameter 1^λ , a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, and polynomial ancillas initialized to $|0\rangle$, $\mathcal{S}$ outputs a pair $(\mathbf{A}, \mathbf{b})$ where $\mathbf{b} \in \mathbb{Z}_q^m$ such that for a uniformly distributed $\mathbf{A}$, the distribution of $\mathcal{S}(1^\lambda, \mathbf{A}, |0\rangle^{\text{poly}(\lambda)})$ is statistically indistinguishable with the distribution $\text{LWE}_{n,m,q,\alpha}$;*
2. *For any valid quantum polynomial-time extractor $\mathcal{E}$ for $\mathcal{S}$, we have that*

$$\Pr \left[\mathbf{s} = \mathbf{s}' \text{ and } \mathbf{e} = \mathbf{e}' : ((\mathbf{A}, \mathbf{b} = \mathbf{A}^T \mathbf{s} + \mathbf{e}), (\mathbf{s}', \mathbf{e}')) \leftarrow \langle \mathcal{S}, \mathcal{E} \rangle \left(\tau_{\mathcal{S}} = (1^\lambda, \mathbf{A}, |0\rangle^{\text{poly}(\lambda)}), \tau_{\mathcal{E}} = |0\rangle^{\text{poly}(\lambda)} \right) \right] \leq \text{negl}(\lambda)$$

where $\langle \mathcal{S}, \mathcal{E} \rangle(\tau_{\mathcal{S}}, \tau_{\mathcal{E}})$ denote the joint output of the sampler $\mathcal{S}$ and the extractor $\mathcal{E}$ on the input $\tau_{\mathcal{S}}$ and $\tau_{\mathcal{E}}$, and an extractor $\mathcal{E}$ is called valid for the sampler $\mathcal{S}$ if and only if $\mathcal{E}$ does not change the state of the sampler $\mathcal{S}$ up to negligible trace distance when $\mathcal{S}$ and $\mathcal{E}$ are running together.

Theorem 2 of [DFS24] presents a generic construction of a witness-oblivious quantum LWE sampler from any algorithm that solves $\text{C}|\text{LWE}\rangle$.

Theorem 43 (Theorem 2, [DFS24]). *Let n, m, q, α be LWE parameters. Assume the quantum hardness of $\text{LWE}_{n,m,q,\alpha}$. If a quantum polynomial algorithm $\mathcal{S}$ solves $\text{C}|\text{LWE}\rangle_{n,m,q,f}$, where the amplitude f satisfies $|f|^2 \propto D_{\mathbb{Z}, \alpha q}$, then $\mathcal{S}$ followed by computational measurements is a witness-oblivious quantum sampler for $\text{LWE}_{n,m,q,\alpha}$.*

Plugging in our $\text{C}|\text{LWE}\rangle$ algorithm in Theorem 35, together with Lemma 36 which shows the state $|\phi\rangle$ in Theorem 35 is negligibly close to $|\phi''\rangle$ in Lemma 36, we immediately get the following witness-oblivious LWE sampler.

Corollary 44. *Let n, m, q, ℓ be positive integers and r be a real number. Suppose that $m = \ell n \cdot \omega(\log n)$, q is a composite number satisfying $q = q_1 q_2 \cdots q_\ell$ where $q_1, q_2, \dots, q_\ell$ are coprime, r satisfies $\frac{q}{2\sqrt{n}} > r > 30n \log n \cdot \max\{q_1, q_2, \dots, q_\ell\}$. Assume the quantum hardness of $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}$, there exists a witness-oblivious quantum sampler for $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}$.*

As noted in [DFS24], once we get an oblivious $\text{LWE}_{n,m,q,\alpha}$ sampler for certain composite modulus q , we can throw away additional samples to get an oblivious $\text{LWE}_{n,m',q,\alpha}$ sampler for $m' \leq m$.

We can also use the modulus switching technique in [BLP⁺13] to get an oblivious LWE sampler for $\text{LWE}_{n,m,q',\alpha'}$ where $q' < q$ is not necessarily composite. Here we spell out the details of the modulus switching part.

To use modulus switching, we need to start from LWE samples where the length of the secret is shorter than q (it is easy to modify the quantum algorithm for $\mathbb{C}|\text{LWE}\rangle$ so that the secret is a superposition of a general distribution). For consistency let us fix the secret to be sampled from $D_{\mathbb{Z}^n,s}$ and adapt from [BLP⁺13, Corollary 3.2]:

Lemma 45. *For any positive integers $n, m, q \geq q' \geq 1$, $\alpha, \alpha' \in (0, 1)$ such that $m, \log q \in \text{poly}(n)$, $(\alpha')^2 \geq \alpha^2 + \left(\frac{s\sqrt{n}}{q'}\right)^2 \cdot \omega(\log n)$, there is a $\text{poly}(n)$ time classical algorithm that takes as input a sample from $\text{LWE}_{n,m,q,\leq\alpha}(D_{\mathbb{Z}^n,s})$, outputs a sample within negligible statistical distance from $\text{LWE}_{n,m,q',\leq\alpha'}(D_{\mathbb{Z}^n,s})$.*

As a corollary of Corollary 44 and Lemma 45:

Corollary 46. *Let n, m, q, ℓ be positive integers and r be a real number. Suppose that $m = \ell n \cdot \omega(\log n)$, q is a composite number satisfying $q = q_1 q_2 \cdots q_\ell$ where $q_1, q_2, \dots, q_\ell$ are coprime, r satisfies $\frac{q}{\sqrt{n}} > r > 30n \log n \cdot \max\{q_1, q_2, \dots, q_\ell\}$. Suppose positive integers $m' \leq m, q' \leq q$ and real numbers $s, r' > 0$ satisfies $\left(\frac{r'}{\sqrt{2}q'}\right)^2 \geq \left(\frac{r}{\sqrt{2}q}\right)^2 + \left(\frac{s\sqrt{n}}{q'}\right)^2 \cdot \omega(\log n)$, and assume the quantum hardness of $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}(D_{\mathbb{Z}^n,s})$, there exists a witness-oblivious quantum sampler for $\text{LWE}_{n,m',q',r'/(\sqrt{2}q')}(D_{\mathbb{Z}^n,s})$.*

For example, to produce an oblivious LWE sample from $\text{LWE}_{n,m',q',r'/(\sqrt{2}q')}(D_{\mathbb{Z}^n,s})$ where $q' \in \tilde{O}(n^2)$ is prime, $r' \in \tilde{O}(n^{1.5})$, $s \in O(\sqrt{n})$, we can start from an oblivious LWE sample from $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}(D_{\mathbb{Z}^n,s})$ where $r \in \tilde{O}(n^{1.5})$, $q \in \tilde{O}(n^2)$ such that $q' < q < 2q'$, $q = q_1 q_2 q_3 q_4$ where $q_1, q_2, q_3, q_4 \in \tilde{O}(\sqrt{n})$ are coprime. The sample complexity required for producing $\text{LWE}_{n,m,q,r/(\sqrt{2}q)}(D_{\mathbb{Z}^n,s})$ from our quantum algorithm in Corollary 44 is in $\tilde{O}(n)$. If we use the quantum oblivious LWE sampler in [DFS24], the sample complexity required for producing $\text{LWE}_{n,m',q',r'/(\sqrt{2}q')}(D_{\mathbb{Z}^n,s})$ is $\tilde{O}(n^{2.5})$.

5 Hardness of $\mathbb{S}|\text{LWE}\rangle$ with Unknown Phase via Extrapolated DCP

In this section, we show how to obtain a quantum reduction from classical LWE to $\mathbb{S}|\text{LWE}\rangle^{\text{phase}}$ with Gaussian amplitude. Our reduction goes through the Extrapolated Dihedral Coset problem (EDCP), derived from a modification of the reduction by Regev [Reg04] and Brakerski et al. [BKSW18]. Our reduction consists of two steps:

Step 1 Given a classical LWE instance, our quantum reduction first generates an Extrapolated DCP state with amplitudes following a Gaussian distribution centered at an unknown value. More precisely, we establish the following theorem:

Theorem 47. *Let $n, m, q \in \mathbb{N}^+$, $\alpha = \Omega(\sqrt{n})$, $\beta, \gamma \in (0, 1)$ satisfy $m \geq n \log q$, $\alpha\gamma\sqrt{m} < \beta < \frac{1}{16\sqrt{m \log(\beta q)}}$. There exists a $\text{poly}(n)$ time quantum algorithm that, given a classical*

LWE instance $\text{LWE}_{n,m,q,\gamma} = (\mathbf{A}, \mathbf{A}^T \mathbf{s} + \mathbf{e})$, generates, with probability $1 - 2^{-\Omega(n)}$, a vector $\mathbf{y} \in \mathbb{Z}_q^m \cap B_{\mathcal{L}_q}(\mathbf{A})$ and an Extrapolated DCP state of form

$$|\text{EDCP}\rangle = \frac{1}{\sqrt{N}} \sum_{j \in \mathbb{Z}_q} \rho_\sigma(j - c) |j\rangle |(\mathbf{v} + j \cdot \mathbf{s}) \bmod q\rangle, \quad (9)$$

for some normalization factor N , where

1. the vector $\mathbf{v}$ is chosen uniformly at random from $\mathbb{Z}_q^n$ and is hidden,
2. the Gaussian width $\sigma = \frac{\alpha\beta q}{\sqrt{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2}}$,
3. the vector $\mathbf{y}$ is sampled by first sampling $\mathbf{x} \in \mathbb{Z}^m \cap B^m(\lambda_1(\mathcal{L}_q(\mathbf{A}))/2)$ with probability $\Pr(\mathbf{x}) \propto \rho_{\beta q \sqrt{\Sigma/2}}(\mathbf{x})$ where $\Sigma = \mathbf{I}_m + \frac{\alpha^2}{\beta^2 q^2} \mathbf{e} \mathbf{e}^T$, then outputting $\mathbf{y} = (\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q$,
4. the center $c = -\frac{\alpha^2 \langle \mathbf{x}, \mathbf{e} \rangle}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2}$ (we don't know how to efficiently compute c with success probability $1 - 2^{-\Omega(n)}$ since we don't know $\mathbf{x}$ and $\mathbf{e}$; we can guess c correctly with non-negligible probability, but the event of guessing correctly is not efficiently checkable).

Step 2 Given an Extrapolated DCP state with amplitudes following a Gaussian distribution centered at an unknown value c , we adapt the quantum reduction proposed by Brakerski et al. [BKS18] to transform it to an $\text{S}|\text{LWE}\rangle^{\text{phase}}$ instance. The resulting $\text{S}|\text{LWE}\rangle^{\text{phase}}$ instance will have amplitudes represented as a Gaussian distribution multiplied by an unknown phase term, due to the unknown center c . More precisely, we establish the following theorem:

Theorem 48. *There exists an efficient quantum algorithm that, given an Extrapolated DCP state of form as Equation (9), generates an $\text{S}|\text{LWE}\rangle^{\text{phase}}$ state of form*

$$\frac{1}{\sqrt{N_{q/\sigma}}} \sum_{e \in \mathbb{Z}_q} \rho_{q/\sigma}(e) \exp(2\pi i \cdot ce/q) |(\langle \mathbf{a}, \mathbf{s} \rangle + e) \bmod q\rangle \quad (10)$$

along with a known vector $\mathbf{a} \leftarrow U(\mathbb{Z}_q^n)$. Here the parameters σ, c correspond to those mentioned in Theorem 47, and $N_{q/\sigma}$ normalizes the state.

In the remaining part of this section, we will provide the detailed proofs for Theorem 47 and Theorem 48. By combining these two theorems, we achieve a quantum reduction from classical LWE to $\text{S}|\text{LWE}\rangle^{\text{phase}}$ with Gaussian amplitude. This establishes that solving the classical LWE problem is as hard as solving the $\text{S}|\text{LWE}\rangle^{\text{phase}}$ with Gaussian amplitude for the phase function defined below. Formally, we define our special parameters and functions, and propose the main theorem for this section here:

Definition 49. *Let $n, m, q \in \mathbb{N}^+$, $\alpha = \Omega(\sqrt{n})$, $\beta, \gamma \in (0, 1)$ satisfy $m \geq n \log q$, $\alpha\gamma\sqrt{m} < \beta < \frac{1}{16\sqrt{m \log(\beta q)}}$. Given a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ and a vector $\mathbf{e}$ such that $\|\mathbf{e}\| \leq \sqrt{m}\gamma q$, we define:*

1. A family of functions $\{f_E : \mathbb{Z} \rightarrow \mathbb{R}\}_{E \in [m\gamma^2 q^2]}$ with $f_E(e) = \rho_{q/\sigma(E)}(e)$ where $\sigma(E) = \frac{\alpha\beta q}{\sqrt{\alpha^2 E + \beta^2 q^2}}$.
2. A distribution $D_{\theta, \mathbf{e}, \mathbf{A}}(\mathbf{y})$ over $\mathbb{Z}_q^m \cap B_{\mathcal{L}_q}(\mathbf{A})$ given by $\Pr(\mathbf{y}) \propto \rho_{\beta q \sqrt{\Sigma/2}}(\mathbf{y}')$ where $\Sigma = \mathbf{I}_m + \frac{\alpha^2}{\beta^2 q^2} \mathbf{e} \mathbf{e}^T$ and $\mathbf{y}' = \mathbf{y} - \kappa_{\mathcal{L}_q}(\mathbf{A})(\mathbf{y})$.

3. A phase function $\theta_{\mathbf{e}, \mathbf{A}} : \mathbb{Z}_q^m \cap B_{\mathcal{L}_q(\mathbf{A})} \rightarrow \mathbb{R}$ with $\theta_{\mathbf{e}}(\mathbf{y}) = -\frac{\alpha^2 \langle \mathbf{y}', \mathbf{e} \rangle}{q(\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2)}$ (It is not efficiently computable when assuming classical LWE is hard).

Theorem 50 (Main theorem, from LWE to $\text{S|LWE}^{\text{phase}}$). *Following the parameters defined in Definition 49. Assume for any $E \in [m\gamma^2 q^2]$, there exists a quantum algorithm that takes $\mathbf{A} \leftarrow U(\mathbb{Z}_q^{n \times m})$ as input, with $1 - 2^{-\Omega(n)}$ probability over $\mathbf{A}$, solves (see Definition 7) $\text{S|LWE}_{n, \ell, q, f_E, \theta_{\mathbf{e}, \mathbf{A}}, D_{\theta, \mathbf{e}, \mathbf{A}}}^{\text{phase}}$ for every $\mathbf{e} \in \mathbb{Z}^m$ such that $\|\mathbf{e}\|_2^2 = E$, with $\ell = 2^{o(n)}$ samples and time complexity $T = 2^{o(n)}$, then there exists a quantum algorithm that takes a classical LWE sample $(\mathbf{A}, \mathbf{A}^T \mathbf{s} + \mathbf{e})$ as input, and outputs the secret vector $\mathbf{s} \in \mathbb{Z}_q^n$ with success probability $1 - 2^{-\Omega(n)}$ and time complexity $O((T + \ell \cdot \text{poly}(n, q)) \cdot m\gamma^2 q^2)$.*

Remark 51. For an example of parameters, let $q \in \tilde{O}(n^2)$, $m \in \Omega(n \log q)$, $\|\mathbf{e}\|_2^2 \in \tilde{O}(n)$, $\alpha \in \tilde{O}(n^{0.5})$, $\beta \in \tilde{O}(n^{-0.5})$, $\gamma \in \tilde{O}(n^{-1.5})$.

Proof. We proceed to address the classical LWE instance $(\mathbf{A}, \mathbf{A}^T \mathbf{s} + \mathbf{e})$ as follows:

1. Enumerate $E \in \{1, 2, \dots, m\gamma^2 q^2\}$ to make a guess for $\|\mathbf{e}\|^2$.
2. Apply Theorem 47 and Theorem 48 ℓ times to generate ℓ instances of $\text{S|LWE}^{\text{phase}}$ in the form of Equation (10).
3. Utilize the quantum algorithm in the assumption for $\text{S|LWE}_{n, \ell, q, f_E, \theta_{\mathbf{e}, \mathbf{A}}, D_{\theta, \mathbf{e}, \mathbf{A}}}^{\text{phase}}$, with those ℓ $\text{S|LWE}^{\text{phase}}$ instances as input, to derive a solution $\mathbf{s}'$.
4. Employ any verification algorithm (e.g., as proposed by Regev [Reg09, Lemma 3.6]) to ascertain whether $\mathbf{s}' = \mathbf{s}$. If this condition holds, output $\mathbf{s}'$ and conclude the process.

It can be easily verified that this algorithm operates with a runtime of $O((T + \ell \cdot \text{poly}(q, n)) \cdot m\gamma^2 q^2)$. Furthermore, as indicated in Theorem 47, the probability of successfully generating ℓ $\text{S|LWE}^{\text{phase}}$ states in step 2 is exponentially close to 1. Thus, when $E = \|\mathbf{e}\|^2$ (i.e., when we guess E correctly), the probability that the solution $\mathbf{s}' = \mathbf{s}$ is exponentially close to 1. Consequently, the aforementioned algorithm achieves success probability exponentially close to 1. $\square$

5.1 Reduce classical LWE to Extrapolated DCP

Our quantum reduction from classical LWE to Extrapolated DCP follows the general design of Regev's reduction [Reg04] and the reduction proposed by Brakerski et al. [BKS18]. In these reductions, the Euclidean space $\mathbb{R}^n$ is divided into grids, with each grid cell having a width that lies between the length of the error vector $\|\mathbf{e}\|$ and the length of the shortest vector in the lattice $\lambda_1(\mathcal{L}_q(\mathbf{A}))$. The key observation is that when randomly selecting a vector $\mathbf{x} \in \mathbb{R}^n$, the vectors $\mathbf{x}, \mathbf{x} + \mathbf{e}, \dots, \mathbf{x} + k \cdot \mathbf{e}$ will be in the same grid cell with high probability, creating a superposition in the quantum world.

We modify the reductions in [Reg04, BKS18] by introducing Gaussian balls around all lattice points in $\mathcal{L}_q(\mathbf{A})$, where the radius of each ball is a quantity smaller than the length of shortest vector in the lattice $\lambda_1(\mathcal{L}_q(\mathbf{A}))$. Note that the reductions in [Reg04, BKS18] use Euclidean balls or cubes.

Here we give the detailed proof of Theorem 47.

Proof of Theorem 47. Following the parameters defined in Theorem 47. Relying on both Lemma 16 and Banaszczyk's tail bound from Lemma 10, we make the assumptions that $\lambda_1(\mathcal{L}_q(\mathbf{A})) \geq \frac{q}{4}$ and $\|\mathbf{e}\| < \sqrt{m}\gamma q$ for the remainder of this proof; these assumptions hold with probability $1 - 2^{-\Omega(n)}$. Our quantum reduction from classical LWE to Extrapolated DCP works as follows (for simplicity, we omit the normalization factors):

1. We start by preparing the superposition using the Gaussian state sampler (see Lemma 28) of form

$$\frac{1}{\sqrt{N_1}} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j) |j\rangle \sum_{\mathbf{v} \in \mathbb{Z}_q^n} |\mathbf{v}\rangle \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_{\beta q}(\mathbf{x}) |\mathbf{x}\rangle. \quad (11)$$

2. We apply a unitary to compute $(j, \mathbf{v}, \mathbf{x}) \rightarrow (\mathbf{A}^T \mathbf{v} - j \cdot (\mathbf{A}^T \mathbf{s} + \mathbf{e}) + \mathbf{x}) \bmod q$ on the third register, and perform a change of variables $\mathbf{v} \leftarrow (\mathbf{v} + j \cdot \mathbf{s}) \bmod q$ and $\mathbf{x} \leftarrow \mathbf{x} + j \cdot \mathbf{e} \bmod q$, to obtain the state

$$\frac{1}{\sqrt{N_1}} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j) |j\rangle \sum_{\mathbf{v} \in \mathbb{Z}_q^n} |(\mathbf{v} + j \cdot \mathbf{s}) \bmod q\rangle \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e} \bmod q) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q\rangle. \quad (12)$$

We next transform the above state to a state where the value of the third register $\mathbf{A}^T \mathbf{v} + \mathbf{x} \bmod q$ determines a unique pair $(\mathbf{v}, \mathbf{x})$, through the following lemma. The proof of this lemma is deferred to Appendix A.4.

Lemma 52. *Under the condition of Theorem 47, the state in Equation (12) is $2^{-\Omega(n)}$ -close to the state*

$$\frac{1}{\sqrt{N_2}} \sum_{\substack{\mathbf{v} \in \mathbb{Z}_q^n, \mathbf{x} \in \mathbb{Z}_q^m, \\ \|\mathbf{x}\| < \lambda_1(\mathcal{L}_q(\mathbf{A}))/2}} \left(\sum_{j \in \mathbb{Z}_q} \rho_\alpha(j) \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) |j\rangle |(\mathbf{v} + j \cdot \mathbf{s}) \bmod q\rangle \right) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q\rangle. \quad (13)$$

3. We measure the state in Equation (13) on the third register and denote the result as $\mathbf{y} = (\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q$ (note that we have $\|\mathbf{x}\| < \lambda_1(\mathcal{L}_q(\mathbf{A}))/2$, so the pair of vectors $(\mathbf{v}, \mathbf{x})$ is unique), then the remaining state on the first two registers is

$$\frac{1}{\sqrt{N_{\mathbf{v}, \mathbf{x}}}} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j) \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) |j\rangle |(\mathbf{v} + j \cdot \mathbf{s}) \bmod q\rangle. \quad (14)$$

The amplitude of this state is computed as follows

$$\begin{aligned} & \rho_\alpha(j) \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) \\ &= \exp \left[-\pi \left(\frac{j^2}{\alpha^2} + \frac{j^2 \|\mathbf{e}\|^2 + 2j \langle \mathbf{x}, \mathbf{e} \rangle + \|\mathbf{x}\|^2}{\beta^2 q^2} \right) \right] \\ &= \exp \left[-\pi \left(\frac{(\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2) j^2 + 2j \alpha^2 \langle \mathbf{x}, \mathbf{e} \rangle}{\alpha^2 \beta^2 q^2} \right) \right] \cdot \exp(-\pi \frac{\|\mathbf{x}\|^2}{\beta^2 q^2}) \\ &= \rho_\sigma(j - c) \cdot \exp(-\pi \frac{\|\mathbf{x}\|^2}{\beta^2 q^2}), \end{aligned} \quad (15)$$

where the Gaussian width $\sigma = \frac{\alpha\beta q}{\sqrt{\alpha^2\|\mathbf{e}\|^2 + \beta^2 q^2}} \in (\alpha/\sqrt{2}, \alpha)$ and the center $c = -\frac{\alpha^2\langle \mathbf{x}, \mathbf{e} \rangle}{\alpha^2\|\mathbf{e}\|^2 + \beta^2 q^2}$. So the amplitude of the remaining state is proportional to Gaussian distribution $\rho_{\sigma, c}$ with a shifted center c , as desired.

It's evident that in the state of Equation (13), the amplitude for every $\mathbf{v} \in \mathbb{Z}_q^n$ is the same, which implies that the distribution of the vector $\mathbf{v}$ is uniformly random. This completes the proof. $\square$

Unfortunately, the center c remains unknown because we have no knowledge of $\mathbf{x}$ other than that it is the error term in the LWE sample $\mathbf{y} = (\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q$. In the following subsection we analyze the distribution of $\mathbf{y}$ and c .

Remark 53. *It seems that our reduction bears similarities to the reduction from classical LWE to G-EDCP (Extrapolated DCP with amplitudes following a Gaussian distribution) proposed by Brakerski et al. [BKS18]. However, our reduction, compared to both Regev's reduction and Brakerski et al.'s reduction, exhibits superior success probability. In the previous reductions, the failure probability is inverse-polynomial, leading to the reduction of classical LWE to only polynomially many (Extrapolated) DCP states. In contrast, our reduction achieves $1 - 2^{-\Omega(n)}$ success probability, allowing for the construction of sub-exponentially many Extrapolated DCP states without failure, the cost is introducing an unknown center in the Gaussian distribution of the amplitude. This novel approach offers the potential for creating sub-exponential time quantum algorithms for the standard LWE problem.*

5.2 Reduce Extrapolated DCP to $S|\text{LWE}\rangle^{\text{phase}}$

The second step of our quantum reduction from classical LWE to $S|\text{LWE}\rangle^{\text{phase}}$ involves reducing the obtained Extrapolated DCP states to $S|\text{LWE}\rangle^{\text{phase}}$. This step is an adaptation of the reduction from G-EDCP to LWE proposed by Brakerski et al. [BKS18]. We give the detailed proof of Theorem 48 here.

Proof of Theorem 48. Suppose we are given an Extrapolated DCP state with the form as Equation (9). Our quantum reduction works as follows (for simplicity, we omit the normalization factors):

1. Apply $\text{QFT}_q^\dagger$ for the second register, obtaining the state

$$\sum_{\mathbf{a} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \omega_q^{\langle \mathbf{a}, \mathbf{v} + j \cdot \mathbf{s} \rangle} \rho_\sigma(j - c) |j\rangle |\mathbf{a}\rangle.$$

2. Measure the second register to get a particular measurement result $\hat{\mathbf{a}}$, which is randomly chosen from $\mathbb{Z}_q^n$ with a uniform distribution. By omitting the global phase term $\omega_q^{\langle \hat{\mathbf{a}}, \mathbf{v} \rangle}$, the remaining state is

$$\sum_{j \in \mathbb{Z}_q} \omega_q^{\langle \hat{\mathbf{a}}, j \cdot \mathbf{s} \rangle} \rho_\sigma(j - c) |j\rangle.$$

3. Apply another Quantum Fourier Transformation on $\mathbb{Z}_q$ and incorporate Gaussian tails of j again, obtaining a state $2^{-\Omega(n)}$ -close to the state

$$\sum_{b \in \mathbb{Z}_q} \sum_{j \in \mathbb{Z}} \omega_q^{j(\langle \hat{\mathbf{a}}, \mathbf{s} \rangle + b)} \rho_\sigma(j - c) |b\rangle.$$

4. Use the Poisson summation formula on the amplitude and change the summation variable to $e \leftarrow \langle \hat{\mathbf{a}}, \mathbf{s} \rangle + b - q \cdot j$, this state can be rewritten as

$$\begin{aligned}
& \sum_{b \in \mathbb{Z}_q} \sum_{j \in \mathbb{Z}} \omega_q^{j(\langle \hat{\mathbf{a}}, \mathbf{s} \rangle + b)} \rho_\sigma(j - c) |b\rangle \\
&= \sum_{b \in \mathbb{Z}_q} \sum_{j \in \mathbb{Z}} \sigma \rho_{1/\sigma} \left(j - \frac{\langle \hat{\mathbf{a}}, \mathbf{s} \rangle + b}{q} \right) \cdot \exp \left(-2\pi i \cdot c \left(j - \frac{\langle \hat{\mathbf{a}}, \mathbf{s} \rangle + b}{q} \right) \right) |b\rangle \\
&\propto \sum_{e \in \mathbb{Z}} \rho_{q/\sigma}(e) \cdot \exp(2\pi i \cdot ce/q) |(\langle -\hat{\mathbf{a}}, \mathbf{s} \rangle + e) \bmod q\rangle.
\end{aligned}$$

Incorporating Gaussian tails of e , this state is $2^{-\Omega(n)}$ -close to the state

$$\sum_{e \in \mathbb{Z}_q} \rho_{q/\sigma}(e) \cdot \exp(2\pi i \cdot ce/q) |(\langle -\hat{\mathbf{a}}, \mathbf{s} \rangle + e) \bmod q\rangle.$$

Finally, this state along with the classical vector $-\hat{\mathbf{a}}$ will be the output $\mathbf{S}|\text{LWE}\rangle^{\text{phase}}$ instance of our quantum reduction. $\square$

5.3 The distribution of unknown center

For additional technical insights, we present a more detailed analysis of the distribution of center c in the Extrapolated DCP states (see Equation (9)) we get. To achieve this, we begin by examining the distribution of $\mathbf{x}$ after measurement on the third register of the state given in Equation (13). It is evident that the probability of obtaining a specific vector $\mathbf{x}$ is proportional to

$$\begin{aligned}
\Pr(\mathbf{x}) &\propto \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e})^2 \\
&= \left(1 \pm 2^{-\Omega(n)}\right) \sum_{j \in \mathbb{Z}} \rho_\alpha(j)^2 \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e})^2 \\
&= \left(1 \pm 2^{-\Omega(n)}\right) \sum_{j \in \mathbb{Z}} \exp \left[-2\pi \left(\frac{j^2}{\alpha^2} + \frac{j^2 \|\mathbf{e}\|^2 + 2j \langle \mathbf{x}, \mathbf{e} \rangle + \|\mathbf{x}\|^2}{\beta^2 q^2} \right) \right] \\
&= \left(1 \pm 2^{-\Omega(n)}\right) \sum_{j \in \mathbb{Z}} \rho_{\sigma/\sqrt{2}}(j - c) \cdot \exp \left[-2\pi \left(\frac{\|\mathbf{x}\|^2}{\beta^2 q^2} - \frac{\alpha^2 \langle \mathbf{x}, \mathbf{e} \rangle^2}{\beta^2 q^2 (\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2)} \right) \right].
\end{aligned}$$

where the second approximation is due to Banaszczyk's tail bound. We observe that $\sigma = \frac{\alpha \beta q}{\sqrt{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2}} > \frac{\alpha}{\sqrt{2}} = \Omega(\sqrt{n})$, which implies that almost all of the weight of $\rho_{\sigma/\sqrt{2}}$ is concentrated at $\rho_{\sigma/\sqrt{2}}(0)$ with exponentially small weight elsewhere. Using the Poisson summation formula, we get that

$$\sum_{j \in \mathbb{Z}} \rho_{\sigma/\sqrt{2}}(j - c) = \sum_{j \in \mathbb{Z}} \frac{\sigma}{\sqrt{2}} \rho_{\sqrt{2}/\sigma}(j) \cdot \exp(-2\pi i \cdot cj) \in \frac{\sigma}{\sqrt{2}} (1 \pm 2^{-\Omega(n)}).$$

Let us denote $\Sigma = \left(\mathbf{I}_m - \frac{\alpha^2}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2} \mathbf{e} \mathbf{e}^T \right)^{-1} = \mathbf{I}_m + \frac{\alpha^2}{\beta^2 q^2} \mathbf{e} \mathbf{e}^T$, the remaining term can be written as

$$\begin{aligned}
\exp \left[-2\pi \left(\frac{\|\mathbf{x}\|^2}{\beta^2 q^2} - \frac{\alpha^2 \langle \mathbf{x}, \mathbf{e} \rangle^2}{\beta^2 q^2 (\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2)} \right) \right] &= \exp \left[-2\pi \cdot \frac{1}{\beta^2 q^2} (\mathbf{x})^T \left(\mathbf{I}_m - \frac{\alpha^2}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2} \mathbf{e} \mathbf{e}^T \right) \mathbf{x} \right] \\
&= \exp \left[-2\pi \cdot \frac{1}{\beta^2 q^2} (\mathbf{x})^T \boldsymbol{\Sigma}^{-1} \mathbf{x} \right] \\
&= \rho_{\beta q \sqrt{\boldsymbol{\Sigma}/2}}(\mathbf{x}).
\end{aligned}$$

This means that the distribution of $\mathbf{x}$ follows the discrete Gaussian distribution with center $\mathbf{0}$ and covariance matrix $\beta^2 q^2 \boldsymbol{\Sigma}/2$. Correspondingly, the distribution of $\mathbf{y} \in \mathbb{Z}_q^m \cap B_{\mathcal{L}_q(\mathbf{A})}$ is given by $\Pr(\mathbf{y}) = \rho_{\beta q \sqrt{\boldsymbol{\Sigma}/2}}(\mathbf{x})$.

To derive the distribution of the unknown center $c = \frac{\alpha^2 \langle \mathbf{x}, \mathbf{e} \rangle}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2}$, we observe that the distribution of $\mathbf{x}$ is smooth enough to be treated as a continuous Gaussian distribution since the eigenvalues of $\beta^2 q^2 \boldsymbol{\Sigma}/2$ are $\beta^2 q^2/2$ and $(\beta^2 q^2 + \alpha^2 \|\mathbf{e}\|^2)/2$. So the distribution of the unknown center c can be approximated by the discrete Gaussian distribution with minimum step $\frac{\alpha^2}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2}$, center 0 and variance

$$\sigma_c^2 = \left(\frac{\alpha^2}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2} \right)^2 \cdot \mathbf{e}^T (\beta^2 q^2 \boldsymbol{\Sigma}/2) \mathbf{e} = \frac{\alpha^4 \|\mathbf{e}\|^2}{2(\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2)}.$$

In conclusion, we propose the following statement for the distribution of the unknown center in the Extrapolated DCP states:

Theorem 54. *The distribution of c in Equation (9) of Theorem 47 approximately follows the discrete Gaussian distribution $D_{\frac{\alpha^2}{\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2} \mathbb{Z}, \sigma_c}$ where $\sigma_c = \frac{\alpha^2 \|\mathbf{e}\|}{\sqrt{2(\alpha^2 \|\mathbf{e}\|^2 + \beta^2 q^2)}}$.*

Remark 55. *As readers may notice, the Gaussian width σ of j and the Gaussian width σ_c of c (the center of the distribution of j) satisfy $\sigma_c = \frac{\alpha \|\mathbf{e}\|}{\sqrt{2}\beta q} \sigma$. In our settings, if we assume $\beta q \gg \alpha \cdot \|\mathbf{e}\|$, then the distribution of j is a discrete Gaussian distribution with a small shift. However, this shift is non-negligible, preventing our $S|\text{LWE}\rangle^{\text{phase}}$ state from being exponentially close to a $S|\text{LWE}\rangle$ state without unknown phase.*

6 Hardness of $S|\text{LWE}\rangle$ with Unknown Phase via Quantizing Regev's Iterative Reduction

In this section, we show how to reduce from the problem of generating discrete Gaussian states ($|\text{DGS}\rangle$, Definition 21) to a variant of $S|\text{LWE}\rangle$ with an unknown phase term on the amplitude ($S|\text{LWE}\rangle^{\text{phase}}$), by modifying Regev's iterative reduction [Reg09] from the problem of generating discrete Gaussian samples (DGS, Definition 20) to LWE. Combined with the known reductions from GapSVP and SIVP to DGS in Lemma 22 and Lemma 23, it gives a quantum reduction from GapSVP $_{\tilde{O}(n^{1.5})}$ and SIVP $_{\tilde{O}(n^{1.5})}$ to $S|\text{LWE}\rangle^{\text{phase}}$.

6.1 Overview of our reduction

As in [Reg09], our proof is iterative. We start from generating discrete Gaussian states with exponentially large widths (in Regev’s reduction, it is classical discrete Gaussian sample with exponentially large widths; both can be done efficiently). Then, each iteration produces discrete Gaussian states (samples) with smaller widths. Repeating the iteration for polynomial number of times gives the discrete Gaussian states (samples) for the $|\text{DGS}\rangle$ (DGS) problem. We illustrate Regev’s reduction in Figure 2a, aligned with our reduction in Figure 2b, and then explain with more details.

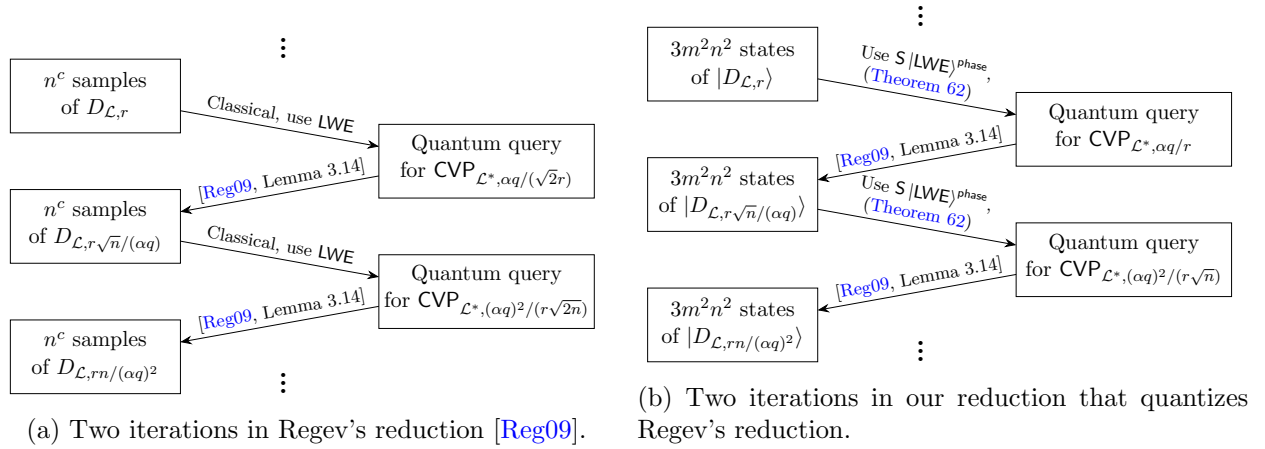


Figure 2: The correspondence between Regev’s reduction (from DGS to LWE) and our reduction (from $|\text{DGS}\rangle$ to $S|\text{LWE}\rangle^{\text{phase}}$).

In order to quantize Regev’s iterative reduction, we focus on quantizing the only classical step in the reduction – solving CVP. Roughly speaking, given a CVP instance $\mathbf{x}$, Regev [Reg09] utilizes LWE oracle to solve CVP by feeding it with samples $\mathbf{a} := \mathcal{L}^{-1}\mathbf{v} \bmod q$ and $\langle \mathbf{x}, \mathbf{v} \rangle \bmod q$ where $\mathbf{v} \leftarrow D_{\mathcal{L}, r}$, which are close to the LWE sample $\mathbf{a}$ and $\langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q$ where $\mathbf{s} = (\mathcal{L}^*)^{-1}\kappa_{\mathcal{L}^*}(\mathbf{x}) \bmod q$ and e is sampled from the Gaussian distribution. To quantize this step, a natural idea is to replace the classical $\mathbf{v}$ with a superposition state of Gaussian samples $\sum_{\mathbf{v} \in \mathcal{L}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle$, measure $\mathbf{a} = \mathcal{L}^{-1}\mathbf{v} \bmod q$, and compute $\langle \mathbf{x}, \mathbf{v} \rangle \bmod q$ in another register, hoping that the register contains an $S|\text{LWE}\rangle$ state. However, we should be careful to make sure that the $\mathbf{v}$ register does not collapse to a classical $\mathbf{v}$. Our solution is to measure the $\mathbf{v}$ register in Fourier basis, which can ensure that each $\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}$ appears in the amplitude of the $\langle \mathbf{x}, \mathbf{v} \rangle$ register. But it also inevitably introduces a phase term that we are unable to compute efficiently from the measurement results. The above discussion ignores the Gaussian distribution to smooth the error distribution. More details can be found in Section 6.2.

The above described reduction leads us to requiring an $S|\text{LWE}\rangle^{\text{phase}}$ oracle with specific parameter. We first formally define our special parameters and functions, and propose the main theorem for this section here:

Definition 56. Let $\mathcal{L}$ be an n -dimensional integer lattice. Given parameters $q, R \in \mathbb{N}^+$ such that $R \in 2^{\text{poly}(n)}$, $\alpha, r \in \mathbb{R}^+$ and a vector $\mathbf{x} \in \mathbb{R}^n$ such that $\text{dist}(\mathbf{x}, \mathcal{L}^*) \leq \lambda_1(\mathcal{L}^*)/2$, we define

1. An amplitude function $f : \mathbb{Z}_{qR}/R \rightarrow \mathbb{R}$ with $f(e) = \rho_{\sqrt{2}\alpha q}(e)$ which is completely known.

2. A family of distribution $D_\theta^{(r,\mathbf{x})}$ over $\mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}$ parameterized by $(r, \mathbf{x})$ and given by $\Pr(\mathbf{y}) \propto \rho_{\sqrt{\Sigma/2}}(\mathbf{z}(\mathbf{y}))$ where $\Sigma := \frac{\mathbf{I}}{r^2} + \frac{\mathbf{x}'\mathbf{x}'^T}{2\alpha^2q^2 - r^2\|\mathbf{x}'\|^2}$, $\mathbf{x}' := \mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{x})$ and $\mathbf{z}(\mathbf{y}) := \mathbf{y}/R - \kappa_{(q\mathcal{L})^*}(\mathbf{y}/R)$.
3. A family of phase function $\theta^{(r,\mathbf{x})} : \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*} \rightarrow \mathbb{R}$ parameterized by $(r, \mathbf{x})$ with $\theta^{(r,\mathbf{x})}(\mathbf{y}) = \frac{r^2\langle \mathbf{x}', \mathbf{z}(\mathbf{y}) \rangle}{2\alpha^2q^2}$, where $\mathbf{x}' = \mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{x})$ and $\mathbf{z}(\mathbf{y}) = \mathbf{y}/R - \kappa_{(q\mathcal{L})^*}(\mathbf{y}/R)$. This function is not known to be efficiently computable since it requires to solve approximate CVP.

Theorem 57 (Main theorem, from $|\text{DGS}\rangle$ to $\text{S}|\text{LWE}\rangle^{\text{phase}}$). Let $\mathcal{L}$ be an n -dimensional integer lattice. Let $\epsilon = \epsilon(n)$ be a negligible function such that $\epsilon(n) < 2^{-n}$, $q = q(n) > 10n$ be an integer of at most $\text{poly}(n)$ bits, $\alpha \in (0, \frac{1}{5\sqrt{n}})$ such that $\alpha q > 2\sqrt{n}$, $R = R(n)$ be an exponentially large integer such that $R > \max\{2^{2n+2}n\lambda_n(\mathcal{L})^2, \frac{2^{4n+1}\sqrt{2}n\lambda_n(\mathcal{L}^*)\lambda_n(\mathcal{L})}{\alpha q}, 2^{3n}\lambda_n(\mathcal{L}^*)\}$. Let $r_0 > 4\sqrt{n}\eta_\epsilon(\mathcal{L})/\alpha$ be the width parameter of the $|\text{DGS}\rangle$ problem.

Assume there exists quantum algorithms that can solve $\text{S}|\text{LWE}\rangle_{n,m,q,f,\theta^{(r,\mathbf{x})},D_\theta^{(r,\mathbf{x})}}^{\text{phase}}$ for any choice of pair $(r, \mathbf{x})$ such that $\alpha q r_0 / \sqrt{n} < r < 2^{2n}\sqrt{2}\lambda_n(\mathcal{L})$, $\mathbf{x} \in \mathcal{L}^*/R$ and $\text{dist}(\mathbf{x}, \mathcal{L}^*) \leq \alpha q / r$, with $m = 2^{o(n)}$ samples and in time complexity T . Then there exists a quantum algorithm that can generate a state that is $2^{-\Omega(n)}$ -close to the discrete Gaussian state $|D_{\mathcal{L},r_0}\rangle = \sum_{\mathbf{v} \in \mathcal{L}} \rho_{r_0}(\mathbf{v}) |\mathbf{v}\rangle$ in time complexity $O((m^4 + m^3T)\text{poly}(n))$.

Then the $|\text{DGS}\rangle$ problem is easily reduced to either GapSVP or SIVP . The connection to GapSVP and SIVP is a Corollary of Theorem 57 and Lemmas 13, 14, 22, and 23.

Corollary 58. Under the same assumption used in Theorem 57, there exists quantum algorithms for solving GapSVP_γ and SIVP_γ for $\gamma \in \tilde{O}(n/\alpha)$ in time complexity $\text{poly}(n, m, T)$.

Remark 59. Readers may think the assumption of Theorem 57 looks too strong because the family of phase functions $\{\theta^{(r,\mathbf{x})}\}$ is a very large family. However, we know the absolute value of $\theta^{(r,\mathbf{x})}(\mathbf{y})$ is small with high probability, when $(r, \mathbf{x})$ follows the setting in Theorem 57 and $\mathbf{y}$ is sampled from the corresponding distribution $D_\theta^{(r,\mathbf{x})}$. This is because when $\mathbf{y} \leftarrow D_\theta^{(r,\mathbf{x})}$, $\mathbf{z}(\mathbf{y}) = \mathbf{y}/R - \kappa_{(q\mathcal{L})^*}(\mathbf{y}/R)$ follows the distribution $\rho_{\sqrt{\Sigma/2}}(\mathbf{z}(\mathbf{y}))$ over support $\mathbb{Z}_R^n/R \cap B_{(q\mathcal{L})^*}$ and thus has ℓ_2 norm at most $\frac{\sqrt{n}\alpha q}{r\sqrt{2\alpha^2q^2 - r^2\|\mathbf{x}'\|^2}}$ with $1 - 2^{-\Omega(n)}$ probability. Then $|\theta^{(r,\mathbf{x})}(\mathbf{y})| \leq \frac{\sqrt{n}}{2\alpha q}$ with $1 - 2^{-\Omega(n)}$ probability. As our algorithm in Section 3 can solve the problem in sub-exponential time if $\theta^{(r,\mathbf{x})}(\mathbf{y})$ is always zero, there might be a way to solve the problem when $\theta^{(r,\mathbf{x})}(\mathbf{y})$ is close to zero.

In what follows, we will display the idea of our proof for Theorem 57, which will be an expansion of the proof idea described earlier with Figure 2. As is described before, our proof is iterative. We start from generating discrete Gaussian states with exponentially large widths. Then, equipped with an $\text{S}|\text{LWE}\rangle^{\text{phase}}$ solver, we iteratively generate discrete Gaussian states with smaller widths in each step. Repeating the iterative step for polynomial number of times gives the discrete Gaussian states we desired. Formally, our initialization and iterative steps are:

Theorem 60 (The initialization step, [Reg09, Lemma 3.12]). There exists an efficient quantum algorithm that given any n -dimensional integer lattice $\mathcal{L}$ and width $r > 2^{2n}\sqrt{2}\lambda_n(\mathcal{L})$, output a state that is $2^{-\Omega(n)}$ -close to the state $|D_{\mathcal{L},r}\rangle = \sum_{\mathbf{v} \in \mathcal{L}} \rho_r(\mathbf{v})$.

Theorem 61 (The iterative step). *Let $\mathcal{L}$ be an n -dimensional integer lattice, $q > 2$ be an integer. Define the parameters $\epsilon \in (0, 2^{-n})$, $\alpha \in (0, \frac{1}{5\sqrt{n}})$, $r > 4q\eta_\epsilon(\mathcal{L})$, and a precision parameter $R > \max\{2\sqrt{nr}\sqrt{\log r}, \frac{2\sqrt{n}}{\alpha q}, \frac{2^{2n+1}nr\lambda_n(\mathcal{L}^*)}{\alpha q}, 2^{3n}\lambda_n(\mathcal{L}^*)\}$ as an integer.*

Assume that there exists a quantum algorithm that solves $S|\text{LWE}\rangle_{n,m,q,f,\theta(r,\mathbf{x}),D_\theta^{(r,\mathbf{x})}}^{\text{phase}}$ for any $\mathbf{x} \in \mathcal{L}^/R$ with $\text{dist}(\mathbf{x}, \mathcal{L}^*) < \alpha q/r$ in time complexity T . Then there exists a quantum algorithm that, given $3m^2n^2$ discrete Gaussian states $|D_{\mathcal{L},r}\rangle = \sum_{\mathbf{v} \in \mathcal{L}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle$, produces $3m^2n^2$ discrete Gaussian states that are $2^{-\Omega(n)}$ -close to $|D_{\mathcal{L},r\sqrt{n}/\alpha q}\rangle$, in time complexity $O((m^4 + m^3T)\text{poly}(n))$.*

The iterative step consists of two steps:

Step 1 Given a CVP instance, we can use a collection of discrete Gaussian states $|D_{\mathcal{L},r}\rangle$ to construct an $S|\text{LWE}\rangle^{\text{phase}}$ instance. Solving the $S|\text{LWE}\rangle^{\text{phase}}$ instance will in return solve the CVP problem. More precisely, we show the following theorem:

Theorem 62. *Let $\mathcal{L}$ be an n -dimensional integer lattice, define the parameters $\epsilon \in (0, 2^{-n})$, $\alpha \in (0, \frac{1}{5\sqrt{n}})$, $r > 4q\eta_\epsilon(\mathcal{L})$, and a precision parameter $R > \max\{2\sqrt{nr}\sqrt{\log r}, \frac{2\sqrt{n}}{\alpha q}, \frac{2^{2n+1}nr\lambda_n(\mathcal{L}^*)}{\alpha q}\}$ as an integer.*

Assume that there exists an quantum algorithm that solves $S|\text{LWE}\rangle_{n,m,q,f,\theta(r,\mathbf{x}),D_\theta^{(r,\mathbf{x})}}^{\text{phase}}$ for any $\mathbf{x} \in \mathcal{L}^/R$ with $\text{dist}(\mathbf{x}, \mathcal{L}^*) < \alpha q/r$ in time complexity T . Then there exists a quantum algorithm that, given $3m^2n^2$ discrete Gaussian states $|D_{\mathcal{L},r}\rangle = \sum_{\mathbf{v} \in \mathcal{L}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle$, answers quantum query to $\text{CVP}_{\mathcal{L}^*, \alpha q/r}$ on the support $\mathcal{L}^*/R$ (denoted by $|\mathbf{x}, \mathbf{y}\rangle \rightarrow |\mathbf{x}, \mathbf{y} + \kappa_{\mathcal{L}^*}(\mathbf{x})\rangle$ with $\mathbf{x} \in \mathcal{L}^*/R$ such that $\text{dist}(\mathbf{x}, \mathcal{L}^*) \leq \alpha q/r$) up to exponentially small error, with exponentially small disturbance to the states $|D_{\mathcal{L},r}\rangle$, and in time $O((m^2 + mT)\text{poly}(n))$.*

Step 2 (Same as the quantum step in Regev's reduction) A query to the CVP oracle can help to generate a discrete Gaussian state with a smaller width. More precisely:

Theorem 63 ([Reg09, Lemma 3.14]). *There exists an efficient quantum algorithm that, given any n -dimensional lattice $\mathcal{L}$, a number $d < \lambda_1(\mathcal{L}^*)/2$ and an integer $R > 2^{3n}\lambda_n(\mathcal{L}^*)$, outputs $|D_{\mathcal{L}, \sqrt{n}/d}\rangle = \sum_{\mathbf{v} \in \mathcal{L}} \rho_{\sqrt{n}/d}(\mathbf{v}) |\mathbf{v}\rangle$, with only one quantum query on the second register of state*

$$\sum_{\mathbf{x} \in \mathcal{L}^*/R, \|\mathbf{x}\| \leq d} \rho_{d/\sqrt{n}}(\mathbf{x}) |\mathbf{x}, \mathbf{x} \bmod \mathcal{P}(\mathcal{L}^*)\rangle,$$

to the $\text{CVP}_{\mathcal{L}^, d}$ oracle, which is on the support $\mathcal{L}^*/R$.*

The full picture of the proof for the main reduction [Theorem 57](#) is illustrated in [Figure 3](#). The proof starts with the initial step ([Theorem 60](#)) then applies the iterative step ([Theorem 61](#)) for $\text{poly}(n)$ times. The iterative step consists of two parts, first the construction of CVP oracle ([Theorem 62](#)) with the help of discrete Gaussian states from the previous iteration and the help of an $S|\text{LWE}\rangle^{\text{phase}}$ solver, and then the generation of a narrower discrete Gaussian state through one query to the CVP oracle ([Theorem 63](#)). Since the discrete Gaussian states generated in the previous iteration are only disturbed by an exponentially small amount upon each query of the CVP oracle, we can reuse them for $3m^2n^2$ times to construct $3m^2n^2$ narrower discrete Gaussian states for use of the next iteration.

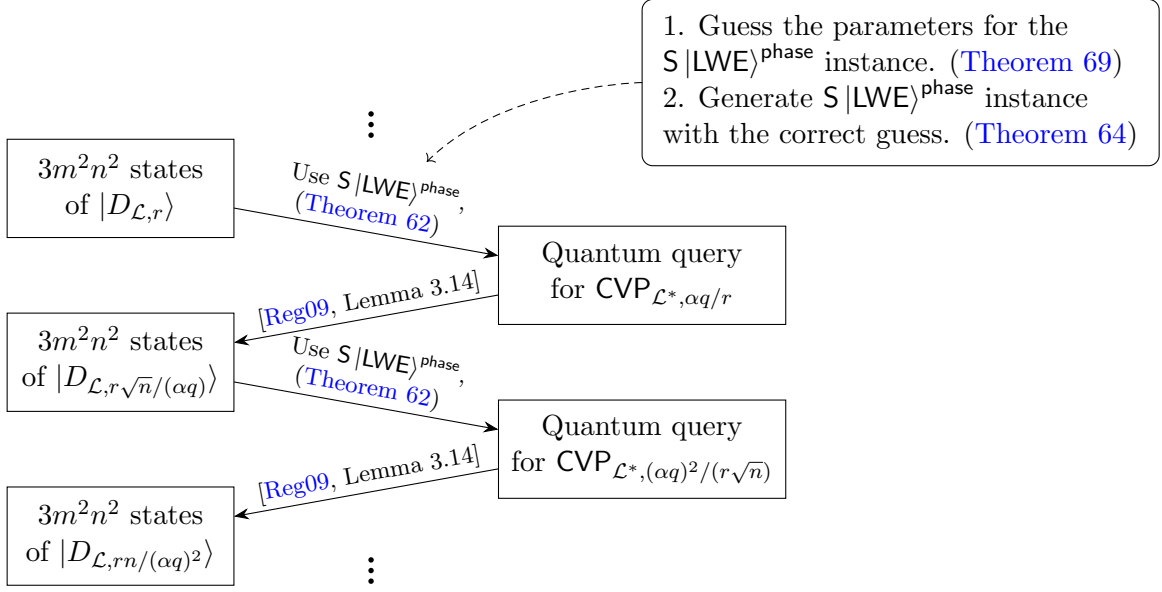


Figure 3: Illustration of two iterations of the reduction algorithm.

It is left to prove [Theorem 62](#), to which the remainder of this section will be devoted. To prove it, we start by generating an $S|LWE\rangle^{\text{phase}}$ instance but with an unknown Gaussian width, instead of the fixed and known width $\sqrt{2}\alpha q$ in the $S|LWE\rangle^{\text{phase}}$ solver, as displayed in [Theorem 64](#) in [Section 6.2](#). We then address and resolve the issue of the unknown width in order to solve the CVP instance using the $S|LWE\rangle^{\text{phase}}$ oracle, in the proof of [Theorem 69](#) in [Section 6.3](#). Finally, we note that the procedure in [Theorem 69](#) answers the CVP quantum query with $1 - 2^{-\Omega(n)}$ probability. Therefore using the idea of gentle measurement, we can answer each CVP quantum query with exponentially small disturbance to the states $|D_{L,r}\rangle$, as discussed in [Section 6.4](#).

6.2 Generating the $S|LWE\rangle^{\text{phase}}$ samples

In this subsection, we show how to create the $S|LWE\rangle^{\text{phase}}$ instance for [Theorem 62](#) but with an unknown Gaussian width. Given a $CVP_{L^*, \alpha q/r}$ instance, the idea is to replace the classical Gaussian samples from $D_{L,r}$ in [\[Reg09\]](#) (that helps to produce the LWE instance) with a superposition state of Gaussian samples $|D_{L,r}\rangle := \sum_{\mathbf{v} \in \mathcal{L}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle$ that helps to produce the $S|LWE\rangle^{\text{phase}}$ instance.

Theorem 64. *Let $\mathcal{L}$ be an n -dimensional integer lattice, define the parameters $\epsilon \in (0, 2^{-n})$, $\alpha \in (0, \frac{1}{5\sqrt{n}})$, $\sigma \in [\alpha q, \sqrt{2}\alpha q]$, $r > 4q\eta_\epsilon(\mathcal{L})$, and a precision parameter $R > 2\sqrt{nr}\sqrt{\log r}$ as an integer. Given a $CVP_{L^*, \alpha q/r}$ instance $\mathbf{x} \in \mathcal{L}^*/R$ and a state $|D_{L,r}\rangle$ as input, there exists an efficient quantum algorithm that generates a random vector $\mathbf{a} \leftarrow \mathcal{U}(\mathbb{Z}_q^n)$ and a state $2^{-\Omega(n)}$ -close to the following state*

$$\gamma_t^{\mathbf{a}} = \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})}^*} \rho_{\sqrt{\Sigma/2}}(\mathbf{z}(\mathbf{y})) |\mathbf{y}\rangle \langle \mathbf{y}| \otimes |\psi_t^{\mathbf{a}, \mathbf{y}}\rangle \langle \psi_t^{\mathbf{a}, \mathbf{y}}|$$

where $t = \sqrt{\sigma^2 + r^2 \|\mathbf{x}'\|^2}$, $\mathbf{s} = (\mathcal{L}^*)^{-1} \kappa_{\mathcal{L}^*}(\mathbf{x}) \bmod q$, $\mathbf{x}' = \mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{x})$, $\mathbf{z}(\mathbf{y}) = \mathbf{y}/R - \kappa_{(q\mathcal{L})^*}(\mathbf{y}/R)$,

$\Sigma = \frac{\mathbf{I}_n}{r^2} + \frac{\mathbf{x}'\mathbf{x}'^T}{\sigma^2}$, and the state $|\psi_t^{\mathbf{a}, \mathbf{y}}\rangle$ is an S |LWE $\rangle^{\text{phase}}$ state

$$|\psi_t^{\mathbf{a}, \mathbf{y}}\rangle := \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u) \exp\left(2\pi i \cdot u \frac{r^2 \langle \mathbf{x}', \mathbf{z}(\mathbf{y}) \rangle}{t^2}\right) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle. \quad (16)$$

Proof. From now on, when it is clear from the context, we use $\mathbf{z}$ to denote the value $\mathbf{z}(\mathbf{y})$, which actually depends on $\mathbf{y}$.

Here is the procedure of our quantum algorithm. For simplicity, we ignore the normalization factors when writing down superposition states.

1. Prepare the initial state

$$|D_{\mathcal{L}, r}\rangle \otimes \sum_{e \in \mathbb{Z}_{qR}/R} \rho_\sigma(e) |e\rangle,$$

which is $2^{-\Omega(n)}$ -close to $\sum_{\mathbf{v} \in \mathcal{L}, \|\mathbf{v}\| \leq \sqrt{nr}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle \otimes \sum_{e \in \mathbb{Z}_{qR}/R} \rho_\sigma(e) |e\rangle$ by Banaszczyk's Gaussian tail bound.

2. Measure $\mathbf{a} := \mathcal{L}^{-1} \mathbf{v} \bmod q$ to get an outcome $\mathbf{a}$ and a result state $2^{-\Omega(n)}$ -close to

$$\sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \|\mathbf{v}\| \leq \sqrt{nr}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle \otimes \sum_{e \in \mathbb{Z}_{qR}/R} \rho_\sigma(e) |e\rangle$$

According to [Lemma 15](#), when $r/\sqrt{2} > q\eta_\epsilon(\mathcal{L})$, the distribution of $\mathbf{a}$ is $2^{-\Omega(n)}$ -close to uniform.

3. Apply a unitary to add the inner product $\langle \mathbf{x}, \mathbf{v} \rangle \bmod q$ to the last register and get

$$\sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \|\mathbf{v}\| \leq \sqrt{nr}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle \otimes \sum_{e \in \mathbb{Z}_{qR}/R} \rho_\sigma(e) |\langle \mathbf{s}, \mathbf{a} \rangle + \langle \mathbf{x}', \mathbf{v} \rangle + e \bmod q\rangle. \quad (17)$$

Note that since we assumed $\mathbf{x} \in \mathcal{L}^*/R$, the second register always has its value in $\mathbb{Z}_{qR}/R$.

Intuitively, since $u := \langle \mathbf{x}', \mathbf{v} \rangle + e \leq \alpha q \sqrt{n} + \sigma \sqrt{n} < q/2$ with high probability and $R > 2\sqrt{nr}$, [Equation \(17\)](#) should be $2^{-\Omega(n)}$ -close to the following state displayed in [Equation \(18\)](#). The proof is deferred to [Appendix A.5](#).

Lemma 65. *Suppose that $q/2 > \alpha q \sqrt{n} + \sigma \sqrt{n}$, $R > 2\sqrt{nr} \sqrt{\log r}$ and $r > \sqrt{2} q \eta_\epsilon(\mathcal{L})$ for $\epsilon < 2^{-n}$, then the state in [Equation \(17\)](#) is $2^{-\Omega(n)}$ -close to the state*

$$\sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}} \rho_r(\mathbf{v}) |\mathbf{v} \bmod R\rangle \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle. \quad (18)$$

Remark 66. *Similar to [Lemma 52](#), the condition of this lemma can be relaxed to $q/2 > \alpha q \sqrt{n} + \sigma \sqrt{n}$ and $R > 2\sqrt{nr}$ if we instead use the proof technique from Claim A.5 in [\[Reg23\]](#), with a slight modification on the lemma statement. Either way, we have the same asymptotic values for the parameters in the end.*

By the assumption that $\sigma \leq \sqrt{2} \alpha q$, we have that $\alpha q \sqrt{n} + \sigma \sqrt{n} \leq (1 + \sqrt{2}) \alpha q \sqrt{n} < q/2$. Therefore, according to the lemma mentioned above, we can obtain a state that is $2^{-\Omega(n)}$ -close to the state displayed in [Equation \(18\)](#).

4. Recall that $\omega_R = e^{2\pi i/R}$. Applying $\text{QFT}_R^\dagger$ to the first register, we can get a state $2^{-\Omega(n)}$ -close to

$$\sum_{\mathbf{y} \in \mathbb{Z}_R^n} \sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}} \rho_r(\mathbf{v}) \cdot \omega_R^{\langle \mathbf{v}, \mathbf{y} \rangle} |\mathbf{y}\rangle \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle. \quad (19)$$

We show that the state in Equation (19) is $2^{-\Omega(n)}$ -close to the following state displayed in Equation (20). The proof is deferred to Appendix A.6.

Lemma 67. *Suppose that $\epsilon < 2^{-n}$, $R > 2\sqrt{n}r\sqrt{\log r}$, $r > 4q\eta_\epsilon(\mathcal{L})$ and $\sigma \geq \alpha q$, then the state in Equation (19) is $2^{-\Omega(n)}$ -close to the state*

$$\begin{aligned} & \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})}^*} \rho_{\sqrt{\Sigma}}(\mathbf{z}) \exp(2\pi i \langle \mathcal{L}\mathbf{a}, \kappa_{(q\mathcal{L})}^*(\mathbf{y}/R) \rangle) |\mathbf{y}\rangle \\ & \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u) \exp\left(2\pi i \cdot u \frac{r^2 \langle \mathbf{x}', \mathbf{z} \rangle}{t^2}\right) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle, \end{aligned} \quad (20)$$

where $t, \mathbf{s}, \mathbf{z}, \Sigma$ are specified in Theorem 64.

By the assumption that $\sigma \in [\alpha q, \sqrt{2}\alpha q]$, we have that $\alpha q\sqrt{n} + \sigma\sqrt{n} < (1 + \sqrt{2})\alpha q\sqrt{n} < (1 + \sqrt{2})q/5 < q/2$. Therefore, according to Lemma 67, we can obtain a state that is $2^{-\Omega(n)}$ -close to the state (recall the definition of $|\psi_t^{\mathbf{a}, \mathbf{y}}\rangle$ in Equation (16))

$$\sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})}^*} \rho_{\sqrt{\Sigma}}(\mathbf{z}) \exp(2\pi i \langle \mathcal{L}\mathbf{a}, \kappa_{(q\mathcal{L})}^*(\mathbf{y}/R) \rangle) |\mathbf{y}\rangle \otimes |\psi_t^{\mathbf{a}, \mathbf{y}}\rangle$$

5. Measure the first register to get a state $2^{-\Omega(n)}$ -close to the state $\gamma_t^{\mathbf{a}}$.¹

Finally, combining the measurement result $\mathbf{a}$ from step 2 and the state $\gamma_t^{\mathbf{a}}$ from step 5 gives the desired sample. $\square$

Remark 68. *An important caveat is that we should not discard the $\mathbf{y}$ register and hope we can solve the problem given only $\mathbf{a}$ and the $\text{S}|\text{LWE}\rangle^{\text{phase}}$ state $|\psi_{\sqrt{2}\alpha q}^{\mathbf{a}, \mathbf{y}}\rangle$ whose error amplitude is Gaussian with a small phase, because such a solver is so strong that it solves LWE directly. This is because such a solver utilizes no information about $\mathbf{y}$ (the seed that generates θ). So it should output $\mathbf{s}$ given samples $\mathbf{a}$ and the second register after step 3 (Note that in the actual procedure the solver is given the second register after step 5, but step 4 and step 5 are both local operations acting on the first register, which should not influence the state of the second register). This leads to an algorithm that outputs $\mathbf{s}$ given samples $(\mathbf{a}, \langle \mathbf{s}, \mathbf{a} \rangle + e \bmod q)$ where $\mathbf{a} \leftarrow \mathcal{U}_{\mathbb{Z}_q^n}$ and $e = \langle \mathbf{x}', \mathbf{v} \rangle$ for $\mathbf{v}$ distributed according to $D_{q\mathcal{L} + \mathcal{L}\mathbf{a}, r/\sqrt{2}}$ and a fixed vector $\|\mathbf{x}'\| \leq \frac{\alpha q}{r}$. As the distribution of e is close to a Gaussian distribution, this will give a surprising method to solve LWE.*

This also explains why we describe the auxiliary information $\mathbf{y}$ carefully in Definition 7 and Definition 56 instead of discarding $\mathbf{y}$ and strengthening the solver in Theorem 57 to solve $\text{S}|\text{LWE}\rangle_{n,m,q,f,\theta,D}^{\text{phase}}$ for any unknown θ such that $|\theta| \leq \frac{\sqrt{n}}{2\alpha q}$ (see Remark 59) and an arbitrary distribution D .

¹We measure the first register only for a better illustration of the residual state. In fact, we'll use the gentle measurement principle to defer all measurements, as described later in Section 6.4.

Given that we must use the information of $\mathbf{y}$, one may hope to compute $\mathbf{z}(\mathbf{y})$ to learn something about the phase $\theta^{(r,\mathbf{x})}(\mathbf{y}) = \frac{r^2 \langle \mathbf{x}', \mathbf{z}(\mathbf{y}) \rangle}{2\alpha^2 q^2}$. However, $\mathbf{z}(\mathbf{y})$ has ℓ_2 norm roughly $\frac{\sqrt{n}}{r}$ (see [Remark 59](#)). So computing $\mathbf{z}(\mathbf{y})$ from $\mathbf{y}$ is a $\text{CVP}_{\mathcal{L}^*, \frac{q\sqrt{n}}{r}}$ instance, which is even harder than the goal of this iteration (a quantum query to $\text{CVP}_{\mathcal{L}^*, \alpha q/r}$). How to utilize $\mathbf{y}$ requires more attempts and is an important step towards our ultimate goal of solving standard LWE efficiently.

6.3 Dealing with the unknown Gaussian width

Now that we know how to generate $(\mathbf{a}, \gamma_t^{\mathbf{a}})$ which resembles an $\text{S|LWE}\rangle^{\text{phase}}$ instance. However, the $\text{S|LWE}\rangle^{\text{phase}}$ solver requires instances with an error distribution of the fixed width $\sqrt{2}\alpha q$. To bridge this gap, we experiment with different values of σ to obtain a suitable width that is sufficiently close to $\sqrt{2}\alpha q$. Equipped with the $\text{S|LWE}\rangle^{\text{phase}}$ solver, we can in turn solve the CVP problem. We realize this idea in the proof of the following theorem:

Theorem 69. *Let $\mathcal{L}$ be an n -dimensional integer lattice, define the parameters $\epsilon \in (0, 2^{-n})$, $\alpha \in (0, \frac{1}{5\sqrt{n}})$, $r > 4q\eta_\epsilon(\mathcal{L})$, and a precision parameter $R > \max\{2\sqrt{nr}\sqrt{\log r}, \frac{2\sqrt{n}}{\alpha q}\}$ as an integer.*

Assume that there exists a quantum algorithm $\mathcal{A}$ that, given m samples of independently uniformly random vector $\mathbf{a} \in \mathbb{Z}_q^n$ and state $\gamma_{\sqrt{2}\alpha q}^{\mathbf{a}}$, solves the secret vector $\mathbf{s}$ in time complexity T with probability $1 - 2^{-\Omega(n)}$. Then there exists a quantum algorithm that, given a $\text{CVP}_{\mathcal{L}^, \alpha q/r}$ instance $\mathbf{x} \in \mathcal{L}^*/R$ and $3m^2n$ states $|D_{\mathcal{L}, r}\rangle$ as input, outputs $\mathbf{s} = (\mathcal{L}^*)^{-1}\kappa_{\mathcal{L}^*}(\mathbf{x}) \bmod q$ with probability $1 - 2^{-\Omega(n)}$ in time $O((m^2 + mT)\text{poly}(n))$.*

Proof. Let $\sigma' = \sqrt{2\alpha^2 q^2 - r^2 \|\mathbf{x}'\|^2}$ and $\sigma_i = \alpha q (1 + (\sqrt{2} - 1)\frac{i}{2m})$, $t_i = \sqrt{\sigma_i^2 + r^2 \|\mathbf{x}'\|^2}$ for $i = 0, 1, \dots, 2m$. In this condition, we have $\sigma_i \in [\alpha q, \sqrt{2}\alpha q]$. Since $r\|\mathbf{x}'\| < \alpha q$, there must exist an index $0 \leq j < 2m$ such that $\sigma_j < \sigma' \leq \sigma_{j+1}$.

Then σ_j is a suitable value of σ to generate samples for the quantum algorithm $\mathcal{A}$. Formally,

Lemma 70. *The quantum algorithm $\mathcal{A}$ can solve $\mathbf{s}$ with probability at least $1/2$, when given m independent samples of vector $\mathbf{a} \in \mathbb{Z}_q^n$ and state $\gamma_{t_j}^{\mathbf{a}}$.*

Before the proof of the lemma, let's first show how to construct a CVP algorithm based on the lemma. Here is the procedure of our quantum algorithm.

1. (Generate classical LWE samples for verification of the solution) Apply [Theorem 64](#) to n states $|D_{\mathcal{L}, r}\rangle$ with $\sigma = \sigma_0$ to obtain n samples of vectors $\mathbf{a} \in \mathbb{Z}_q^n$ and states $\gamma_{t_0}^{\mathbf{a}}$. Then, measure the second register of $\gamma_{t_0}^{\mathbf{a}}$ to obtain n classical LWE samples of the form $\langle \mathbf{a}, \mathbf{s} \rangle + u \bmod q$.
2. Enumerate σ from the set $\{\sigma_i : i \in \{0, 1, \dots, 2m - 1\}\}$.
3. For each $\sigma = \sigma_i$, apply [Theorem 64](#) to mn states $|D_{\mathcal{L}, r}\rangle$ to obtain mn samples of vectors $\mathbf{a} \in \mathbb{Z}_q^n$ and states $\gamma_{t_i}^{\mathbf{a}}$ with a precision of $1 - 2^{-\Omega(n)}$.
4. Utilize the quantum algorithm $\mathcal{A}$ on a group of m samples of vectors $\mathbf{a} \in \mathbb{Z}_q^n$ and states $\gamma_{t_i}^{\mathbf{a}}$ to derive a solution $\mathbf{s}'$.

5. Employ any verification process (e.g., as proposed by Regev in [Reg09, Lemma 3.6]) with the assistance of the n classical LWE samples obtained in step 1 to check whether $\mathbf{s}' = \mathbf{s}$. If this condition holds, output $\mathbf{s}'$ and conclude the process.

This procedure will use a maximum of $2m^2n + n < 3m^2n$ samples of $|D_{\mathcal{L},r}\rangle$ and operates with a time complexity of $O((m^2 + mT)\text{poly}(n))$. Moreover, it will output the correct $\mathbf{s}$ with a probability of at least $1 - 2^{-\Omega(n)}$ when $\sigma = \sigma_j$, by Lemma 70. $\square$

Lemma 70 follows from the fact that $\gamma_{t_j}^{\mathbf{a}}$ is close to $\gamma_{\sqrt{2}\alpha q}^{\mathbf{a}}$ as t_j is close to $\sqrt{2}\alpha q$. For completeness, let's provide the formal proof here.

Proof of Lemma 70. We label the m samples to be $\{(\mathbf{a}_i, \gamma_{t_j}^{\mathbf{a}_i})\}_{i \in [m]}$. By assumption, the quantum algorithm $\mathcal{A}$ can solve $\mathbf{s}$ with probability at least $1 - 2^{-\Omega(n)}$, when given m samples $\{(\mathbf{a}_i, \gamma_{\sqrt{2}\alpha q}^{\mathbf{a}_i})\}_{i \in [m]}$. The trace distance between the states in these two different types of samples is given by

$$\begin{aligned}
\delta\left(\bigotimes_{i=1}^m \gamma_{t_j}^{\mathbf{a}_i}, \bigotimes_{i=1}^m \gamma_{\sqrt{2}\alpha q}^{\mathbf{a}_i}\right) &\leq \sum_{i=1}^m \delta\left(\gamma_{t_j}^{\mathbf{a}_i}, \gamma_{\sqrt{2}\alpha q}^{\mathbf{a}_i}\right) \\
&\leq \sum_{i=1}^m \delta\left(|D_{q\mathcal{L}+\mathcal{L}\mathbf{a}_i,r}\rangle \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\sigma_j}(e) |e\rangle, |D_{q\mathcal{L}+\mathcal{L}\mathbf{a}_i,r}\rangle \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\sigma'}(e) |e\rangle\right) + 2^{-\Omega(n)} \\
&= \sum_{i=1}^m \delta\left(\sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\sigma_j}(e) |e\rangle, \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\sigma'}(e) |e\rangle\right) + 2^{-\Omega(n)} \\
&\leq_{(*)} m \sqrt{\frac{(\sigma' - \sigma_j)^2}{\sigma_j^2 + \sigma'^2}} (1 + 2^{-\Omega(n)}) + 2^{-\Omega(n)} \\
&\leq m \cdot \sqrt{\frac{(\alpha q / (2m))^2}{2(\alpha q)^2}} (1 + 2^{-\Omega(n)}) + 2^{-\Omega(n)} \\
&\leq \frac{1}{2\sqrt{2}} + 2^{-\Omega(n)},
\end{aligned}$$

where $(*)$ is according to the following lemma with $\sigma_j, \sigma' \in [\alpha q, \sqrt{2}\alpha q]$, $R > \frac{2\sqrt{n}}{\alpha q}$.

Lemma 71. When $q > 2\sqrt{n} \max(\beta_1, \beta_2)$ and $R \geq \frac{2\sqrt{n}}{\min(\beta_1, \beta_2)}$, the trace distance between

$$|\phi_1\rangle = \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\beta_1}(e) |e\rangle \quad \text{and} \quad |\phi_2\rangle = \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\beta_2}(e) |e\rangle$$

is at most $\sqrt{\frac{(\beta_1 - \beta_2)^2}{\beta_1^2 + \beta_2^2}} (1 + 2^{-\Omega(n)})$.

Proof.

$$\begin{aligned}
\frac{\langle \phi_1 | \phi_2 \rangle}{\| |\phi_1\rangle \| \| |\phi_2\rangle \|} &= \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\frac{\beta_1 \beta_2}{\sqrt{\beta_1^2 + \beta_2^2}}}(e) \Big/ \sqrt{\sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\beta_1/\sqrt{2}}(e) \sum_{e \in \mathbb{Z}_{qR}/R} \rho_{\beta_2/\sqrt{2}}(e)} \\
&= \sum_{e \in \mathbb{Z}/R} \rho_{\frac{\beta_1 \beta_2}{\sqrt{\beta_1^2 + \beta_2^2}}}(e) \Big/ \sqrt{\sum_{e \in \mathbb{Z}/R} \rho_{\beta_1/\sqrt{2}}(e) \sum_{e \in \mathbb{Z}/R} \rho_{\beta_2/\sqrt{2}}(e) (1 + 2^{-\Omega(n)})} \\
&= \frac{\beta_1 \beta_2}{\sqrt{\beta_1^2 + \beta_2^2}} \cdot \frac{\sqrt{2}}{\sqrt{\beta_1 \beta_2}} \sum_{e \in R\mathbb{Z}} \rho_{\frac{\sqrt{\beta_1^2 + \beta_2^2}}{\beta_1 \beta_2}}(e) \Big/ \sqrt{\sum_{e \in R\mathbb{Z}} \rho_{\sqrt{2}/\beta_1}(e) \sum_{e \in R\mathbb{Z}} \rho_{\sqrt{2}/\beta_2}(e) (1 + 2^{-\Omega(n)})} \\
&= \frac{\sqrt{2\beta_1 \beta_2}}{\sqrt{\beta_1^2 + \beta_2^2}} (1 + 2^{-\Omega(n)}),
\end{aligned}$$

where we use the Poisson summation formula and Banaszczyk's tail bound.

So their trace distance is at most

$$\sqrt{1 - \left(\frac{|\langle \phi_1 | \phi_2 \rangle|}{\| |\phi_1\rangle \| \| |\phi_2\rangle \|} \right)^2} \leq \sqrt{\frac{(\beta_1 - \beta_2)^2}{\beta_1^2 + \beta_2^2}} (1 + 2^{-\Omega(n)}). \quad \square$$

Therefore, the quantum algorithm $\mathcal{A}$ will output $\mathbf{s}$ when given $\{(\mathbf{a}_i, \gamma_{t_j}^{\mathbf{a}_i})\}_{i \in [m]}$ with probability at least

$$1 - \frac{1}{2\sqrt{2}} - 2^{-\Omega(n)} > \frac{1}{2}. \quad \square$$

One last gap between [Theorem 69](#) and the theorem we need to prove ([Theorem 62](#)) is that, in [Theorem 69](#) the algorithm only outputs $\mathbf{s} = (\mathcal{L}^*)^{-1} \kappa_{\mathcal{L}^*}(\mathbf{x}) \bmod q$, which answers a modulo version of $\text{CVP}_{\mathcal{L}^*, \alpha q/r}$ for instance $\mathbf{x} \in \mathcal{L}^*/R$. However, as [\[Reg09\]](#) shows, CVP is efficiently reducible to its modulo version, which closes this final gap. Formally,

Theorem 72 ([\[Reg09\]](#), a slight modification of Lemma 3.5). *Let $\mathcal{L}$ be an n -dimensional integer lattice, define distance parameter $d \in (0, \lambda_1(\mathcal{L})/2)$, and integers $q > 2, R > \frac{2^{2n+1} n \lambda_n(\mathcal{L})}{d}$. Assume there exists an algorithm $\mathcal{A}$ that, on input $\mathbf{x} \in \mathcal{L}/R$ with the guarantee $\text{dist}(\mathbf{x}, \mathcal{L}) \leq d$, outputs $\mathbf{s} = \mathcal{L}^{-1} \kappa_{\mathcal{L}}(\mathbf{x}) \bmod q$ with probability $1 - 2^{-\Omega(n)}$, then there exists a polynomial-time algorithm that, on input $\mathbf{x} \in \mathcal{L}/R$ with guarantee $\text{dist}(\mathbf{x}, \mathcal{L}) \leq d$, outputs $\kappa_{\mathcal{L}}(\mathbf{x})$ with probability $1 - 2^{-\Omega(n)}$ using at most n calls to $\mathcal{A}$.*

Proof. It is merely the same as the proof of Lemma 3.5 in [\[Reg09\]](#), but with a slight modification. Compute a sequence $\mathbf{x}_1, \mathbf{x}_2, \dots$ where $\mathbf{x}_1$ is the input $\mathbf{x}$ and $\mathbf{x}_{i+1}$ is given by the following: for $\mathbf{x}_i \in \mathcal{L}/R$, call $\mathcal{A}$ to compute $\mathbf{s}_i = \mathcal{L}^{-1} \kappa_{\mathcal{L}}(\mathbf{x}_i) \bmod q$, and then compute $\mathbf{y}_i := (\mathbf{x}_i - \mathcal{L} \mathbf{s}_i)/q$. Now that $\mathbf{y}_i \in \mathcal{L}/(Rq)$ and $\text{dist}(\mathbf{y}_i, \mathcal{L}) = \text{dist}(\mathbf{x}_i, \mathcal{L})/q$, we can apply Babai's nearest plane algorithm [\[Bab86\]](#) to find a point $\mathbf{x}_{i+1} \in \mathcal{L}/R$ such that $\text{dist}(\mathbf{x}_{i+1}, \mathbf{y}_i) \leq 2^n \text{dist}(\mathbf{y}_i, \mathcal{L}/R) \leq 2^{n-1} n \lambda_n(\mathcal{L}/R) < d/2^{n+2}$. Then $\text{dist}(\mathbf{x}_{i+1}, \mathcal{L}) \leq \text{dist}(\mathbf{y}_i, \mathcal{L}) + \text{dist}(\mathbf{x}_{i+1}, \mathbf{y}_i) < \text{dist}(\mathbf{x}_i, \mathcal{L})/q + d/2^{n+2}$.

Thus $\text{dist}(\mathbf{x}_i, \mathcal{L}) \leq \frac{1}{q^{i-1}} (\text{dist}(\mathbf{x}_1, \mathcal{L}) - dq/(2^{n+2}(q-1))) + dq/(2^{n+2}(q-1)) \leq \frac{d}{q^{i-1}} + \frac{d}{2^{n+1}}$. Then $\kappa_{\mathcal{L}}(\mathbf{x}_{i+1}) = \kappa_{\mathcal{L}}(\mathbf{y}_i)$ because $\text{dist}(\mathbf{y}_i, \mathcal{L}) + \text{dist}(\mathbf{x}_{i+1}, \mathbf{y}_i) < \text{dist}(\mathbf{x}_i, \mathcal{L})/q + d/2^{n+2} < d < \lambda_1(\mathcal{L})/2$.

After n steps, we have a point $\mathbf{x}_{n+1}$ such that $\text{dist}(\mathbf{x}_{n+1}, \mathcal{L}) \leq \frac{d}{q^n} + \frac{d}{2^{n+1}} < \frac{d}{2^n}$. Hence we can apply Babai's nearest plane algorithm [Bab86] to recover $\kappa_{\mathcal{L}}(\mathbf{x}_{n+1})$.

Note that $\kappa_{\mathcal{L}}(\mathbf{x}_i) = q\kappa_{\mathcal{L}}(\mathbf{y}_i) + \mathcal{L}\mathbf{s}_i = q\kappa_{\mathcal{L}}(\mathbf{x}_{i+1}) + \mathcal{L}\mathbf{s}_i$. We can recover $\kappa_{\mathcal{L}}(\mathbf{x}_1)$ step by step. Notice that each call to $\mathcal{A}$ has failure probability $2^{-\Omega(n)}$ and we use n calls to $\mathcal{A}$, so our algorithm has failure probability at most $2^{-\Omega(n)}$, which completes the proof. $\square$

As a direct corollary of Theorem 69 and Theorem 72, we can reduce CVP to $\text{S|LWE}\rangle^{\text{phase}}$ given a sufficient number of states $|D_{\mathcal{L},r}\rangle$. Formally,

Corollary 73. *Let $\mathcal{L}$ be an n -dimensional integer lattice, define the parameters $\epsilon \in (0, 2^{-n})$, $\alpha \in (0, \frac{1}{5\sqrt{n}})$, $r > 4q\eta_{\epsilon}(\mathcal{L})$, and a precision parameter $R > \max\{2\sqrt{nr}\sqrt{\log r}, \frac{2\sqrt{n}}{\alpha q}, \frac{2^{2n+1}nr\lambda_n(\mathcal{L}^*)}{\alpha q}\}$ as an integer. Assume that there exists a quantum algorithm $\mathcal{A}$ that, given m samples of uniformly random vector $\mathbf{a} \in \mathbb{Z}_q^n$ and state $\gamma_{\sqrt{2}\alpha q}^{\mathbf{a}}$, solves the secret vector $\mathbf{s}$ in time complexity T . Then there exists an algorithm that given a $\text{CVP}_{\mathcal{L}^*, \alpha q/r}$ instance $\mathbf{x} \in \mathcal{L}^*/R$ and $3m^2n^2$ states $|D_{\mathcal{L},r}\rangle$ as input, outputs $\kappa_{\mathcal{L}^*}(\mathbf{x})$ with probability $1 - 2^{-\Omega(n)}$ in time $O((m^2 + mT)\text{poly}(n))$.*

6.4 Answering the quantum CVP query with small disturbance on $|D_{\mathcal{L},r}\rangle$

We are now ready to conclude the proof of Theorem 62. Corollary 73 provides a method to answer classical queries for $\text{CVP}_{\mathcal{L}^*, \alpha q/r}$ on instance $\mathbf{x} \in \mathcal{L}^*/R$ with $1 - 2^{-\Omega(n)}$ probability. By deferred measurement principle and gentle measurement principle, we expect that it can help us to answer quantum query $|\mathbf{x}, \mathbf{y}\rangle \rightarrow |\mathbf{x}, \mathbf{y} + \kappa_{\mathcal{L}^*}(\mathbf{x})\rangle$ with $\text{dist}(\mathbf{x}, \mathcal{L}^*) \leq \alpha q/r$, using $3m^2n^2$ states $|D_{\mathcal{L},r}\rangle$ while introducing only an exponentially small disturbance on them. However, it is important to note that the gentle measurement principle is primarily suitable for producing measurement results (which is classical) rather than answering quantum query. Therefore, we shall provide a formal proof for using gentle measurement principle to answer quantum query:

Proof of Theorem 62. The existence of the quantum algorithm $\mathcal{A}$ in Corollary 73 is a direct consequence of the assumption that a quantum algorithm can solve the $\text{S|LWE}\rangle_{n,m,q,f,\theta(r,\mathbf{x}),D_{\theta}^{(r,\mathbf{x})}}^{\text{phase}}$ problem. Specifically, let's recall the expression

$$\gamma_{\sqrt{2}\alpha q}^{\mathbf{a}} = \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \rho_{\sqrt{\Sigma/2}}(\mathbf{z}(\mathbf{y})) |\mathbf{y}\rangle \langle \mathbf{y}| \otimes |\psi_{\sqrt{2}\alpha q}^{\mathbf{a},\mathbf{y}}\rangle \langle \psi_{\sqrt{2}\alpha q}^{\mathbf{a},\mathbf{y}}|.$$

The quantum algorithm $\mathcal{A}$ proceeds by first measuring the result of $|\mathbf{y}\rangle$ to obtain a specific $\mathbf{y}$ and an $\text{S|LWE}\rangle^{\text{phase}}$ state $|\psi_{\sqrt{2}\alpha q}^{\mathbf{a},\mathbf{y}}\rangle$. The parameters and functions of this state corresponds to the parameters and functions defined in Definition 56. Subsequently, $\mathcal{A}$ applies the $\text{S|LWE}\rangle_{n,m,q,f,\theta(r,\mathbf{x}),D_{\theta}^{(r,\mathbf{x})}}^{\text{phase}}$ solver to compute $\mathbf{s}$.

We defer all the measurements in the algorithm in Corollary 73 including those in the $\text{S|LWE}\rangle^{\text{phase}}$ solver $\mathcal{A}$ to obtain a unitary $U : |\mathbf{x}\rangle |D_{\mathcal{L},r}\rangle^{\otimes (3m^2n^2)} |0^{\text{Aux}}\rangle \rightarrow |\mathbf{x}\rangle |\phi_{\mathbf{x}}\rangle$, where the first register of $|\phi_{\mathbf{x}}\rangle$ contains the solution $\kappa_{\mathcal{L}^*}(\mathbf{x})$, and the size of U is $O((m^2 + mT)\text{poly}(n))$. As the algorithm in Corollary 73 outputs $\kappa_{\mathcal{L}^*}(\mathbf{x})$ with probability at least $1 - 2^{-\Omega(n)}$ whenever $\text{dist}(\mathbf{x}, \mathcal{L}^*) \leq \alpha q/r$ and $\mathbf{x} \in \mathcal{L}^*/R$, by gentle measurement principle [Win99], there exists state $|\phi_{\mathbf{x}}^{\text{Aux}}\rangle$ such that the state $|\phi_{\mathbf{x}}\rangle$ is $2^{-\Omega(n)}$ -close to the state $|\kappa_{\mathcal{L}^*}(\mathbf{x})\rangle |\phi_{\mathbf{x}}^{\text{Aux}}\rangle$ in ℓ_2 -norm. Our quantum algorithm answers the query $|\mathbf{x}, \mathbf{y}\rangle \rightarrow |\mathbf{x}, \mathbf{y} + \kappa_{\mathcal{L}^*}(\mathbf{x})\rangle$ with $\text{dist}(\mathbf{x}, \mathcal{L}^*) \leq \alpha q/r$ using the following procedure:

1. Prepare the initial state

$$|\mathbf{x}, \mathbf{y}\rangle |D_{\mathcal{L},r}\rangle^{\otimes(3m^2n^2)} |0^{\text{Aux}}\rangle.$$

2. Apply U to the registers containing $|\mathbf{x}\rangle |D_{\mathcal{L},r}\rangle^{\otimes(3m^2n^2)} |0^{\text{Aux}}\rangle$ to get a state $2^{-\Omega(n)}$ -close to the state

$$|\mathbf{x}, \mathbf{y}\rangle |\kappa_{\mathcal{L}^*}(\mathbf{x})\rangle |\phi_{\mathbf{x}}^{\text{Aux}}\rangle$$

in ℓ_2 -norm.

3. Apply a unitary to add the value of $\kappa_{\mathcal{L}^*}(\mathbf{x})$ to $\mathbf{y}$ to get a state $2^{-\Omega(n)}$ -close to the state

$$|\mathbf{x}, \mathbf{y} + \kappa_{\mathcal{L}^*}(\mathbf{x})\rangle |\kappa_{\mathcal{L}^*}(\mathbf{x})\rangle |\phi_{\mathbf{x}}^{\text{Aux}}\rangle$$

in ℓ_2 -norm.

4. Apply $U^\dagger$ to the registers containing $|\mathbf{x}\rangle |\kappa_{\mathcal{L}^*}(\mathbf{x})\rangle |\phi_{\mathbf{x}}^{\text{Aux}}\rangle$ to get a state $2^{-\Omega(n)}$ -close to the state

$$|\mathbf{x}, \mathbf{y} + \kappa_{\mathcal{L}^*}(\mathbf{x})\rangle |D_{\mathcal{L},r}\rangle^{\otimes(3m^2n^2)} |0^{\text{Aux}}\rangle$$

in ℓ_2 -norm.

Therefore, we can answer the quantum query up to exponentially small error with exponentially small disturbance on the states $|D_{\mathcal{L},r}\rangle$ in time $O((m^2 + mT)\text{poly}(n))$, which ends up the proof of [Theorem 62](#). $\square$

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A Appendix

A.1 Proof of [Lemma 16](#)

The lemma is proven when q is a prime in [[GPV08](#), Lemma 5.3]. Here we extend the proof to a general q .

For any fixed non-zero $\mathbf{s} \in \mathbb{Z}_q^n$, wlog assuming s_1 is a non-zero entry of $\mathbf{s}$. Then for any $\mathbf{a} \in \mathbb{Z}_q^n$, $y := \langle \mathbf{a}, \mathbf{s} \rangle \bmod q$ can be written as $y = s_1 a_1 + v \bmod q$ for some $v \in \mathbb{Z}_q$. We observe that for any $q \in \mathbb{N}$, for any $v \in \mathbb{Z}_q$, for any non-zero $s_1 \in \mathbb{Z}_q$,

$$\Pr_{a_1 \in \mathbb{Z}_q} [s_1 a_1 + v \bmod q \in (-q/4, q/4) \cap \mathbb{Z}] \leq 2/3,$$

where the equality holds when $q \in 3^k \cdot \mathbb{N}$ for some $k \geq 1$, $s_1 \in (q/3) \cdot \mathbb{Z}/q\mathbb{Z}$, $s_1 \neq 0$, and for some $v \in \mathbb{Z}_q$ (for example, when $q = 15$, $s_1 = 5$, and $v = 2$).

Therefore, over the randomness of $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, the probability that $\mathbf{A}^T \mathbf{s} = \mathbf{y} \bmod q$ for some $\mathbf{y} \in \mathbb{Z}^m$ such that $\|\mathbf{y}\|_\infty < q/4$ is at most $(2/3)^m \leq (3/2)^{-2n \log_2 q} \leq q^{-1.16n}$. Applying a union bound over all $\mathbf{s} \in \mathbb{Z}_q^n$ completes the proof of [Lemma 16](#).

A.2 Proof of [Fact 38](#)

We prove the claim for the special case of $|\varphi_0\rangle$ v.s. $|\varphi_c\rangle$ for $c \in \mathbb{Z}_q, c > 0$, and the proof can be easily generalized to any pair of distinct centers c_1, c_2 .

The overlap of $|\varphi_0\rangle$ v.s. $|\varphi_c\rangle$ can be written as

$$|\langle \varphi_c | \varphi_0 \rangle| = \frac{1}{N_r} \left| \sum_{x \in \mathbb{Z}_q} \rho_r(x) \rho_r((x - c) \bmod q) e^{-\pi i (x^2 - ((x - c) \bmod q)^2) / q'} \right|. \quad (21)$$

We then consider the following two cases focusing on the size of c .

- **Case 1.** $c > \frac{1}{4}r\sqrt{n}$ is large. Then for any $x \in \mathbb{Z}_q$, at least one of $x, (x - c) \bmod q$ has absolute value greater than $c/2 > \frac{1}{8}r\sqrt{n}$, thus

$$|\langle \varphi_c | \varphi_0 \rangle| \leq \frac{1}{N_r} \sum_{x \in \mathbb{Z}_q} \rho_r(x) \rho_r((x - c) \bmod q) \leq \frac{q}{N_r} \rho_r(r\sqrt{n}/8) = 2^{-\Omega(n)}.$$

- **Case 2.** $c \leq \frac{1}{4}r\sqrt{n}$ is small. Then we can apply Banaszczyk's tail bound on the summation in [Equation \(21\)](#) to change the term $(x - c) \bmod q$ to $x - c$, since when $(x - c) \bmod q = x - c + q$, the absolute value of $x - c, x - c + q, x$ are all at least $q/2 - c > \frac{1}{4}r\sqrt{n}$. Therefore,

$$\begin{aligned} |\langle \varphi_c | \varphi_0 \rangle| &\leq \frac{1 + 2^{-\Omega(n)}}{N_r} \left| \sum_{x \in \mathbb{Z}_q} \rho_r(x) \rho_r(x - c) e^{-\pi i (x^2 - (x - c)^2) / q'} \right| \\ &\leq_{(a)} \frac{1 + 2^{-\Omega(n)}}{N_r} \rho_{r/\sqrt{2}}(c/2) \left| \sum_{x \in \mathbb{Z}} \rho_{r/\sqrt{2}}(x - c/2) e^{2\pi i c x / q'} \right| \\ &\leq_{(b)} \frac{1 + 2^{-\Omega(n)}}{N_r} \rho_{r/\sqrt{2}}(c/2) \left| \sum_{y \in \mathbb{Z}} \rho_{\sqrt{2}/r}(y - c/q') e^{2\pi i (y - c/q') \cdot c/2} \right| \\ &\leq_{(c)} \frac{1 + 2^{-\Omega(n)}}{N_r} \rho_{r/\sqrt{2}}(c/2) \left(\rho_{\sqrt{2}/r}((c \bmod q')/q') + 2^{-\Omega(n)} \right), \end{aligned}$$

where (a) is from Banaszczyk's tail bound; (b) uses the Poisson summation formula. (c) holds since $r/q' = \Omega(\sqrt{n})$ by condition, and thus the summation $\sum_{y \in \mathbb{Z}} \rho_{\sqrt{2}/r}(y - c/q')$ (with some phase term that does not affect the norm) is concentrated at $\rho_{\sqrt{2}/r}((c \bmod q')/q')$, with exponentially small weight elsewhere.

Therefore, when $c \not\equiv 0 \bmod q'$, the overlap of $|\varphi_0\rangle$ and $|\varphi_c\rangle$ can be bounded by

$$|\langle \varphi_c | \varphi_0 \rangle| \leq \frac{1 + 2^{-\Omega(n)}}{N_r} \rho_{r/\sqrt{2}}(c/2) \left(\rho_{\sqrt{2}/r}(1/q') + 2^{-\Omega(n)} \right) = 2^{-\Omega(n)}.$$

A.3 Upper bounds on Gaussian tails

Lemma 74. *Let $\mathcal{L} \subseteq \mathbb{R}^n$ be a lattice, $\mathbf{u} \in \mathbb{R}^n$ be a fixed vector, $\epsilon \in (0, 1)$ be a small error parameter, σ be a positive real number with $\sigma > 2\eta_\epsilon(\mathcal{L})$. Then we have*

$$\sum_{\mathbf{x} \in \mathcal{L}^*} \rho_{1/\sigma}(\mathbf{x} - \mathbf{u}) < \rho_{1/\sigma}(\mathbf{u} - \kappa_{\mathcal{L}^*}(\mathbf{u})) + \epsilon.$$

If $\mathbf{u}$'s closest vectors in $\mathcal{L}^$ are not unique, then $\kappa_{\mathcal{L}^*}(\mathbf{u})$ can be an arbitrary one of the closest vectors.*

Proof. It suffices to prove that

$$\sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\kappa_{\mathcal{L}^*}(\mathbf{u})\}} \rho_{1/\sigma}(\mathbf{x} - \mathbf{u}) < \epsilon.$$

For any $\mathbf{x} \in \mathcal{L}^*$, we have that

$$\begin{aligned} \|\mathbf{x} - \mathbf{u}\|^2 &\geq \frac{1}{2} (\|\mathbf{x} - \mathbf{u}\|^2 + \|\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}\|^2) && \text{(by the definition of } \kappa_{\mathcal{L}^*}(\mathbf{u})\text{)} \\ &\geq \frac{1}{4} (\|\mathbf{x} - \mathbf{u}\| + \|\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}\|)^2 \\ &\geq \frac{1}{4} \|\mathbf{x} - \mathbf{u} - (\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u})\|^2 && \text{(triangle inequality)} \\ &= \frac{1}{4} \|\mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{u})\|^2, \end{aligned}$$

so

$$\begin{aligned} \sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\kappa_{\mathcal{L}^*}(\mathbf{u})\}} \rho_{1/\sigma}(\mathbf{x} - \mathbf{u}) &\leq \sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\kappa_{\mathcal{L}^*}(\mathbf{u})\}} \rho_{2/\sigma}(\mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{u})) \\ &= \sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\mathbf{0}\}} \rho_{2/\sigma}(\mathbf{x}) \\ &< \epsilon, \end{aligned}$$

as desired. □

Lemma 75. *Let $\mathcal{L} \subseteq \mathbb{R}^n$ be a lattice, $\mathbf{u} \in \mathbb{R}^n$ be a fixed vector, $\epsilon \in (0, 1)$ be a small error parameter, σ be a positive real number with $\sigma > 2\sqrt{2}\eta_\epsilon(\mathcal{L})$. Then we have*

$$\sum_{\mathbf{x} \in \mathcal{L}^*} \rho_{1/\sigma}(\mathbf{x} - \mathbf{u}) < \rho_{1/\sigma}(\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}) + \epsilon \cdot \rho_{\sqrt{2}/\sigma}(\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}).$$

If $\mathbf{u}$'s closest vectors in $\mathcal{L}^$ are not unique, then $\kappa_{\mathcal{L}^*}(\mathbf{u})$ can be an arbitrary one of the closest vectors.*

Proof. It suffices to prove that

$$\sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\kappa_{\mathcal{L}^*}(\mathbf{u})\}} \rho_{1/\sigma}(\mathbf{x} - \mathbf{u}) < \epsilon \cdot \rho_{\sqrt{2}/\sigma}(\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}).$$

For any $\mathbf{x} \in \mathcal{L}^*$, we have that

$$\begin{aligned}
\|\mathbf{x} - \mathbf{u}\|^2 - \frac{1}{2}\|\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}\|^2 &\geq \frac{1}{4}(\|\mathbf{x} - \mathbf{u}\|^2 + \|\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}\|^2) && \text{(by the definition of } \kappa_{\mathcal{L}^*}(\mathbf{u})\text{)} \\
&\geq \frac{1}{8}(\|\mathbf{x} - \mathbf{u}\| + \|\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}\|)^2 \\
&\geq \frac{1}{8}\|\mathbf{x} - \mathbf{u} - (\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u})\|^2 && \text{(triangle inequality)} \\
&= \frac{1}{8}\|\mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{u})\|^2,
\end{aligned}$$

so

$$\begin{aligned}
\sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\kappa_{\mathcal{L}^*}(\mathbf{u})\}} \rho_{1/\sigma}(\mathbf{x} - \mathbf{u}) &\leq \rho_{\sqrt{2}/\sigma}(\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}) \sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\kappa_{\mathcal{L}^*}(\mathbf{u})\}} \rho_{2\sqrt{2}/\sigma}(\mathbf{x} - \kappa_{\mathcal{L}^*}(\mathbf{u})) \\
&= \rho_{\sqrt{2}/\sigma}(\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}) \sum_{\mathbf{x} \in \mathcal{L}^* \setminus \{\mathbf{0}\}} \rho_{2\sqrt{2}/\sigma}(\mathbf{x}) \\
&< \epsilon \cdot \rho_{\sqrt{2}/\sigma}(\kappa_{\mathcal{L}^*}(\mathbf{u}) - \mathbf{u}),
\end{aligned}$$

as desired. $\square$

A.4 Proof of Lemma 52

Proof of Lemma 52. Note that if $j < \alpha\sqrt{m}$, by assuming $\|\mathbf{e}\| < \sqrt{m}\gamma q$ and the condition of Theorem 47, $\|j \cdot \mathbf{e}\| < \alpha\gamma q m < q/16$. It follows that for $\mathbf{x} \in \mathbb{Z}_q^m$, either $\|\mathbf{x}\| > \frac{15}{16}q$ or $(\mathbf{x} + j \cdot \mathbf{e} \bmod q) = \mathbf{x} + j \cdot \mathbf{e}$. Then by Banaszczyk's tail bound and Lemma 24, the following two states

$$\begin{aligned}
&\frac{1}{\sqrt{N}} \sum_{\mathbf{x} \in \mathbb{Z}_q^m} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e} \bmod q) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q\rangle, \\
&\frac{1}{\sqrt{N'}} \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q\rangle
\end{aligned} \tag{22}$$

are $2^{-\Omega(n)}$ -close, where N, N' are the corresponding normalization factors. Then, the state Equation (12) is $2^{-\Omega(n)}$ -close to the state

$$\frac{1}{\sqrt{N_3}} \sum_{j \in \mathbb{Z}_q} \rho_{\alpha}(j) |j\rangle \sum_{\mathbf{v} \in \mathbb{Z}_q^n} |(\mathbf{v} + j \cdot \mathbf{s}) \bmod q\rangle \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q\rangle. \tag{23}$$

where N_3 is the normalization factor. Denote this state as $|\Phi\rangle = \frac{1}{\sqrt{N_3}} \mathbf{u}_3$.

Let's rewrite the state in Equation (13) as

$$\frac{1}{\sqrt{N_2}} \sum_{j \in \mathbb{Z}_q} \rho_{\alpha}(j) |j\rangle \sum_{\mathbf{v} \in \mathbb{Z}_q^n} |(\mathbf{v} + j \cdot \mathbf{s}) \bmod q\rangle \sum_{\substack{\mathbf{x} \in \mathbb{Z}^m, \\ \|\mathbf{x}\| < \lambda_1(\mathcal{L}_q(\mathbf{A}))/2}} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q\rangle. \tag{24}$$

Denote this state as $|\Phi'\rangle = \frac{1}{\sqrt{N_2}} \mathbf{u}_2$. Now it suffices to show that $|\Phi\rangle$ and $|\Phi'\rangle$ are $2^{-\Omega(n)}$ -close, under the condition of Theorem 47 that implies:

$$(1) (\beta q \sqrt{m} + \alpha \gamma q m) \sqrt{\log(\beta q)} < 2\beta q \sqrt{m \log(\beta q)} < q/8 < \lambda_1(\mathcal{L}_q(\mathbf{A}))/2.$$

$$(2) \beta q > \alpha \gamma q \sqrt{m} > \sqrt{m}.$$

Then we have

$$\begin{aligned} & \|\mathbf{u}_3 - \mathbf{u}_2\|^2 \\ &= \sum_{\mathbf{v} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \left\| \sum_{\substack{\mathbf{x} \in \mathbb{Z}^m, \\ \|\mathbf{x}\| \geq \lambda_1(\mathcal{L}_q(\mathbf{A}))/2}} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) |(\mathbf{A}^T \mathbf{v} + \mathbf{x}) \bmod q| \right\|^2 \\ &\leq_{(1)} q^n \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \left(\sum_{\substack{\mathbf{x} \in \mathbb{Z}^m, \\ \|\mathbf{x}\| \geq \lambda_1(\mathcal{L}_q(\mathbf{A}))/2}} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e}) \right)^2 \\ &\leq_{(2)} q^n \sum_{\substack{j \in \mathbb{Z}_q, \\ |j| < \alpha \sqrt{m \log(\beta q)}}} \rho_\alpha(j)^2 \left(\sum_{\substack{\mathbf{x} \in \mathbb{Z}^m, \\ \|\mathbf{x}\| \geq \beta q \sqrt{m \log(\beta q)}}} \rho_{\beta q}(\mathbf{x}) \right)^2 + q^n \sum_{\substack{j \in \mathbb{Z}_q, \\ |j| \geq \alpha \sqrt{m \log(\beta q)}}} \rho_\alpha(j)^2 \left(\sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x}) \right)^2, \end{aligned}$$

where in (1) we absorb the summation over $\mathbf{v} \in \mathbb{Z}_q^n$ in q^n , and use the fact that ρ is non-negative; in (2), the first term uses the following: when $\|\mathbf{x}\| \geq \lambda_1(\mathcal{L}_q(\mathbf{A}))/2$ and $|j| < \alpha \sqrt{m \log(\beta q)}$, we have $\|\mathbf{x} + j \cdot \mathbf{e}\| > \lambda_1(\mathcal{L}_q(\mathbf{A}))/2 - \alpha \sqrt{m \log(\beta q)} \cdot \gamma q \sqrt{m} > \beta q \sqrt{m \log(\beta q)}$, the second term uses for any $\mathbf{c} \in \mathbb{R}^n$, $\sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x} + \mathbf{c}) \leq \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x})$.

From Banaszczyk's tail bound (Lemma 10), we have that

$$\sum_{\substack{\mathbf{x} \in \mathbb{Z}^m, \\ \|\mathbf{x}\| \geq \beta q \sqrt{m \log(\beta q)}}} \rho_{\beta q}(\mathbf{x}) < \beta^{-3m} q^{-3m} \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x}), \quad \sum_{\substack{j \in \mathbb{Z}_q, \\ |j| \geq \alpha \sqrt{m \log(\beta q)}}} \rho_{\alpha/\sqrt{2}}(j) < \beta^{-6m} q^{-6m} \sum_{j \in \mathbb{Z}_q} \rho_{\alpha/\sqrt{2}}(j).$$

Notice that $\beta q > \sqrt{m}$, so from Lemma 11, $\sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x}) < (1 + 2^{-\Omega(m)}) \beta^m q^m$. Therefore,

$$\begin{aligned} \|\mathbf{u}_1 - \mathbf{u}_2\|^2 &< 2q^n (\beta q)^{-6m} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \left(\sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x}) \right)^2 \\ &< 2(1 + 2^{-\Omega(n)}) q^n (\beta q)^{-4m} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \end{aligned}$$

On the other hand, notice that from Lemma 11, $\sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q/\sqrt{2}}(\mathbf{x}) > (\beta q/\sqrt{2})^m$, we have that

$$\begin{aligned} N_1 &\geq \sum_{\mathbf{v} \in \mathbb{Z}_q^n} \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x} + j \cdot \mathbf{e})^2 \\ &= q^n \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \sum_{\mathbf{x} \in \mathbb{Z}^m} \rho_{\beta q}(\mathbf{x})^2 \\ &> q^n \left(\frac{\beta q}{\sqrt{2}} \right)^m \sum_{j \in \mathbb{Z}_q} \rho_\alpha(j)^2 \end{aligned}$$

Hence, we get that $\frac{\|\mathbf{u}_1 - \mathbf{u}_2\|^2}{N_1} < 2^{-\Omega(n)}$, this implies that $|\Phi\rangle$ and $|\Phi'\rangle$ are $2^{-\Omega(n)}$ -close to each other, by [Lemma 24](#), as desired. $\square$

Remark 76. *The condition of this lemma can be relaxed if we instead use the proof technique from Claim A.5 in [Reg23]. To be more specific, the above lemma holds as long as $\beta q\sqrt{m} + \alpha\gamma qm < \lambda_1(\mathcal{L}_q(\mathbf{A}))/2$ and $q/2 > \alpha\sqrt{m}$. However, the relaxed condition provides similar parameters in the overall reduction, and therefore, we do not emphasize it here.*

A.5 Proof of [Lemma 65](#)

Proof of Lemma 65. For any vector $\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}$ such that $\|\mathbf{v}\| \leq \sqrt{nr}$, as $q/2 > \sigma\sqrt{n}$, the state

$$\sum_{e \in \mathbb{Z}_{qR}/R} \rho_\sigma(e) |\langle \mathbf{s}, \mathbf{a} \rangle + \langle \mathbf{x}', \mathbf{v} \rangle + e \bmod q\rangle \quad (25)$$

is $2^{-\Omega(n)}$ -close to the state

$$\sum_{e \in [-\sigma\sqrt{n}, \sigma\sqrt{n}] \cap (\mathbb{Z}/R)} \rho_\sigma(e) |\langle \mathbf{s}, \mathbf{a} \rangle + \langle \mathbf{x}', \mathbf{v} \rangle + e \bmod q\rangle$$

in ℓ_2 norm by Banaszczyk's tail bound (see [Lemma 10](#)), which is $2^{-\Omega(n)}$ -close to the state

$$\sum_{e \in \mathbb{Z}_{qR}/R - \langle \mathbf{x}', \mathbf{v} \rangle} \rho_\sigma(e) |\langle \mathbf{s}, \mathbf{a} \rangle + \langle \mathbf{x}', \mathbf{v} \rangle + e \bmod q\rangle$$

in ℓ_2 norm due to Banaszczyk's tail bound and the fact that $|\langle \mathbf{x}', \mathbf{v} \rangle| \leq \frac{\alpha q}{r} \cdot \sqrt{nr} < q/2 - \sigma\sqrt{n}$, which implies $[-\sigma\sqrt{n}, \sigma\sqrt{n}] \cap (\mathbb{Z}/R) \subseteq \mathbb{Z}_{qR}/R - \langle \mathbf{x}', \mathbf{v} \rangle$.

We can perform a change of variable $u \leftarrow \langle \mathbf{x}', \mathbf{v} \rangle + e$ to write the last state as

$$\sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle. \quad (26)$$

Observe that [Equation \(25\)](#) and [Equation \(26\)](#) are $2^{-\Omega(n)}$ -close to each other in ℓ_2 norm for every $\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}$ such that $\|\mathbf{v}\| \leq \sqrt{nr}$. Therefore, the given state

$$\sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \leq \sqrt{nr}}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle \otimes \sum_{e \in \mathbb{Z}_{qR}/R} \rho_\sigma(e) |\langle \mathbf{s}, \mathbf{a} \rangle + \langle \mathbf{x}', \mathbf{v} \rangle + e \bmod q\rangle$$

is $2^{-\Omega(n)}$ -close to the state

$$|\Psi\rangle := \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \leq \sqrt{nr}}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle.$$

We will show $|\Psi\rangle$ is $2^{-\Omega(n)}$ -close to the state

$$|\Phi\rangle := \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| < R/2}} \rho_r(\mathbf{v}) |\mathbf{v}\rangle \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle.$$

The proof is similar to the proof of [Lemma 52](#). We first give an upper bound for $\| |\Psi\rangle - |\Phi\rangle \|^2$:

$$\begin{aligned}
\| |\Psi\rangle - |\Phi\rangle \|^2 &= \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \sqrt{n}r < \|\mathbf{v}\| < R/2}} \rho_{r/\sqrt{2}}(\mathbf{v}) \sum_{u \in \mathbb{Z}_{qR}/R} \rho_{\sigma/\sqrt{2}}(u - \langle \mathbf{x}', \mathbf{v} \rangle) \\
&\leq \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| > \sqrt{n}r}} \rho_{r/\sqrt{2}}(\mathbf{v}) \sum_{u \in \mathbb{Z}/R} \rho_{\sigma/\sqrt{2}}(u) \\
&\leq 2^{-\Omega(n)} \sum_{\mathbf{v} \in q\mathcal{L}} \rho_{r/\sqrt{2}}(\mathbf{v}) \sum_{u \in \mathbb{Z}/R} \rho_{\sigma/\sqrt{2}}(u)
\end{aligned}$$

by Banaszczyk's tail bound.

On the other hand, notice that

$$\begin{aligned}
\| |\Psi\rangle \|^2 &= \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \leq \sqrt{n}r}} \rho_{r/\sqrt{2}}(\mathbf{v}) \sum_{u \in \mathbb{Z}_{qR}/R} \rho_{\sigma/\sqrt{2}}(u - \langle \mathbf{x}', \mathbf{v} \rangle) \\
&\geq \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \leq \sqrt{n}r}} \rho_{r/\sqrt{2}}(\mathbf{v}) \sum_{u \in [-\sigma\sqrt{n}, \sigma\sqrt{n}] \cap \mathbb{Z}/R} \rho_{\sigma/\sqrt{2}}(u) \\
&\geq (1 - 2^{-\Omega(n)}) \sum_{\mathbf{v} \in q\mathcal{L}} \rho_{r/\sqrt{2}}(\mathbf{v}) \sum_{u \in \mathbb{Z}/R} \rho_{\sigma/\sqrt{2}}(u)
\end{aligned}$$

by Banaszczyk's tail bound and [Lemma 15](#) together with the fact that $r/\sqrt{2} > q\eta_\epsilon(\mathcal{L})$ for $\epsilon < 2^{-n}$.

So $\frac{\| |\Phi\rangle - |\Psi\rangle \|^2}{\| |\Phi\rangle \|^2} < 2^{-\Omega(n)}$, which implies that $|\Phi\rangle$ and $|\Psi\rangle$ are $2^{-\Omega(n)}$ -close to each other by [Lemma 24](#).

It remains to prove that $|\Phi\rangle$ is $2^{-\Omega(n)}$ -close to

$$|\Phi'\rangle := \sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}} \rho_r(\mathbf{v}) |\mathbf{v} \bmod R\rangle \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle.$$

We again use a similar argument as the proof of [Lemma 52](#).

We first give an upper bound for $\| |\Phi\rangle - |\Phi'\rangle \|^2$:

$$\begin{aligned}
\| |\Phi\rangle - |\Phi'\rangle \|^2 &\leq_{(1)} \left(\sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \geq R/2}} \rho_r(\mathbf{v}) \left\| |\mathbf{v} \bmod R\rangle \otimes \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle \right\| \right)^2 \\
&\leq_{(2)} \left(\sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| > \sqrt{nr}\sqrt{\log r}}} \rho_r(\mathbf{v}) \sqrt{\sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle)^2} \right)^2 \\
&\leq_{(3)} r^{-6n} \left(\sum_{u \in \mathbb{Z}/R} \rho_\sigma(u)^2 \right) \left(\sum_{\mathbf{v} \in q\mathcal{L}} \rho_r(\mathbf{v}) \right)^2 \\
&\leq_{(4)} r^{-4n} \left(\sum_{u \in \mathbb{Z}/R} \rho_\sigma(u)^2 \right) \det((q\mathcal{L})^*)^2 (1 + 2^{-\Omega(n)})
\end{aligned}$$

where (1) is due to triangle inequality, (2) uses that $R/2 > \sqrt{nr}\sqrt{\log r}$, (3) is due to Banaszczyk's tail bound, and (4) is due to [Lemma 15](#) and $r \geq \eta_\epsilon(q\mathcal{L})$ for $\epsilon < 2^{-n}$.

On the other hand, notice that

$$\begin{aligned}
\| |\Phi\rangle \|^2 &\geq \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \leq \sqrt{nr}}} \rho_r(\mathbf{v})^2 \sum_{u \in \mathbb{Z}_{qR}/R} \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle)^2 \\
&\geq_{(1)} (1 - 2^{-\Omega(n)}) \sum_{\substack{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}, \\ \|\mathbf{v}\| \leq \sqrt{nr}}} \rho_r(\mathbf{v})^2 \sum_{u \in \mathbb{Z}/R} \rho_\sigma(u)^2 \\
&\geq_{(2)} (1 - 2^{-\Omega(n)}) (r/\sqrt{2})^n \det((q\mathcal{L})^*) \sum_{u \in \mathbb{Z}/R} \rho_\sigma(u)^2
\end{aligned}$$

where (1) is due to Banaszczyk's tail bound and the fact that $\sigma\sqrt{n} < q/2 - |\langle \mathbf{x}', \mathbf{v} \rangle|$ when $\|\mathbf{v}\| \leq \sqrt{nr}$, and (2) is due to Banaszczyk's tail bound, [Lemma 15](#) and $r/\sqrt{2} \geq \eta_\epsilon(q\mathcal{L})$ for $\epsilon < 2^{-n}$.

Hence, we have that $\frac{\| |\Phi\rangle - |\Phi'\rangle \|^2}{\| |\Phi\rangle \|^2} < 2^{-\Omega(n)}$, this implies that $|\Phi\rangle$ and $|\Phi'\rangle$ are $2^{-\Omega(n)}$ -close to each other, by [Lemma 24](#), as desired. $\square$

A.6 Proof of [Lemma 67](#)

Proof of [Lemma 67](#). The amplitude of $|\mathbf{y}\rangle |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle$ in the given state ([Equation \(19\)](#)) is

$$\begin{aligned}
\sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}} \rho_r(\mathbf{v}) \rho_\sigma(u - \langle \mathbf{x}', \mathbf{v} \rangle) \omega_R^{\langle \mathbf{v}, \mathbf{y} \rangle} &= \rho_t(u) \sum_{\mathbf{v} \in q\mathcal{L} + \mathcal{L}\mathbf{a}} \rho_{\sqrt{\Sigma^{-1}}} \left(\mathbf{v} - \frac{r^2 u}{t^2} \mathbf{x}' \right) \omega_R^{\langle \mathbf{v}, \mathbf{y} \rangle} \\
&\propto \rho_t(u) \sum_{\mathbf{w} \in (q\mathcal{L})^* - \mathbf{y}/R} \rho_{\sqrt{\Sigma}}(\mathbf{w}) \exp \left(2\pi i \left\langle \mathcal{L}\mathbf{a} - \frac{r^2 u}{t^2} \mathbf{x}', \mathbf{w} \right\rangle \right) \omega_R^{\langle \mathcal{L}\mathbf{a}, \mathbf{y} \rangle},
\end{aligned}$$

where $t = \sqrt{\sigma^2 + r^2 \|\mathbf{x}'\|^2} \in [\sigma, \sqrt{2}\sigma)$, matrix $\Sigma = \frac{\mathbf{I}_n}{r^2} + \frac{\mathbf{x}'\mathbf{x}'^T}{\sigma^2}$ with eigenvalues $1/r^2, (t/r\sigma)^2$, and we use the Poisson Summation Formula to compute the last equality. We define the amplitude as a function

$$\phi(\mathbf{y}, u) := \rho_t(u) \sum_{\mathbf{w} \in (q\mathcal{L})^* - \mathbf{y}/R} \rho_{\sqrt{\Sigma}}(\mathbf{w}) \exp \left(2\pi i \left\langle \mathcal{L}\mathbf{a} - \frac{r^2 u}{t^2} \mathbf{x}', \mathbf{w} \right\rangle \right) \omega_R^{\langle \mathcal{L}\mathbf{a}, \mathbf{y} \rangle}.$$

Meanwhile, for $\mathbf{y}/R \in B_{(q\mathcal{L})^*}$, we define another amplitude function $\phi'(\mathbf{y}, u)$ as

$$\phi'(\mathbf{y}, u) := \rho_t(u) \rho_{\sqrt{\Sigma}}(\mathbf{z}) \exp \left(2\pi i \left\langle \mathcal{L}\mathbf{a} - \frac{r^2 u}{t^2} \mathbf{x}', -\mathbf{z} \right\rangle \right) \omega_R^{\langle \mathcal{L}\mathbf{a}, \mathbf{y} \rangle},$$

where we recall that $\mathbf{z} = \mathbf{z}(\mathbf{y}) = \mathbf{y}/R - \kappa_{(q\mathcal{L})^*}(\mathbf{y}/R)$. $\phi'(\mathbf{y}, u)$ is the leading term in $\phi(\mathbf{y}, u)$, as we will implicitly show below.

We prove that the following (unnormalized) states

$$|\Phi\rangle := \sum_{\mathbf{y} \in \mathbb{Z}_R^n} \sum_{u \in \mathbb{Z}_{qR}/R} \phi(\mathbf{y}, u) |\mathbf{y}\rangle |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle \quad (\text{the given state})$$

$$|\Phi'\rangle := \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \sum_{u \in \mathbb{Z}_{qR}/R} \phi'(\mathbf{y}, u) |\mathbf{y}\rangle |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle \quad (\text{the target state in Equation (20)})$$

are $2^{-\Omega(n)}$ -close to each other. To prove it, we need in addition the following (unnormalized) state

$$|\Phi''\rangle := \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \sum_{u \in \mathbb{Z}_{qR}/R} \phi(\mathbf{y}, u) |\mathbf{y}\rangle |\langle \mathbf{s}, \mathbf{a} \rangle + u \bmod q\rangle.$$

Let's begin by establishing upper bounds for both $\| |\Phi''\rangle - |\Phi'\rangle \|^2$ and $\| |\Phi\rangle - |\Phi''\rangle \|^2$. The first term is relatively simpler to bound: for any $\epsilon < 2^{-n}$, according to Lemma 75 and the assumption that $\frac{r\sigma}{t} > \frac{r}{\sqrt{2}} > 2\sqrt{2}\eta_\epsilon(q\mathcal{L})$, we get that

$$\begin{aligned} \| |\Phi''\rangle - |\Phi'\rangle \|^2 &= \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \sum_{u \in \mathbb{Z}_{qR}/R} |\phi(\mathbf{y}, u) - \phi'(\mathbf{y}, u)|^2 \\ &\leq \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \sum_{u \in \mathbb{Z}_{qR}/R} \left(\rho_t(u) \sum_{\mathbf{w} \in (q\mathcal{L})^* \setminus \{\mathbf{0}\}} \rho_{t/r\sigma}(\mathbf{w} - \mathbf{z}) \right)^2 \\ &< \epsilon^2 \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u)^2 \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \rho_{t/r\sigma}(\mathbf{z}). \end{aligned} \tag{27}$$

To establish an upper bound for the latter term $\| |\Phi\rangle - |\Phi''\rangle \|^2$, according to Lemma 75 and

the fact that $\frac{r\sigma}{t} > 2\sqrt{2}\eta_\epsilon(q\mathcal{L})$, we get that for $\mathbf{y} \in \mathbb{Z}_R^n \setminus R \cdot B_{(q\mathcal{L})^*}$,

$$\begin{aligned} \sum_{\mathbf{w} \in (q\mathcal{L})^* - \mathbf{y}/R} \rho_{\sqrt{\Sigma}}(\mathbf{w}) &\leq \sum_{\mathbf{w} \in (q\mathcal{L})^* - \mathbf{y}/R} \rho_{t/r\sigma}(\mathbf{w}) \\ &\leq \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(t/r\sigma)^2}\right) + \epsilon \cdot \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{2(t/r\sigma)^2}\right) \\ &\leq \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{2(t/r\sigma)^2}\right) \left(\exp\left(-\pi \frac{(\lambda_1((q\mathcal{L})^*)/2)^2}{4/r^2}\right) + \epsilon\right) \\ &\leq 2^{-n+1} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{2(t/r\sigma)^2}\right) \end{aligned}$$

where the last inequality holds since $\lambda_1((q\mathcal{L})^*) \geq \sqrt{\frac{n \ln 2}{\pi}} \cdot \frac{1}{q\eta_\epsilon(\mathcal{L})} \geq \sqrt{\frac{n \ln 2}{\pi}} \cdot \frac{4}{r}$ by [Lemma 14](#). Therefore,

$$\begin{aligned} \|\Phi\rangle - \Phi''\rangle\|^2 &= \sum_{\mathbf{y} \in \mathbb{Z}_R^n \setminus R \cdot B_{(q\mathcal{L})^*}} \sum_{u \in \mathbb{Z}_{qR}/R} |\phi(\mathbf{y}, u)|^2 \\ &\leq \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u)^2 \sum_{\mathbf{y} \in \mathbb{Z}_R^n \setminus R \cdot B_{(q\mathcal{L})^*}} \left(\sum_{\mathbf{w} \in (q\mathcal{L})^* - \mathbf{y}/R} \rho_{\sqrt{\Sigma}}(\mathbf{w}) \right)^2 \\ &\leq 2^{-2n+2} \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u)^2 \sum_{\mathbf{y} \in \mathbb{Z}_R^n \setminus R \cdot B_{(q\mathcal{L})^*}} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(t/r\sigma)^2}\right). \end{aligned} \tag{28}$$

Combine the two upper bounds in [Equation \(27\)](#) and [Equation \(28\)](#), we get that

$$\begin{aligned} \|\Phi\rangle - \Phi'\rangle\|^2 &\leq 2\|\Phi\rangle - \Phi''\rangle\|^2 + 2\|\Phi''\rangle - \Phi'\rangle\|^2 \\ &\leq 2^{-2n+3} \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u)^2 \sum_{\mathbf{y} \in \mathbb{Z}_R^n} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(t/r\sigma)^2}\right). \end{aligned} \tag{29}$$

On the other hand, we can compute that

$$\begin{aligned} \|\Phi'\rangle\|^2 &= \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \sum_{u \in \mathbb{Z}_{qR}/R} |\phi'(\mathbf{y}, u)|^2 \\ &\geq \sum_{u \in \mathbb{Z}_{qR}/R} \rho_t(u)^2 \sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(1/\sqrt{2}r)^2}\right). \end{aligned} \tag{30}$$

Combining with the bounds in [Equation \(29\)](#), we get that

$$\frac{\|\Phi\rangle - \Phi'\rangle\|^2}{\|\Phi'\rangle\|^2} \leq 2^{-2n+3} \frac{\sum_{\mathbf{y} \in \mathbb{Z}_R^n} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(t/r\sigma)^2}\right)}{\sum_{\mathbf{y} \in \mathbb{Z}_R^n \cap R \cdot B_{(q\mathcal{L})^*}} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(1/\sqrt{2}r)^2}\right)}. \tag{31}$$

Now let's establish an upper bound on the ratio between the two summations on the right-hand side of the above inequality. We accomplish this through the following steps:

1. Expanding the support of $\mathbf{y}$. Given our assumption that $\mathcal{L}$ is an integer lattice, we have $(q\mathcal{L})^* + \mathbf{k} = (q\mathcal{L})^*$ for any $\mathbf{k} \in \mathbb{Z}^n$. Consequently, $\text{dist}((\mathbf{y} - R\mathbf{k})/R, (q\mathcal{L})^*) = \text{dist}(\mathbf{y}/R, (q\mathcal{L})^* + \mathbf{k}) = \text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)$. Therefore, we can expand the support of $\mathbf{y}$ from $\mathbb{Z}_R^n$ to $\mathbb{Z}_{RN}^n$ for any $N \in \mathbb{N}_+$, which is a combination of N^n hypercubes, each in the form $\mathbb{Z}_R^n + R\mathbf{k}$, i.e.

$$\frac{\|\Phi\rangle - \|\Phi'\rangle\|^2}{\|\Phi'\rangle\|^2} \leq 2^{-2n+3} \lim_{N \rightarrow +\infty} \frac{\sum_{\mathbf{y} \in \mathbb{Z}_{RN}^n} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(t/r\sigma)^2}\right)}{\sum_{\mathbf{y} \in \mathbb{Z}_{RN}^n \cap R \cdot B_{(q\mathcal{L})^*}} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(1/\sqrt{2}r)^2}\right)}.$$

2. Bound the numerator and denominator as N approaches infinity. Assume that N is a sufficiently large integer, and denote $\ell_{\max} = \max_{\mathbf{y} \in \mathbb{R}^n} \text{dist}(\mathbf{y}, (q\mathcal{L})^*) < +\infty$. For the numerator, when considering $\mathbf{y} \in \mathbb{Z}_{RN}^n$, the closest vector from $\mathbf{y}/R$ to the lattice $(q\mathcal{L})^*$ will have ℓ_{∞} norm at most $RN/2 + \ell_{\max}$. Thus

$$\sum_{\mathbf{y} \in \mathbb{Z}_{RN}^n} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(t/r\sigma)^2}\right) \leq \sum_{\substack{\mathbf{u} \in (q\mathcal{L})^*, \\ \|\mathbf{u}\|_{\infty} \leq RN/2 + \ell_{\max}}} \sum_{\mathbf{y} \in \mathbb{Z}^n} \rho_{t/r\sigma}(\mathbf{y}/R - \mathbf{u}).$$

Similarly, for the denominator, when considering $\mathbf{u} \in (q\mathcal{L})^*$ with ℓ_{∞} norm at most $RN/2 - \ell_{\max}$, the vectors $\mathbf{y} \in \mathbb{Z}^n \cap R \cdot B_{(q\mathcal{L})^*}$ for which the closest vector to $(q\mathcal{L})^*$ is $\mathbf{u}$ will certainly have an ℓ_{∞} norm at most $RN/2$. Thus

$$\begin{aligned} & \sum_{\mathbf{y} \in \mathbb{Z}_{RN}^n \cap R \cdot B_{(q\mathcal{L})^*}} \exp\left(-\pi \frac{\text{dist}(\mathbf{y}/R, (q\mathcal{L})^*)^2}{(1/\sqrt{2}r)^2}\right) \\ & \geq \sum_{\substack{\mathbf{u} \in (q\mathcal{L})^*, \\ \|\mathbf{u}\|_{\infty} \leq RN/2 - \ell_{\max}}} \sum_{\substack{\mathbf{y} \in \mathbb{Z}^n, \\ \|\mathbf{y}/R - \mathbf{u}\| < \lambda_1((q\mathcal{L})^*)/2}} \rho_{1/\sqrt{2}r}(\mathbf{y}/R - \mathbf{u}) \\ & \geq \sum_{\substack{\mathbf{u} \in (q\mathcal{L})^*, \\ \|\mathbf{u}\|_{\infty} \leq RN/2 - \ell_{\max}}} \left(\sum_{\mathbf{y} \in \mathbb{Z}^n} \rho_{1/\sqrt{2}r}(\mathbf{y}/R - \mathbf{u}) - 2^{-\Omega(n)} \rho_{1/\sqrt{2}r}(\mathbb{Z}^n/R) \right), \end{aligned}$$

where the final inequality is based on Banaszczyk's tail bound ([Lemma 10](#)), with the guarantee that $\lambda_1((q\mathcal{L})^*)/2 > \sqrt{\frac{n \ln 2}{\pi}} \cdot \frac{2}{r} > \frac{2}{\sqrt{2\pi}} \cdot \frac{1}{\sqrt{2}r} \sqrt{n}$.

According to [Lemma 13](#), it follows that $\eta_{2^{-n}}(\mathbb{Z}^n/R) \leq \frac{1}{R} \cdot \sqrt{\frac{2n}{\pi}} < \frac{1}{\sqrt{2}r} < \frac{t}{r\sigma}$. Therefore, by using [Lemma 15](#), we can conclude that

$$\begin{aligned} \sum_{\mathbf{y} \in \mathbb{Z}^n} \rho_{t/r\sigma}(\mathbf{y}/R - \mathbf{u}) & \leq (1 + 2^{-\Omega(n)}) \cdot R^n \cdot (t/r\sigma)^n, \\ \sum_{\mathbf{y} \in \mathbb{Z}^n} \rho_{1/\sqrt{2}r}(\mathbf{y}/R - \mathbf{u}) - 2^{-\Omega(n)} \rho_{1/\sqrt{2}r}(\mathbb{Z}^n/R) & \geq (1 - 2^{-\Omega(n)}) \cdot R^n \cdot (1/\sqrt{2}r)^n. \end{aligned}$$

3. Establish an upper bound for the right hand side of [Equation \(31\)](#). By combining the in-

equalities in the previous step, we obtain that

$$\begin{aligned}
\frac{\| |\Phi\rangle - |\Phi'\rangle \|^2}{\| |\Phi'\rangle \|^2} &\leq 2^{-2n+3}(1 + 2^{-\Omega(n)}) \cdot (\sqrt{2}t/\sigma)^n \cdot \lim_{N \rightarrow +\infty} \frac{\#\{\mathbf{u} \in (q\mathcal{L})^* : \|\mathbf{u}\|_\infty \leq RN/2 + \ell_{\max}\}}{\#\{\mathbf{u} \in (q\mathcal{L})^* : \|\mathbf{u}\|_\infty \leq RN/2 - \ell_{\max}\}} \\
&\leq 2^{-n+3}(1 + 2^{-\Omega(n)}) \cdot \lim_{N \rightarrow +\infty} \frac{\#\{\mathbf{u} \in (q\mathcal{L})^* : \|\mathbf{u}\|_\infty \leq RN/2 + \ell_{\max}\}}{\#\{\mathbf{u} \in (q\mathcal{L})^* : \|\mathbf{u}\|_\infty \leq RN/2 - \ell_{\max}\}} \\
&= 2^{-n+3}(1 + 2^{-\Omega(n)}) \cdot \lim_{N \rightarrow +\infty} \frac{(RN/2 + \ell_{\max})^n}{(RN/2 - \ell_{\max})^n} \\
&= 2^{-n+3}(1 + 2^{-\Omega(n)}).
\end{aligned}$$

Consequently, we can infer that the provided state $|\Phi\rangle$ is $2^{-\Omega(n)}$ -close to $|\Phi'\rangle$, according to [Lemma 24](#), as desired. $\square$

Remark 77. *The condition of this lemma can be relaxed if we instead use the proof technique from Claim A.5 in [\[Reg23\]](#). To be more specific, the above lemma holds as long as $\beta q\sqrt{m} + \alpha\gamma qm < \lambda_1(\mathcal{L}_q(\mathbf{A}))/2$ and $q/2 > \alpha\sqrt{m}$. However, the relaxed condition provides similar parameters in the overall reduction, and therefore, we do not emphasize it here.*